

PMath 453: Functional Analysis with Prof. Camila Sehnem

The following are notes I took for PMath 453 at the University of Waterloo in Fall 2024. They are largely “as-is” from the lectures, with some details of some arguments filled in by me. I would like to thank my good friend Liam Benoit for sharing his notes with me for lectures that I missed, and for his help working out some of the arguments, I thank Liam Gardner for sharing his notes with me to help fill in some details, and I thank Alwin Chen and Michael Alexander for pointing out numerous typos. Any errors that remain are entirely my fault, please contact me if you find any.

1: Banach Spaces

$\mathbb{F}$ denotes a field that is either $\mathbb{R}$ or $\mathbb{C}$.

Definition 1.1

Let V be a vector space over $\mathbb{F}$. A norm is a function $\|\cdot\| : V \rightarrow \mathbb{R}$ satisfying for all $v, w \in V, \alpha \in \mathbb{F}$:

(N1): $\|v\| \geq 0$ with equality if and only if $v = 0$. (Strict positivity)

(N2): $\|\alpha v\| = |\alpha| \|v\|$. (Absolute homogeneity)

(N3): $\|v + w\| \leq \|v\| + \|w\|$. (The triangle inequality)

Properties (N2) and (N3) imply that $\|\cdot\|$ satisfies for all $v, w \in V$:

(N4) $\|v\| - \|w\| \leq \|v - w\|$. (Reverse triangle inequality)

If $\|\cdot\|$ is a norm on V , we call $(V, \|\cdot\|)$ a normed space.

A norm $\|\cdot\|$ on V induces a metric on V given by

$$d(v, w) = \|v - w\|$$

for all $v, w \in V$. The topology induced by this metric is called the norm topology on V .

Definition 1.2

We say that $(V, \|\cdot\|)$ is a Banach space if V is complete with respect to the metric induced by the norm.

Examples of Banach Spaces

(a) The Euclidean space $(\mathbb{F}^n, \|\cdot\|)$ where $\|x\| = \sqrt{|x_1|^2 + \cdots + |x_n|^2}$ for $x = (x_1, \dots, x_n) \in \mathbb{F}^n$. We can also use the p -norm ($1 \leq p < \infty$), $(\mathbb{F}^n, \|\cdot\|_p)$ with $\|x\|_p = (|x_1|^p + \cdots + |x_n|^p)^{1/p}$ and the ∞ -norm $\|x\|_\infty = \max_{1 \leq i \leq n} |x_i|$.

(b) The space $\ell^p = \{(x_n)_{n=1}^\infty \subseteq \mathbb{F} : \sum_{n=1}^\infty |x_n|^p < \infty\}$. The norm on ℓ^p is $\|(x_n)_{n=1}^\infty\| = (\sum_{n=1}^\infty |x_n|^p)^{1/p}$. Similarly ℓ^∞ (the space of bounded sequences) is a Banach space with $\|(x_n)_{n=1}^\infty\| = \sup_{n \geq 1} |x_n|$.

(c) Let $X \subseteq \mathbb{R}^n$ be Lebesgue measurable. Let

$$L^p(X) = \{f : X \rightarrow \mathbb{R} \text{ measurable} : \int_X |f|^p < \infty\} / \sim$$

where $f \sim g$ if $f = g$ a.e. The norm on $L^p(X)$ is $\|f\|_p = (\int_X |f|^p)^{1/p}$. Similarly, we have

$$L^\infty(X) := \{f : X \rightarrow \mathbb{R} \text{ measurable} : f \text{ is essentially bounded}\} / \sim$$

where f is essentially bounded if there exists $c > 0$ such that $m(\{x \in X : |f(x)| \geq c\}) = 0$, where m is the Lebesgue measure (so functions that are bounded except to a set of measure zero), where the norm on $L^\infty(X)$ is

$$\|f\|_\infty = \inf\{c > 0 : m(\{x \in X : |f(x)| \geq c\}) = 0\}$$

(d) Let X be a topological space, and let

$$C_b(X) := \{f : X \rightarrow \mathbb{F} : f \text{ continuous and bounded}\}$$

where its norm is the supremum norm $\|f\|_\infty = \sup_{x \in X} |f(x)|$.

(e) We say that a continuous function $f : X \rightarrow \mathbb{F}$ vanishes at infinity if for every $\varepsilon > 0$, the set $\{x \in X : |f(x)| \geq \varepsilon\}$ is compact. We let $C_o(X)$ denote the space of continuous functions that vanish at infinity.

Lemma

$C_o(X) \subseteq C_b(X)$ and $C_o(X)$ is a Banach space with $\|\cdot\|_\infty$.

Proof:

The inclusion $C_o(X) \subseteq C_b(X)$ follows because continuous functions on compact sets are bounded (Extreme Value Theorem), so we can take an arbitrary $\varepsilon_1 > 0$, and take the upper bound given by the extreme value theorem on the set $\{x \in X : |f(x)| \geq \varepsilon_1\}$, since $|f|$ is smaller than ε_1 outside this set.

To see that $C_o(X)$ is a Banach space, since $C_b(X)$ is a Banach space (by the $3 \times \varepsilon/3$ argument), it suffices to show it is closed in $C_b(X)$ (closed subspaces of Banach spaces are themselves Banach spaces by 351). Let $(f_n)_{n=1}^\infty \subseteq C_o(X)$ with $\lim_{n \rightarrow \infty} f_n = f$ for $f \in C_b(X)$. Let $\varepsilon > 0$ and $n \in \mathbb{N}$ such that $\|f - f_n\|_\infty < \varepsilon/2$ for all $n \geq N$.

Suppose $|f(x)| \geq \varepsilon$. Then

$$|f(x)| \leq |f(x) - f_N(x)| + |f_N(x)|$$

so

$$|f_N(x)| \geq |f(x)| - |f(x) - f_N(x)| \geq \varepsilon - \varepsilon/2 = \varepsilon/2$$

Hence

$$\{x \in X : |f(x)| \geq \varepsilon\} \subseteq \{x \in X : |f_N(x)| \geq \varepsilon/2\}$$

The right hand side is compact, and the left hand side is closed by continuity, which shows the left hand side is compact, hence $f \in C_o(X)$. ■

Non-examples of Banach Spaces

(a) $P[0, 1] := \{p : [0, 1] \rightarrow \mathbb{F} : p \text{ polynomial}\}$ with the norm $\|\cdot\|_\infty$ is a normed space but not a Banach space: it has Cauchy sequences that do not converge to polynomials, as seen by taking any Taylor series that does not converge to polynomials, for example $f(x) = \sum_{n=0}^\infty \frac{x^n}{2^n}$ for $x \in [0, 1]$ satisfies $f = \lim_{m \rightarrow \infty} p_m$, where $p_m(x) = \sum_{n=0}^m \frac{x^n}{2^n}$ (by the Weierstrass M-test), but is not a polynomial since its n th derivative is not zero for any $n \in \mathbb{N}$.

(b) Let $C^1([0, 1])$ be the space of continuously differentiable functions on $[0, 1]$. Then with the supremum norm $\|\cdot\|_\infty$, $(C^1([0, 1]), \|\cdot\|_\infty)$ is a normed space but not a Banach space. However with the norm

$$\|f\|_{C^1([0, 1])} = \max\{\|f\|_\infty, \|f'\|_\infty\}$$

it is a Banach space because:

Theorem

If $(f_n)_{n=1}^\infty$ is a sequence of differentiable functions on $[0, 1]$ such that if $(f_n(x_o))_{n=1}^\infty$ converges for some $x_o \in [0, 1]$, and $(f'_n)_{n=1}^\infty$ converges uniformly to some g , then $(f_n)_{n=1}^\infty$ converges uniformly and $(\lim_{n \rightarrow \infty} f_n)' = g$.

Proof:

See Section 6.3 of Abbott's book Understanding Analysis. ■

Example

Consider the space $C_o(X)$ with $X = \mathbb{R}$. Then $C_o(\mathbb{R}) = \{f(x) \in C(\mathbb{R}) : \lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow -\infty} f(x) = 0\}$ and more generally for $-\infty \leq a < b \leq \infty$, $C_o((a, b)) = \{f \in C((a, b)) : \lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow b} f(x) = 0\}$.

Lemma 1.3 (Equivalence of norms)

Let V be a vector space over $\mathbb{F}$. Let $\|\cdot\|_1$ and $\|\cdot\|_2$ be norms on V . Then $\|\cdot\|_1$ and $\|\cdot\|_2$ induce the

same topology on V if and only if there exists a constant $c > 0$ such that

$$\frac{1}{c} \|v\|_2 \leq \|v\|_1 \leq c \|v\|_2$$

for all $v \in V$.

Proof:

In the forward direction, suppose $\|\cdot\|_1$ and $\|\cdot\|_2$ induce the same topology. Then the unit open ball with respect to $\|\cdot\|_1$, $B_1^{\|\cdot\|_1}(0)$, is open with respect to $\|\cdot\|_2$. Thus there exists $r > 0$ such that $B_r^{\|\cdot\|_2}(0) \subseteq B_1^{\|\cdot\|_1}(0)$. Let $0 \neq v \in V$. We have

$$\frac{r}{2} \frac{v}{\|v\|_2} \in B_r^{\|\cdot\|_2}(0) \subseteq B_1^{\|\cdot\|_1}(0)$$

and so

$$\frac{r}{2} \frac{\|v\|_1}{\|v\|_2} < 1$$

implying

$$\|v\|_1 < \frac{2}{r} \|v\|_2$$

setting $c_1 = \frac{2}{r}$, we get $\|v\|_1 \leq c_1 \|v\|_2$ for all $v \in \mathbb{R}$. Reversing the roles of $\|\cdot\|_1$ and $\|\cdot\|_2$, we get a constant $c_2 > 0$ such that $\|v\|_2 \leq c_2 \|v\|_1$ for all $v \in V$. Setting $c = \max\{c_1, c_2\} > 0$, we obtain

$$\frac{1}{c} \|v\|_2 \leq \|v\|_1 \leq c \|v\|_2$$

for all $v \in V$, as desired.

Conversely, suppose such a constant $c > 0$ exists. Let $v \in V$ be arbitrary. The inequality $\frac{1}{c} \|v\|_2 \leq \|v\|_1$ implies that for all $\varepsilon > 0$, $B_\varepsilon^{\|\cdot\|_1}(v) \subseteq B_{c\varepsilon}^{\|\cdot\|_2}(v)$. (To see this inequality, if $\|y - v\|_1 < \varepsilon$, then $\|y - v\|_2 \leq c\|y - v\|_1 < c\varepsilon$). The other inequality $\|v\|_1 \leq c\|v\|_2$ implies that for all $\varepsilon > 0$, $B_\varepsilon^{\|\cdot\|_2}(v) \subseteq B_{c\varepsilon}^{\|\cdot\|_1}(v)$. Therefore, the metric topologies of $(V, \|\cdot\|_1)$ and $(V, \|\cdot\|_2)$ have the same sets of open balls, and are thus equivalent. ■

If any of the equivalent conditions in Lemma 1.3 hold, we say that $\|\cdot\|_1$ and $\|\cdot\|_2$ are equivalent.

Proposition 1.4

Any two norms on $\mathbb{F}^n$ are equivalent.

Proof:

Let $|||\cdot|||$ be a norm on $\mathbb{F}^n$. It suffices to show $|||\cdot|||$ is equivalent to the Euclidean norm $\|\cdot\|$. Let $x = (x_1, \dots, x_n) \in \mathbb{F}^n$. Then

$$|||x||| \leq \sum_{i=1}^n |x_i| |||e_i|||$$

where e_i is the i th standard basis vector. Then

$$\sum_{i=1}^n |x_i| |||e_i||| \leq \sum_{i=1}^n |x_i| \max_{i=1}^n |||e_i||| \stackrel{(\dagger)}{\leq} n \|x\| \max_{i=1}^n |||e_i|||$$

where $(\dagger)$ holds because $|x_i| = \sqrt{|x_i|^2} \leq \sqrt{\sum_{j=1}^n |x_j|^2} = \|x\|$. Letting $c_1 = n \max_{i=1}^n |||e_i|||$, we thus get $|||x||| \leq c_1 \|x\|$.

Now consider $S = \partial B_1(0)$ (with respect to $\|\cdot\|$). Then S is compact with respect to the Euclidean topology on $\mathbb{F}^n$. Consider the function $x \in S \mapsto |||x|||$. This is continuous with respect to $\|\cdot\|$ because

$$|||x||| - |||y||| \leq |||x - y||| \leq c_1 \|x - y\|$$

for all $x, y \in S$. Therefore, by the Extreme Value Theorem, there exists $p \in S$ such that $\|p\| \leq \|x\|$ for all $x \in S$. Let $x \neq 0 \in \mathbb{F}^n$ be arbitrary. Then $\frac{x}{\|x\|} \in S$ implying $\frac{\|x\|}{\|x\|} \geq \|p\|$ giving $\|x\| \leq c_2 \|x\|$ where $c_2 = 1/\|p\|$. Taking $c = \max\{c_1, c_2\}$ gives the result. ■

Corollary 1.5

Let $(V, \|\cdot\|)$ be a normed space and $W \subseteq V$ be a finite-dimensional subspace of V . Then W is closed.

Proof:

Let $\{w_1, \dots, w_n\}$ be a basis for W , and let $\alpha_1, \dots, \alpha_n \in \mathbb{F}$ be arbitrary. Consider the norm on W defined by $\|\sum_{i=1}^n \alpha_i w_i\|_1 = \sum_{i=1}^n |\alpha_i|$. By Proposition 1.4, there exists $c > 0$ such that

$$\frac{1}{c} \sum_{i=1}^n |\alpha_i| \leq \|\sum_{i=1}^n \alpha_i w_i\| \leq c \sum_{i=1}^n |\alpha_i|$$

With this, we prove W is closed in V . Let $(w^{(k)})_{k \in \mathbb{N}}$ be a sequence in W converging to some $v \in V$. Then $(w^{(k)})_{k \in \mathbb{N}}$ is a Cauchy sequence. For each $k \in \mathbb{N}$, write $w^{(k)} = \alpha_1^{(k)} w_1 + \dots + \alpha_n^{(k)} w_n$. We claim $(\alpha_j^{(k)})_{k \in \mathbb{N}}$ is a Cauchy sequence in $\mathbb{F}$. Indeed, for $m, N \in \mathbb{N}$ and fixed $j \in \{1, \dots, n\}$, by the left hand side of the displayed inequality,

$$|\alpha_j^{(m)} - \alpha_j^{(N)}| \leq \sum_{i=1}^n |\alpha_i^{(m)} - \alpha_i^{(N)}| \leq c \|\sum_{i=1}^n (\alpha_i^{(m)} - \alpha_i^{(N)}) w_i\| = c \|w^{(m)} - w^{(N)}\| \rightarrow 0$$

using the fact that $(w^{(k)})_{k \in \mathbb{N}}$ is Cauchy. Therefore, by completeness of $\mathbb{F}$, we get some $\alpha_j := \lim_{k \rightarrow \infty} \alpha_j^{(k)} \in \mathbb{F}$. Set $w := \alpha_1 w_1 + \dots + \alpha_n w_n$. We claim $(w^{(k)})_{k \in \mathbb{N}}$ converges to w . Indeed, for $k \in \mathbb{N}$,

$$\|w^{(k)} - w\| = \|\sum_{i=1}^n \alpha_i^{(k)} w_i - \sum_{i=1}^n \alpha_i w_i\| = \|\sum_{i=1}^n (\alpha_i^{(k)} - \alpha_i) w_i\| \leq c \sum_{i=1}^n |\alpha_i^{(k)} - \alpha_i| \rightarrow 0$$

by convergence of $(\alpha_i^{(k)})_{k \in \mathbb{N}}$ to α_i . Therefore, by uniqueness of limits, we have $v = w \in W$, so W is closed. ■

Quotient Spaces

Let $(V, \|\cdot\|)$ be a normed space and let $W \subseteq V$ be a closed subspace. Consider the quotient space $V/W = \{v+W : v \in V\}$. This is a vector space under the operations $(v_1+W) + (v_2+W) = (v_1+v_2)+W$, $\alpha(v+W) = \alpha v+W$. We define a norm $\|\cdot\|_{V/W}$ on V/W by $\|v+W\|_{V/W} = \inf_{w \in W} \|v+w\|$.

Theorem 1.6

$(V/W, \|\cdot\|_{V/W})$ is a normed space, and if $(V, \|\cdot\|)$ is Banach, then so is $(V/W, \|\cdot\|_{V/W})$.

Proof:

To prove $\|\cdot\|_{V/W}$ is well-defined, suppose $v_1+W = v_2+W$. Then $v_1 - v_2 \in W$. For each $w \in W$, we have

$$\|v_1+w\| = \|v_2+(v_1-v_2)+w\| \geq \inf_{w \in W} \|v_2+w\| = \|v_2+W\|_{V/W}$$

so taking infimums, we have $\|v_1+W\|_{V/W} \geq \|v_2+W\|_{V/W}$. Interchanging v_1 and v_2 , we obtain $\|v_1+W\|_{V/W} = \|v_2+W\|_{V/W}$, so $\|\cdot\|_{V/W}$ is well-defined. Now, we verify it is a norm.

To show (N1), it is clear by definition that $\|v+W\|_{V/W} \geq 0$ for all $v \in V$, so suppose $\|v+W\|_{V/W} = 0$. Then $\inf_{w \in W} \|v+w\| = 0$, which tells us that for each $n \in \mathbb{N}$, there exists $w_n \in W$ such that $\|v+w_n\| < 1/n$ and $(w_n)_{n \in \mathbb{N}}$ converges, so $w_n \rightarrow v$. By closure of W , this shows $v \in W$, so $v+W = 0+W$.

To show (N2), let $v \in V$ and $\alpha \in \mathbb{F}$. If $\alpha = 0$, this is clear, otherwise, if $\alpha \neq 0$, then

$$\|\alpha(v+W)\|_{V/W} = \inf_{w \in W} \|\alpha v+w\| = \inf_{w \in W} |\alpha| \|v + \frac{w}{\alpha}\| = |\alpha| \inf_{w' \in W} \|v+w'\| = |\alpha| \|v+W\|_{V/W}$$

as desired.

For (N3), let $v_1, v_2 \in V$. Then for any $w_1, w_2 \in W$,

$$\|v_1 + v_2 + W\|_{V/W} = \inf_{w \in W} \|v_1 + v_2 + w\| \leq \|(v_1 + w_1) + (v_2 + w_2)\| \leq \|v_1 + w_1\| + \|v_2 + w_2\|$$

So taking infimums of the right hand side, we obtain

$$\|v_1 + v_2 + W\|_{V/W} \leq \|v_1 + W\|_{V/W} + \|v_2 + W\|_{V/W}$$

Now, suppose $(V, \|\cdot\|)$ is a Banach space. Let $(v_n + W)_{n \in \mathbb{N}} \subseteq V/W$ be a Cauchy sequence. It suffices to show $(v_n + W)_{n \in \mathbb{N}}$ has a convergent subsequence. Passing to a subsequence, we may assume

$$\|(v_n + W) - (v_{n+1} + W)\|_{V/W} < \frac{1}{2^n}$$

for all $n \in \mathbb{N}$. The idea is to find a sequence in W such that $(v_n + w_n)_{n \in \mathbb{N}}$ is Cauchy and use the fact that V is Banach. Set $w_1 = 0$. Since

$$\inf_{w \in W} \|v_1 - v_2 + W\| = \|(v_1 + W) - (v_2 + W)\|_{V/W} < \frac{1}{2}$$

we can find $w_2 \in W$ such that $\|v_1 - (v_2 + w_2)\| < \frac{1}{2}$. Applying the same argument with $v_2 - v_3 + W = v_2 + w_2 - v_3 + W$ we can find $w_3 \in W$ such that $\|v_2 + w_2 - (v_3 + w_3)\| < \frac{1}{2^2}$. More generally, the same argument shows for $n \geq 3$ that we can find a sequence $(w_n)_{n \in \mathbb{N}}$ contained in W such that $\|(v_n + w_n) - (v_{n+1} + w_{n+1})\| < \frac{1}{2^n}$ for all $n \in \mathbb{N}$. Hence $(v_n + w_n)_{n \in \mathbb{N}}$ is a Cauchy sequence in V , so since it is Banach, there exists $v := \lim_{n \rightarrow \infty} (v_n + w_n) \in V$. We claim $v_n + W \rightarrow v + W$. Indeed, let $\varepsilon > 0$ and $N \in \mathbb{N}$ large enough so that $\|v - (v_n + w_n)\| < \varepsilon$ for all $n \geq N$. Then,

$$\|v + W - (v_n + W)\|_{V/W} \stackrel{\inf}{\leq} \|v - (v_n + w_n)\| < \varepsilon$$

for all $n \geq N$. Hence, $(V/W, \|\cdot\|_{V/W})$ is a Banach space. ■

Remark

The above proof shows that (N1) is what will not hold for $\|\cdot\|_{V/W}$ if W is not closed, making $\|\cdot\|_{V/W}$ into a seminorm.

Product Spaces

Let $(V, \|\cdot\|)$ and $(W, \|\cdot\|)$ be normed spaces. Their direct sum is $V \oplus W = \{(v, w) : v \in V, w \in W\}$ which has p -norm $\|(v, w)\|_p = (\|v\|^p + \|w\|^p)^{1/p}$ for $1 \leq p < \infty$. More generally,

Definition

Let $(V_\lambda, \|\cdot\|)_{\lambda \in \Lambda}$ be a collection of normed spaces. The ℓ^p direct sum/product of $(V_\lambda, \|\cdot\|)_{\lambda \in \Lambda}$ is

$$\oplus_p V_\lambda := \{(v_\lambda)_{\lambda \in \Lambda} \in \prod_{\lambda \in \Lambda} V_\lambda : \left(\sum_{\lambda \in \Lambda} \|v_\lambda\|^p \right)^{1/p} < \infty\}$$

for $1 \leq p < \infty$ (where the sum in the definition to make sense requires $\|v_\lambda\|$ to have countable support as λ ranges over Λ). With the p -norm $\|(v_\lambda)_{\lambda \in \Lambda}\|_p = \left(\sum_{\lambda \in \Lambda} \|v_\lambda\|^p \right)^{1/p}$, this is a normed space.

For the case $p = \infty$, we instead use the norm $\sup_{\lambda \in \Lambda} \|v_\lambda\|$ (this doesn't require countable support of $\|v_\lambda\|$ over Λ). If $\Lambda = \mathbb{N}$, we define,

$$\oplus_0 V_n = \{(v_n)_{n \in \mathbb{N}} \in \prod_{n=1}^{\infty} V_n : \lim_{n \rightarrow \infty} \|v_n\| = 0\}$$

with the $\|\cdot\|_\infty$ norm.

Proposition 1.8

$(\oplus_p V_\lambda, \|\cdot\|_p)$ for $1 \leq p \leq \infty$ and $(\oplus_0 V_\lambda, \|\cdot\|_0)$ are normed spaces, which are Banach spaces if and only if each $(V_\lambda, \|\cdot\|_\lambda)_{\lambda \in \Lambda}$ are Banach spaces.

Proof:

Exercise. ■

Linear operators on Banach spaces

Definition 1.9

Let $(V, \|\cdot\|)$ and $(W, \|\cdot\|)$ be normed spaces. Let $\mathcal{L}(V, W)$ be the set of all linear maps from V to W (called linear operators often in this context).

For $T \in \mathcal{L}(V, W)$ we define its operator norm,

$$\|T\|_{op} = \sup\{\|Tv\| : v \in V, \|v\| = 1\} \in [0, \infty]$$

We say that T is bounded if $\|T\|_{op} < \infty$, and denote the set of bounded linear operators from V to W by $\mathcal{B}(V, W)$. If $V = W$, we just denote this by $\mathcal{B}(V)$.

Note

There are many equivalent definitions of the operator norm.

$$\begin{aligned} \|T\|_{op} &= \sup\{\|Tv\| : v \in V, \|v\| = 1\} \\ &= \sup\{\|Tv\| : v \in V, \|v\| \leq 1\} \\ &= \sup\{\|Tv\| : v \in V, \|v\| < 1\} \end{aligned}$$

Exercise (Assignment 1)

$T \in \mathcal{L}(V, W)$ is bounded if and only if there exists a constant $M > 0 \in \mathbb{R}$ such that $\|Tv\| \leq M\|v\|$ for all $v \in V$, in which case $\|T\|_{op}$ is the smallest such constant.

Theorem 1.10

Let $T \in \mathcal{L}(V, W)$. The following are equivalent:

- (1) T is continuous.
- (2) T is continuous at some $p_0 \in V$.
- (3) T is bounded.

Proof:

(1) implies (2) is clear, to see (2) implies (1), let $x_0 \in V$ be arbitrary and let $\varepsilon > 0$. By continuity at p_0 , there exists $\delta > 0$ such that if $p \in V$ satisfies $\|p - p_0\| < \delta$, then $\|T(p) - T(p_0)\| < \varepsilon$. Thus, if we let $x \in V$ such that $\|x - x_0\| < \delta$, then $\|p_0 - (p_0 + (x - x_0))\| = \|x - x_0\| < \delta$, so

$$\|T(x) - T(x_0)\| = \|T(p_0) - T(p_0 + (x - x_0))\| < \varepsilon$$

so we have continuity everywhere.

To see (1) implies (3), we use continuity of T at 0: there exists $\delta > 0$ such that $\|x\| < \delta$ implies $\|T(x)\| < 1$. Thus for $\|x\| < 1$ we have

$$\|T(x)\| = \left\| \frac{T(\delta x)}{\delta} \right\| < \frac{1}{\delta}$$

so $\|T\|_{op} \leq 1/\delta$, which proves T is bounded.

To see (3) implies (1), suppose T is bounded. Then for any $v \in V$, $\|T(v)\| \leq \|T\|_{op}\|v\|$, so for any $x, y \in V$,

$$\|T(x) - T(y)\| = \|T(x - y)\| \leq \|T\|_{op}\|x - y\|$$

so for arbitrary $\varepsilon > 0$, taking $\|x - y\| < \frac{\varepsilon}{\|T\|_{op}}$ shows continuity. ■

Every linear operator on a finite dimensional space is continuous. This is not true for infinite dimensional spaces.

Example

Let $V = C([0, 1])$, and let $T : C([0, 1]) \rightarrow P$ be the map $f \mapsto f(0)$, which is linear. T is continuous with respect to the supremum norm, but not continuous with respect to the norm $\|\cdot\|_p$. Indeed, via a sequence of step functions $(f_n)_{n \in \mathbb{N}}$ converging to 0, we have $f_n \rightarrow 0$ with $\|f_n\| = 1$ for all $n \in \mathbb{N}$.

Theorem 1.11

Let $(V, \|\cdot\|)$ and $(W, \|\cdot\|)$ be normed spaces. Then $(\mathcal{B}(V, W), \|\cdot\|_{op})$ is a normed space, and is Banach if W is.

Proof:

To prove this is a vector space, we can use the triangle inequality to show that it is closed under vector addition, and absolute homogeneity to show it is closed under scalar multiplication. Also, $0 \in \mathcal{B}(V, W)$ since $\|0\|_{op} = 0 < \infty$. All these proofs end up showing that $\|\cdot\|_{op}$ is a seminorm. To show it is a norm, we must show $\|T\|_{op} = 0$ implies $T = 0$. Indeed,

$$0 = \|T\|_{op} = \sup_{\|x\|=1} \|T(x)\|$$

which implies $\|T(x)\| = 0$ for all $x \in S_V$ (the unit sphere), so $T(x) = 0$ for all $x \in S_V$, which shows by rescaling $T(x) = 0$ for all $x \in V$. Now, we prove completeness of $\mathcal{B}(V, W)$ when W is complete. Let (T_n) be a Cauchy sequence in $\mathcal{B}(V, W)$. Then for $v \in V$, we have for $m, n \in \mathbb{N}$,

$$\|T_n(v) - T_m(v)\| = \|(T_n - T_m)(v)\| \leq \|T_n - T_m\|_{op} \|v\|$$

which shows $(T_n(v))$ is a Cauchy sequence in W , so by completeness of W , $\lim_{n \rightarrow \infty} T_n(v)$ exists. Define an operator $T : V \rightarrow W$ by $T(v) = \lim_{n \rightarrow \infty} T_n(v)$ for $v \in V$. Since linearity is preserved by limits, T is linear. To show it is bounded, by the reverse triangle inequality, for $m, n \in \mathbb{N}$,

$$|\|T_n\|_{op} - \|T_m\|_{op}| \leq \|T_n - T_m\|_{op}$$

Thus $(\|T_n\|_{op})$ is a Cauchy sequence in $\mathbb{F}$, so it converges to some $M \in \mathbb{F}$. Then,

$$\|T(v)\| = \|\lim_{n \rightarrow \infty} T_n(v)\| = \lim_{n \rightarrow \infty} \|T_n(v)\| \leq \lim_{n \rightarrow \infty} \|T_n\|_{op} \|v\| = M \|v\|$$

so T is bounded. Now we show that (T_n) converges to T . Choose $\varepsilon > 0$ and $N \in \mathbb{N}$ such that for $n, m \geq N$,

$$\|T_n - T_m\|_{op} < \varepsilon$$

so for $v \in V$ of norm 1,

$$\|T_n(v) - T_m(v)\| \leq \|T_n - T_m\|_{op} \|v\| < \varepsilon$$

so taking the limit $m \rightarrow \infty$ we get

$$\|T_n(v) - T(v)\| \leq \varepsilon$$

which shows $T = \lim_{n \rightarrow \infty} T_n$. ■

Examples

(a) The quotient map $\pi : V \rightarrow V/W$ defined by $\pi(v) = v + W$ is linear and bounded. The projection onto any component of a product space is linear and bounded.

(b) Suppose $\frac{1}{p} + \frac{1}{q} = 1$ and $A \subseteq \mathbb{R}^n$ is measurable. Then for $f \in L^q(A)$, the map $\mu_f : L^p(A) \rightarrow L^1(A)$ defined by $\mu_f(g) = fg$ is a bounded linear operator.

Linear Functionals

Definition 1.12

Let V be a vector space over $\mathbb{F}$. A linear functional is a linear operator $\varphi : V \rightarrow \mathbb{F}$. Suppose $(V, \|\cdot\|)$ is a normed space. The dual space of V is $V^* := \mathcal{B}(V, \mathbb{F})$ with the operator norm $\|\cdot\|_{op}$.

Theorem 1.13

$(V^*, \|\cdot\|_{op})$ is a Banach space.

Proof:

This is Theorem 1.11 with $W = \mathbb{F}$. ■

The big issue here is we don't know how to identify the dual space V^* : how do we know $V^* \neq 0$ when $V \neq 0$? This is what the Hahn-Banach theorem is about.

Examples

(a) Let $1 \leq p < \infty$. Then the dual space of ℓ^p is ℓ^q where $\frac{1}{p} + \frac{1}{q} = 1$. The isometric isomorphism is $M : \ell^q \rightarrow (\ell^p)^*$ by mapping $(a_n) = a \in \ell^q \mapsto M(a) \in (\ell^p)^*$ where $M(a) : \ell^p \rightarrow \mathbb{F}$ is $(x_n) \mapsto \sum_{n=1}^{\infty} a_n x_n$. For $p = \infty$, there is an isometric embedding $M : \ell^1 \rightarrow (\ell^\infty)^*$, using the map above, but it is not surjective.

(b) Let $1 \leq p < \infty$ and A a Lebesgue measurable set. Then the dual space of L^p is L^q where $\frac{1}{p} + \frac{1}{q} = 1$, where the isometric isomorphism is $M : L^q \rightarrow (L^p)^*$, which sends $f \in L^p$ to $M_f(g) = \int_A f g dm$, where m is Lebesgue measure.

(c) If $c = \{(x_n) \in \ell^\infty : \lim_{n \rightarrow \infty} x_n \text{ exists}\}$, then $c^* \cong \ell^1 \cong c_0^*$ (by Assignment 1). The isomorphism is as in (a).

Let V be a vector space, and $f : V \rightarrow \mathbb{F}$ a nonzero linear functional on V . Then $\dim(V/\ker f) = 1$ by the first isomorphism theorem.

Definition

A hyperplane of a vector space V is a subspace $H \subseteq V$ such that $\dim(V/H) = 1$.

Proposition 1.14

(1) $H \subseteq V$ is a hyperplane if and only if $H = \ker f$ for some nonzero linear functional $f : V \rightarrow \mathbb{F}$.

(2) If $f_1, f_2 \neq 0$ are linear functionals on V , then $\ker f_1 = \ker f_2$ if and only if $f_2 = \alpha f_1$ for some $\alpha \in \mathbb{F}$.

Proof:

(1) $\ker f$ is a hyperplane by the first isomorphism theorem via the argument above. Conversely, suppose H is a hyperplane. Let $v \in V$ such that $v + H \neq 0$, so $V/H = \text{Span}_{\mathbb{F}}\{v + H\}$. We define a linear functional $f : V \rightarrow \mathbb{F}$ by composing the quotient map $\pi : V \rightarrow V/H$ with the isomorphism $V/H \cong \mathbb{F}$, $\alpha(v + H) \mapsto \alpha$, which gives a functional $f : V \rightarrow \mathbb{F}$ such that $\ker f = H$.

(2) That $f_2 = \alpha f_1$ implies $\ker f_2 = \ker f_1$ is clear. Suppose $\ker f_1 = \ker f_2 = H$. Let $v \in V \setminus H$ (which exists since $f_1, f_2 \neq 0$). Then $f_1(v), f_2(v) \neq 0$. Let $\alpha := \frac{f_2(v)}{f_1(v)}$. Then for any $u \in V$, there exists $\lambda \in \mathbb{F}$ such that $u - \lambda v \in H$ (because V/H is 1-dimensional and v is a representative of the equivalence class, so we can subtract multiples of it so that $u - \lambda v + H = 0 + H$). Hence

$$f_2(u) = f_2(\lambda v) = \lambda f_2(v) = \lambda \alpha f_1(v) = \alpha f_1(\lambda v) = \alpha f_1(u)$$

giving that $f_2 = \alpha f_1$. ■

Proposition 1.15

Suppose $(V, \|\cdot\|)$ is a normed space. Then $0 \neq f$ is bounded if and only if $\ker f$ is closed.

Proof:

If f is bounded, f is continuous, and hence $\ker f = f^{-1}(\{0\})$ is closed. Suppose $\ker f$ is closed. Let $v \in V$ such that $v \notin \ker f$. Then the map $V/\ker f \rightarrow \mathbb{F}$, $\alpha v + \ker f \mapsto f(\alpha v)$ is continuous since $V/\ker f$

and $\mathbb{F}$ are finite dimensional and all linear maps on finite dimensional spaces are continuous ($V/\ker f$ is well-defined as a normed quotient space because $\ker f$ is closed).

By continuity, let M be such that $|f(\alpha v)| \leq M\|\alpha v + \ker f\|_{V/\ker f}$. To see f is bounded, take $u \in V$ and let $\alpha \in \mathbb{F}$ such that $u - \alpha v \in \ker f$. Then

$$|f(u)| = |f(\alpha v)| \leq M\|\alpha v + \ker f\|_{V/\ker f} = M\|u + \ker f\|_{V/\ker f} \leq M\|u\|$$

so f is bounded. ■

Corollary 1.16

In a normed space, a hyperplane is either closed or dense.

Proof:

If it is not closed, the hyperplane is the kernel of an unbounded linear functional $f : V \rightarrow \mathbb{F}$. Because f is nontrivial and unbounded, we can find a sequence (x_n) with $(x_n) \rightarrow 0$ but $f(x_n) = 1$ for all $n \in \mathbb{N}$. Now pick any $y \in V$. Then $y - f(y)x_n \in \ker f$, and $y = \lim_{n \rightarrow \infty} (y - f(y)x_n)$. ■

Examples

- (a) On c_0 , define $f : c_0 \rightarrow \mathbb{F}$ by $f((x_n)) = x_1$. Then f is continuous, and $\ker f = \{(x_n) \in c_0 : x_1 = 0\}$ is closed.
- (b) Let $\phi : (C([0, 1]), \|\cdot\|_p) \rightarrow \mathbb{F}$ for $1 \leq p < \infty$ be defined by $\phi(f) = f(0)$. We have seen that ϕ is not continuous. Then $\ker \phi = \{f \in C([0, 1]) : f(0) = 0\}$ is a dense hyperplane.

Hahn-Banach Theorem

We will see that $V^* \neq 0$ whenever $V \neq 0$.

Definition 1.17

Let V be a vector space. A sublinear functional on V is a function $q : V \rightarrow \mathbb{R}$ such that $q(v + w) \leq q(v) + q(w)$ for all $v, w \in V$, and $q(\alpha v) = \alpha q(v)$ for all $\alpha \geq 0$.

Examples

- (a) Norms and seminorms.
- (b) Take ℓ^∞ over $\mathbb{R}$. Then $q((x_n)) = \limsup_{n \rightarrow \infty} x_n$ is a sublinear functional that is not a seminorm. The same works for ℓ^∞ over $\mathbb{C}$ by taking $\limsup_{n \rightarrow \infty} \operatorname{Re}(x_n)$ or $\limsup_{n \rightarrow \infty} \operatorname{Im}(x_n)$.

Theorem 1.18 (Hahn-Banach theorem - real version)

Let V be a vector space over $\mathbb{R}$ and $q : V \rightarrow \mathbb{R}$ be a sublinear functional on V . Suppose $W \subseteq V$ is a subspace of V and $f : W \rightarrow \mathbb{R}$ is a linear functional on W such that $f(w) \leq q(w)$ for all $w \in W$. Then there exists a linear functional $F : V \rightarrow \mathbb{R}$ that extends f (i.e. $F(w) = f(w)$ for $w \in W$) and such that $F(v) \leq q(v)$ for all $v \in V$.

Proof below.

Corollary 1.19 (Hahn-Banach theorem - complex version)

Let V be a vector space over $\mathbb{C}$ and let q be a seminorm on V . Suppose $W \subseteq V$ is a subspace and $f : W \rightarrow \mathbb{C}$ is a linear functional such that $|f(w)| \leq q(w)$ for all $w \in W$. Then there exists a linear functional $F : V \rightarrow \mathbb{C}$ that extends f and such that $|F(v)| \leq q(v)$ for all $v \in V$.

Proof below.

For the proof of Theorem 1.18, we will need Zorn's Lemma.

Lemma (Zorn)

Let $(S, \leq)$ be a poset. Suppose for every $L \subseteq S$ such that $(L, \leq)$ is totally ordered, there exists some $a \in S$ such that $b \leq a$ for all $b \in L$. Then, there exists a maximal element $m \in S$. That is, if $m \leq s$ for

some $s \in S$, then $s = m$.

Proof of Theorem 1.18:

First, we assume $\dim(V/W) = 1$. The second step is to use Zorn's lemma to get a subspace U and a functional F on U such that U is maximal with the properties from the statement of the theorem. Using step 1, we will conclude $U = V$. Now, pick $v_0 \in V$ such that $v_0 \notin W$. Then, every $v \in V$ can be written as $\lambda v_0 + w$ with $\lambda \in \mathbb{R}, w \in W$. Hence, we can define a linear functional $F : V \rightarrow \mathbb{R}$ by $F(v) = f(w) + \lambda \alpha_0$ for some $\alpha_0 \in \mathbb{R}$. We need to show we can choose $\alpha_0 \in \mathbb{R}$ such that $F \leq q$. For $\lambda > 0$ and $w \in W$, we want:

$$F(\lambda v_0 + w) = \lambda \alpha_0 + f(w) \leq q(\lambda v_0 + w)$$

Equivalently, we want

$$F(\lambda v_0 + \lambda w) = \lambda \alpha_0 + \lambda f(w) \leq q(\lambda v_0 + \lambda w)$$

which implies by dividing by λ that

$$\alpha_0 + f(w) \leq q(v_0 + w)$$

and so we want $\alpha_0 \leq q(v_0 + w) - f(w)$.

Now, for $\lambda < 0$ and $w \in W$, we want:

$$F(\lambda v_0 + \lambda w) = \lambda \alpha_0 + \lambda f(w) \leq q(\lambda v_0 + \lambda w)$$

which is equivalently

$$\lambda \alpha_0 + \lambda f(w) \leq -\lambda q(-v_0 - w)$$

so dividing by λ (which is negative), we want $\alpha_0 \geq -q(-v_0 - w) - f(w)$.

Taking $w' = -w$, we want $\alpha_0 \geq -q(-v_0 + w') + f(w')$. Thus, combining our two inequalities, for all $w, w' \in W$, we want

$$-q(-v_0 + w') + f(w') \leq \alpha_0 \leq q(v_0 + w) - f(w)$$

which is equivalent to

$$\sup_{w' \in W} (-q(-v_0 + w') + f(w')) \leq \alpha_0 \leq \inf_{w \in W} (q(v_0 + w) - f(w))$$

Thus such an α_0 exists if and only if

$$\sup_{w' \in W} (-q(-v_0 + w') + f(w')) \leq \inf_{w \in W} (q(v_0 + w) - f(w))$$

To prove this is true, note that

$$\begin{aligned} \sup_{w' \in W} (-q(-v_0 + w') + f(w')) &\leq \inf_{w \in W} (q(v_0 + w) - f(w)) \\ \iff -q(-v_0 + w') + f(w') &\leq q(v_0 + w) - f(w) \quad \forall w, w' \in W \\ \iff f(w) + f(w') &\leq q(v_0 + w) + q(-v_0 + w') \quad \forall w, w' \in W \end{aligned}$$

But by linearity of f and subadditivity of q ,

$$\begin{aligned} f(w) + f(w') &= f(w + w') = f((-v_0 + w') + (v_0 + w)) \\ &\leq q((-v_0 + w') + (v_0 + w)) \leq q(w + v_0) + q(w' - v_0) \end{aligned}$$

as desired, which concludes this step. For the second step, let S be the pairs (U, g) where $U \subseteq V$ is a subspace containing W and $g : U \rightarrow \mathbb{R}$ is a linear function such that $g \leq q$ and $g|_W = f$. We define a partial order $(U_1, g_1) \leq (U_2, g_2)$ if $U_1 \subseteq U_2$ and $g_2|_{U_1} = g_1$. Suppose $L \subseteq S$ is totally ordered. Write $L = \{(U_\lambda, g_\lambda)\}_{\lambda \in \Lambda}$. Let $U = \bigcup_{\lambda \in \Lambda} U_\lambda$. Then, $W \subseteq U$ and $U \subseteq V$, and U is still a linear subspace. We define $g : U \rightarrow \mathbb{R}$ by $g(v) = g_\lambda(v)$ for $v \in U_\lambda$. This is well-defined by definition of our partial order. Thus, (U, g) is an upper bound of L . Thus by Zorn's lemma, S admits a maximal element (U, F) .

We must have $U = V$, otherwise step 1 would allow us to extend F to a bigger subspace U' satisfying $U \subsetneq U' \subseteq V$, which contradicts maximality. ■

To prove Corollary 1.19, we need:

Lemma 1.20

Let V be a complex vector space.

- (1) If $f : V \rightarrow \mathbb{R}$ is $\mathbb{R}$ -linear, then $g(v) = f(v) - if(iv)$ is $\mathbb{C}$ -linear and $f = \operatorname{Re}(g)$.
- (2) If $g : V \rightarrow \mathbb{C}$ is $\mathbb{C}$ -linear and we let $f = \operatorname{Re}(g)$, then $g(v) = f(v) - if(iv)$.
- (3) If $f : V \rightarrow \mathbb{R}$ is $\mathbb{R}$ -linear and $q : V \rightarrow \mathbb{R}$ is a seminorm, then $|f(v)| \leq q(v)$ for all $v \in V$ if and only if $|g(v)| \leq q(v)$ for all $v \in V$.
- (4) If V is normed, $\|f\|_{op} = \|g\|_{op}$.

Proof:

We prove (3) and (4). If $|g(v)| \leq q(v)$ for all $v \in V$, then

$$|f(v)| = |\operatorname{Re}(g(v))| \leq |g(v)| \leq q(v)$$

for all $v \in V$. Conversely, suppose $|f(v)| \leq q(v)$ for all $v \in V$. Suppose $g(v) \neq 0$. Then, set $\alpha = \frac{\overline{g(v)}}{|g(v)|}$ so that $|\alpha| = 1$ and $\alpha g(v) = |g(v)|$. Then,

$$|g(v)| = \alpha g(v) = g(\alpha v) = \operatorname{Re}(g(\alpha v)) = f(\alpha v) \leq q(\alpha v) = q(v)$$

To prove (4), to prove $\|f\|_{op} \leq \|g\|_{op}$, apply (3) with $q(v) = \|g\|_{op}\|v\|$. By (3), $|f(v)| \leq q(v) = \|g\|_{op}\|v\|$, hence $\|f\|_{op} \leq \|g\|_{op}$. Conversely, apply (3) with $q(v) = \|f\|_{op}\|v\|$ to get $\|g\|_{op} \leq q(v) = \|f\|_{op}\|v\|$, so $\|g\|_{op} \leq \|f\|_{op}$. ■

Proof of Corollary 1.19

Let $h = \operatorname{Re}(f)$. Then, $|h(w)| \leq q(w)$ for all $w \in W$ by the above lemma. Apply the real version of Hahn-Banach to h to get an extension $H : V \rightarrow \mathbb{R}$ such that $H(v) \leq q(v)$ for all $v \in V$. Then, $|H(v)| \leq q(v)$ for all $v \in V$. Set F to be the g obtained from H in Lemma 1.20(1), where by the lemma, $F|_W = f$, and moreover $|F(v)| \leq q(v)$ for all $v \in V$ by Lemma 1.20(3). ■

Consequences of the Hahn-Banach theorem

Theorem 1.21

Let $(V, \|\cdot\|)$ a normed space. Let $W \subseteq V$ and $f : W \rightarrow \mathbb{F}$ a bounded linear functional. Then there exists a linear functional $F \in V^*$ which extends f and $\|F\|_{op} = \|f\|_{op}$.

Proof:

Apply the Hahn-Banach theorem with $q(v) = \|f\|_{op}\|v\|$. ■

Corollary 1.22 (Separation)

Let $(V, \|\cdot\|)$ be a normed space with $V \neq 0$. Then for every $v \in V$, there exists $f \in V^*$ with $\|f\|_{op} = 1$ and $f(v) = \|v\|$. In particular, if $w \neq v$, there exists $f \in V^*$ such that $f(v) \neq f(w)$.

Proof:

If $v = 0$, then any $f \in V^*$ such that $\|f\|_{op} = 1$ works (which we can do by rescaling one of these we obtain from the theorem). Thus, assume $v \neq 0$. Let $W = \operatorname{Span} v$. Define $f : W \rightarrow \mathbb{F}$ by $f(\alpha v) = \alpha\|v\|$ for any $\alpha v \in W$. Then f is linear and $\|f\|_{op} = 1$. By Theorem 1.21, there exists $F \in V^*$ with $\|F\|_{op} = \|f\|_{op} = 1$ which extends f . Hence $F(v) = \|v\|$. To get the result for $w \neq v$, apply the above result to $v - w$. ■

Corollary 1.23

Let $(V, \|\cdot\|)$ be a normed space and let $W \subseteq V$ be a proper closed subspace. Let $v_0 \in V \setminus W$ and set

$d := \inf_{w \in W} \|v_0 - w\|$. Then there exists $f \in V^*$ with $\|f\|_{op} = 1/d$, $f(v_0) = 1$, and $f(w) = 0$ for all $w \in W$.

Proof:

Notice first that $d = \|v_0 + W\|_{V/W}$. By Corollary 1.22, there exists a linear functional $g' : V/W \rightarrow \mathbb{F}$ such that $g'(v_0 + W) = \|v_0 + W\|_{V/W} = d$ and $\|g'\|_{op} = 1$. Define $g = g' \circ \pi$, where $\pi : V \rightarrow V/W$ is the quotient map and set $f = \frac{1}{d}g$. Then

$$f(v_0) = \frac{1}{d}g(v_0) = \frac{1}{d}g'(v_0 + W) = 1$$

and

$$\|f\|_{op} = \frac{1}{d}\|g' \circ \pi\|_{op} \leq \frac{1}{d}\|g'\|_{op}\|\pi\|_{op} = \frac{1}{d}$$

Also, $f(w) = 0$ for all $w \in W$. It remains to show $\|f\|_{op} = \frac{1}{d}$. Since $\|g'\|_{op} = 1$, let $(x_n + W)$ be a sequence in $B_1(0)$ such that $|g'(x_n + W)| \rightarrow 1$. For each $n \in \mathbb{N}$, take $w_n \in W$ such that $\|x_n + w_n\| < 1$ (which we can do by definition of the quotient norm and the elements $(x_n + W)$ we took contained in $B_1(0)$). Then $|f(x_n + w_n)| \leq \|f\|_{op}$ for all $n \in \mathbb{N}$, and

$$|f(x_n + w_n)| = \left| \frac{1}{d}g'(\pi(x_n + w_n)) \right| = \frac{1}{d}|g'(x_n + W)| \rightarrow \frac{1}{d}$$

This gives that $\|f\|_{op} = 1/d$. ■

Corollary 1.24

Let $(V, \|\cdot\|)$ be a normed space and $W \subseteq V$ be a subspace. Then

$$\overline{W} = \bigcap \{\ker f : f \in V^*, f|_W = 0\}$$

Proof:

Recall that $f \in V^*$ if and only if $\ker f$ is closed, so since $\bigcap \{\ker f : f \in V^*, f|_W = 0\}$ is a closed set containing W (as the intersection of closed sets), we get $\overline{W} \subseteq \bigcap \{\ker f : f \in V^*, f|_W = 0\}$ by minimality of the closure. Conversely, suppose $x_0 \in V$ with $x_0 \notin \overline{W}$. By Corollary 1.23, there exists $f \in V^*$ such that $f(x_0) = 1$ and $f|_{\overline{W}} = 0$. Hence $x_0 \notin \bigcap \{\ker f : f \in V^*, f|_{\overline{W}} = 0\}$, giving the inclusion $\bigcap \{\ker f : f \in V^*, f|_{\overline{W}} = 0\} \subseteq \overline{W}$. ■

Theorem 1.25

Let $(V, \|\cdot\|)$ be a normed space and suppose that V^* is separable. Then V is also separable.

Proof:

Let (f_n) be a dense countable set in the unit sphere of V^* . For each $n \in \mathbb{N}$, choose $x_n \in V$ with $\|x_n\| = 1$ such that $|f_n(x_n)| > \frac{1}{2}$. Set $W = \overline{\text{Span}\{x_n : n \in \mathbb{N}\}}$, which is separable. It suffices to show that $W = V$. Suppose not, so there exists $v_0 \in V$ with $v_0 \notin W$. Then there exists $f \in V^*$ such that $f(v_0) \neq 0$ with $f|_W = 0$ by Corollary 1.23. Let $N \in \mathbb{N}$ be such that

$$\left\| \frac{f}{\|f\|_{op}} - f_N \right\|_{op} < \frac{1}{2}$$

Then because x_N is of norm 1,

$$\left| \frac{f}{\|f\|_{op}}(x_N) - f_N(x_N) \right| < \frac{1}{2}$$

but this is a contradiction since $f(x_N) = 0$, so this is just $|f_N(x_N)| < \frac{1}{2}$, so we have contradictory bounds. ■

The power of the Hahn-Banach theorem lies in the fact that we can extend bounded linear functionals and keep the same norm.

Definition

Let V be a vector space. A Hamel basis $\mathcal{B}$ for V is a maximal (under inclusion) linearly independent set. By maximality, every element of V can be written as a finite linear combination of elements of $\mathcal{B}$.

Proposition 1.26

Every vector space admits a Hamel basis. If $W \subseteq V$ is a subspace and $\mathcal{B}_W$ is a Hamel basis for W , then there exists a Hamel basis $\mathcal{B}$ for V such that $\mathcal{B}_W \subseteq \mathcal{B}$.

Proof:

Existence follows by an application of Zorn's lemma to the set of all linearly independent subsets of V with the order given by inclusion of sets. If $W \subseteq V$ is a subspace, the same argument with all linearly independent sets containing $\mathcal{B}_W$ works. ■

Suppose $(V, \|\cdot\|)$ is a normed space and $W \subseteq V$ is a subspace. Let $f \in W^*$, with $\mathcal{B}_W = \{w_\lambda\}$ a Hamel basis for W and $\mathcal{B}$ a Hamel basis for V such that $\mathcal{B}_W \subseteq \mathcal{B}$. Define $f : V \rightarrow \mathbb{F}$ by

$$f(v) = \sum_{j=1}^m \alpha_j w_{\lambda(j)}$$

where we write $v = \sum_{j=1}^m \alpha_j w_{\lambda(j)} + \sum_{i=1}^n \beta_i v_{\lambda(i)}$ for $v_{\lambda(i)} \in \mathcal{B} \setminus \mathcal{B}_W$. This is a linear functional but not necessarily a continuous one.

Duals of ℓ^p Spaces

Recall:

Theorem (Hölder's Inequality)

Let $p, q \in [1, \infty]$ with $\frac{1}{p} + \frac{1}{q} = 1$. Then for $(x_n) \in \ell^p, (y_n) \in \ell^q$, then

$$\sum_{n=1}^{\infty} |x_n y_n| \leq \|(x_n)\|_p \|(y_n)\|_q$$

In particular, $(x_n y_n) \in \ell^1$.

Theorem 1.27

Let $1 \leq p < \infty$ with $\frac{1}{p} + \frac{1}{q} = 1$. The map $M : \ell^q \rightarrow (\ell^p)^*$ that sends $a = (a_n) \mapsto M_a$ with $M_a((x_n)) = \sum_{n=1}^{\infty} a_n x_n$ is an isometric isomorphism.

Proof:

M_a is clearly linear, and by Hölder's inequality, $\|M_a((x_n))\| \leq \|a\|_q \|(x_n)\|_p$, so M_a is a bounded linear map. First, we claim $\|M_a\|_{op} = \|a\|_q$. The fact $\|M_a\|_{op} \leq \|a\|_q$ is immediate from Hölder's inequality. For the opposite inequality, we define for each $n \in \mathbb{N}$, $x_n = |a_n|^{q/p} \operatorname{sgn}(a_n)$, where $\operatorname{sgn}(a_n) = \frac{a_n}{|a_n|}$ if $a_n \neq 0$, and 0 if $a_n = 0$. Then for each $N \in \mathbb{N}$, we define $x^{(N)} = \sum_{n=1}^N x_n e_n$. Then,

$$\|x^{(N)}\|_p = \left(\sum_{n=1}^N |a_n|^q \right)^{1/p} = \|a^{(N)}\|_q^{q/p}$$

where $a^{(N)} = \sum_{n=1}^N a_n e_n$. Then,

$$\|a^{(N)}\|_q^q = |M_a(x^{(N)})| \leq \|M_a\|_{op} \|x^{(N)}\|_p = \|M_a\|_{op} \|a^{(N)}\|_q^{q/p}$$

Dividing on both sides gives

$$\|a^{(N)}\|_q \leq \|M_a\|_{op}$$

so taking $N \rightarrow \infty$ gives $\|a\|_q \leq \|M_a\|_{op}$, as desired. Next, we claim M is surjective. Indeed, let $f \in (\ell^p)^*$. For each $n \in \mathbb{N}$, define $a_n = f(e_n)$ and $a = (a_n)$. To ensure $a \in \ell^q$, we proceed as we did previously, defining $x^{(N)} = (x_1, \dots, x_N, 0, \dots)$ with $x_n = |a_n|^{q/p} \operatorname{sgn}(a_n)$. Then,

$$\|(x^{(N)})\|_p = \|a^{(N)}\|_q^{q/p}$$

so

$$\|a^{(N)}\|_q^q = |f(x^{(N)})| \leq \|f\|_{op} \|x^{(N)}\|_p = \|f\|_{op} \|a^{(N)}\|_q^{q/p}$$

and hence $\|a^{(N)}\|_q \leq \|f\|_{op}$ for all $N \in \mathbb{N}$, so $a \in \ell^q$. The fact that $M_a = f$ follows because M_a and f are both continuous linear functionals that coincide on the vectors $e_n, n \in \mathbb{N}$, which are dense. Thus, M is surjective. ■

Theorem 1.28

Take $p = \infty, q = 1$. Then the map M in the above theorem is an isometric embedding (but not an isomorphism).

Proof:

The same argument to show M is bounded and linear applies, we need only show it is an isometry by showing the inequality $\|a\|_q \leq \|M_a\|_{op}$. For $N \in \mathbb{N}$, define $x^{(N)} = x_N e_n$, where $x_N = \operatorname{sgn}(a_n)$ (with the same setup as the above proof). Then $\|x^{(N)}\|_1 \leq 1$, and hence

$$|a_N| = |M_a(x^{(N)})| \leq \|M_a\|_{op}$$

Since this holds for all $N \in \mathbb{N}$, we obtain $\|a\|_\infty \leq \|M_a\|_{op}$. ■

2: Hilbert Spaces

Definition

Let V be a vector space. An inner product on V is a function $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{F}$ such that:

- (1) $\langle v, v \rangle \in \mathbb{R}$ for all $v \in V$, with $\langle v, v \rangle \geq 0$ with equality if and only if $v = 0$ (positive definiteness).
- (2) $\langle \alpha v + w, z \rangle = \alpha \langle v, z \rangle + \langle w, z \rangle$ for all $v, w, z \in V, \alpha \in \mathbb{F}$.
- (3) $\langle v, w \rangle = \overline{\langle w, v \rangle}$ for all $v, w \in V$.

Note that (2) and (3) imply that $\langle \cdot, \cdot \rangle$ is conjugate-linear in the second argument.

We say that $\langle \cdot, \cdot \rangle$ is a semi-inner product if all aspects of the definition hold except the equality condition in (1), in which case the condition is referred to as positive semidefiniteness.

We call a vector space V with an inner product $\langle \cdot, \cdot \rangle$ an inner product space.

For Hilbert spaces, we are looking for spaces that serve as appropriate infinite-dimensional generalizations of Euclidean spaces.

Examples

(1) Any finite dimensional inner product space is isometrically isomorphic to $\mathbb{F}^N$ with the standard inner product.

(2) For $\emptyset \neq X$, take $\ell^2(X) = \{u_x : X \rightarrow \mathbb{C} : \sum |u_x|^2 < \infty\}$. Its inner product is given by $\langle u_x, u_y \rangle = \sum u_x u_y$.

(3) The space $L^2([0, 1])$ of measurable functions $f : [0, 1] \rightarrow \mathbb{C}$ define up to equivalence a.e. with $\int_{[0, 1]} |f|^2 dm < \infty$, where m is Lebesgue measure, is a Hilbert space with inner product $\langle f, g \rangle = \int_{[0, 1]} f g dm$.

Really, the reason why we require the conditions on our infinite sums is so that our algebraic operations are well-defined, think of trying to multiply an infinite matrix of 1s by a column vector of 1s: that isn't well-defined, and so doesn't really give an appropriate infinite-dimensional generalization of Euclidean spaces.

Definition

Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space. Then V has a norm induced by the inner product given by $\|v\| = \sqrt{\langle v, v \rangle}$ for all $v \in V$. We'll see below that this is in fact a norm.

Theorem 2.3 (Cauchy-Schwarz Inequality)

Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space. Then

$$|\langle v, w \rangle| \leq \|v\| \|w\|$$

for all $v, w \in V$, with equality if and only if v and w are linearly dependent.

Proof:

If $v = 0$ or $w = 0$ the statement holds for free, so assume $v, w \neq 0$. For $t \in \mathbb{R}$, we have

$$\begin{aligned} 0 \leq \|v + tw\|^2 &= \langle v + tw, v + tw \rangle = \langle v, v \rangle + t\langle w, v \rangle + \bar{t}\langle v, w \rangle + |t|^2\langle w, w \rangle \\ &= |v|^2 + 2\operatorname{Re}(t\langle w, v \rangle) + |t|^2|w|^2 \end{aligned}$$

This is a real quadratic in $|t|$ which is greater than or equal to 0, and thus has only one root. Now, assume $\langle v, w \rangle \neq 0$ (since the desired inequality holds in that case), and let $\alpha = t \frac{\overline{\langle w, v \rangle}}{|\langle w, v \rangle|} \in \mathbb{R}$. Then

$$0 \leq \|v\|^2 + 2\alpha|\langle w, v \rangle| + \alpha^2|w|^2$$

as a quadratic polynomial in a real variable with at most 1 root, by looking at its discriminant we obtain

$$4|\langle w, v \rangle|^2 - 4|w|^2|v|^2 \leq 0$$

which is our desired inequality. To see the equality condition, if u and v are linearly dependent, this follows by writing one as a multiple of the other and pulling the scalar multiple out. Conversely, if $|\langle v, w \rangle| = \|v\| \|w\|$, then we get the discriminant above is zero for some $t_0 \in \mathbb{R}$, so for $\alpha := t_0 \frac{\overline{\langle w, v \rangle}}{|\langle w, v \rangle|}$ we get $\|v + \alpha w\|^2 = 0$, so $v = -\alpha w$, so v and w are linearly dependent, as desired. ■

Corollary

The norm $\|\cdot\|$ in an inner product space is indeed a norm.

Proof:

We verify the triangle inequality, the rest follow easily from the other properties of the norm. Then, for $v, w \in V$,

$$\begin{aligned} \|v + w\|^2 &= \langle v + w, v + w \rangle = \|v\|^2 + 2\operatorname{Re}(\langle v, w \rangle) + \|w\|^2 \\ &\stackrel{\operatorname{Re} \leq |\cdot|}{\leq} \|v\|^2 + 2|\langle v, w \rangle| + \|w\|^2 \\ &\stackrel{CS}{\leq} \|v\|^2 + 2\|v\| \|w\| + \|w\|^2 = (\|v\| + \|w\|)^2 \quad \blacksquare \end{aligned}$$

Definition

An inner product space $(V, \langle \cdot, \cdot \rangle)$ is a Hilbert space if it is a Banach space with respect to the induced norm $\|\cdot\|$.

Given a norm, can we recover an inner product from it? Yes:

Lemma (Polarization Identity)

Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space.

Over $\mathbb{R}$, for $x, y \in V$, we have $4\langle x, y \rangle = \|x + y\|^2 - \|x - y\|^2$.

Over $\mathbb{C}$, for $x, y \in V$ we have $4\langle x, y \rangle = \sum_{k=0}^3 i^k \|x + i^k y\|^2$.

Proof:

Expand everything in the definitions. ■

Lemma (Parallelogram Law)

Let $(V, \|\cdot\|)$ be a normed space. Then $\|\cdot\|$ is induced by an inner product if and only if, for all $x, y \in V$,

$$\|x + y\|^2 + \|x - y\|^2 = 2(\|x\|^2 + \|y\|^2)$$

Lemma 2.5

Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space. Then the completion $(\mathcal{H}, \|\cdot\|)$ with respect to the induced norm $\|\cdot\|$ is a Hilbert space, whose inner product agrees with $\langle \cdot, \cdot \rangle$ on V .

Proof:

Using the fact that the parallelogram law holds on a dense subset V of $\mathcal{H}$, since all the operations in the parallelogram law are continuous with respect to the norm, we get that the parallelogram law holds on $\mathcal{H}$, so it is an inner product space. But the inner product is determined by the norm by the polarization identities, so since the norms of $\mathcal{H}$ and V agree where they are defined, the inner products must agree as well. ■

Definition

Let $(\mathcal{H}, \langle \cdot, \cdot \rangle)$ be an inner product space. We say $\eta, \xi \in \mathcal{H}$ are orthogonal (denoted $\eta \perp \xi$) if $\langle \eta, \xi \rangle = 0$. Given a subset $S \subseteq \mathcal{H}$, its orthogonal complement $S^\perp$ is $S^\perp := \{\xi \in \mathcal{H} : \forall \zeta \in S, \langle \xi, \zeta \rangle = 0\}$.

Remark

For a given subset $S \subseteq \mathcal{H}$, $S^\perp$ is closed in $\mathcal{H}$.
If $\mathcal{H}$ is a Hilbert space, then $(S^\perp)^\perp = \overline{\text{Span } S}$.

Theorem 2.7 (Pythagorean Theorem)

If $\{\xi_1, \dots, \xi_n\} \subseteq \mathcal{H}$ is an orthogonal set (consists of pairwise orthogonal vectors), then

$$\left\| \sum_{i=1}^n \xi_i \right\|^2 = \sum_{i=1}^n \|\xi_i\|^2$$

Proof:

Induct on n , using the base case

$$\|\xi_1 + \xi_2\|^2 = \langle \xi_1 + \xi_2, \xi_1 + \xi_2 \rangle = \|\xi_1\|^2 + 2 \underbrace{\text{Re}(\langle \xi_1, \xi_2 \rangle)}_{=0} + \|\xi_2\|^2 = \|\xi_1\|^2 + \|\xi_2\|^2$$

The inductive step is the same argument using associativity of n vectors into $n - 1$ and 1 vector. ■

Projection onto Closed Convex Sets

Definition

Let V be a vector space. We say a subset $C \subseteq V$ is convex if for all $t \in [0, 1]$, $x, y \in C$, we have $tx + (1 - t)y \in C$.

Example

Any linear subspace of a vector space is convex.

Theorem 2.9

Let $(\mathcal{H}, \langle \cdot, \cdot \rangle)$ be a Hilbert space. Let $C \subseteq \mathcal{H}$ be a nonempty closed convex subset, and let $\xi \in \mathcal{H}$. Then there exists a unique point $\kappa \in C$ such that $\|\xi - \kappa\| = \inf_{\zeta \in C} \|\xi - \zeta\|$.

Proof:

By translating, without loss of generality take $\xi = 0$. Define $\alpha := \inf_{\zeta \in C} \|\zeta\|$. By definition of the infimum, take a sequence (u_n) in C whose norms converge to α . Then, for all $u, v \in C$,

$$\begin{aligned} 2(\|u\|^2 + \|v\|^2) &= \|u + v\|^2 + \|u - v\|^2 = 4 \left\| \underbrace{\frac{u + v}{2}}_{\text{in } C \text{ by conv.}} \right\|^2 + \|u - v\|^2 \\ &\geq 4\alpha^2 + \|u - v\|^2 \end{aligned}$$

and so

$$2(\|u\|^2 + \|v\|^2) - 4\alpha^2 \geq \|u - v\|^2$$

for all $u, v \in C$. Therefore, take $u = u_n, v = v_m$ for $n, m \in \mathbb{N}$, so

$$2(\|u_n\|^2 + \|u_m\|^2) - 4\alpha^2 \geq \|u_n - u_m\|^2 \geq 0$$

so taking $n, m \rightarrow \infty$, by definition of (u_n) , we obtain by the squeeze theorem that $\|u_n - u_m\|^2 \rightarrow 0$, so (u_n) is Cauchy. By completeness of $\mathcal{H}$, this converges. Let $\kappa := \lim_{n \rightarrow \infty} u_n$, which is in C because it is closed. This proves existence.

For uniqueness, suppose there exist $\kappa, \kappa' \in C$ such that $\|\kappa\| = \|\kappa'\| = \alpha$. Plug $\kappa = u, \kappa' = v$ in our above estimate to obtain

$$0 = 2(\|\kappa\|^2 + \|\kappa'\|^2) - 4\alpha^2 \geq \|\kappa - \kappa'\|^2 = 0$$

so $\kappa = \kappa'$. ■

Corollary 2.10 (Orthogonal Decomposition)

Let H be a Hilbert space and $Y \subseteq H$ be a closed subspace. Then $H = Y \oplus Y^\perp$. Moreover, $(Y^\perp)^\perp = Y$. *Proof:*

If $h \in H$ is equal to both $y + z$ and $y' + z'$ for $y, y' \in Y, z, z' \in Y^\perp$, then $y - y' = z - z' \in Y \cap Y^\perp = \{0\}$, so $y = y', z = z'$, so such a decomposition will be unique if it exists. Now, since Y is closed and is convex, there exists $y \in Y$ closest to h (the projection of h onto Y). Let $z = h - y$. Then for $t \in \mathbb{R}$ and an arbitrary $v \in Y$,

$$\|z\|^2 = \|h - y\|^2 \stackrel{\text{min. dist.}}{\leq} \|h - (y + tv)\|^2 = \|z - tv\|^2 = \langle z - tv, z - tv \rangle = \|z\|^2 - 2t \operatorname{Re}(\langle z, v \rangle) + t^2 \|v\|^2$$

Letting $P(t) = \|z\|^2 - 2t \operatorname{Re}(\langle z, v \rangle) + t^2 \|v\|^2$, we see it has a minimum at $t = 0$. Thus $0 = P'(0) = -2 \operatorname{Re}(\langle z, v \rangle)$, implying $\operatorname{Re}(\langle z, v \rangle) = 0$. Now replace tv with itv in the same calculation to obtain $\operatorname{Im}(\langle z, v \rangle) = 0$, so overall $z \in Y^\perp$, finishing this part of the proof. To see the moreover, we have

$$(Y^\perp)^\perp = \{x \in H : \langle z, x \rangle = 0, \forall z \in Y^\perp\} \supseteq Y$$

Conversely, if $x \in (Y^\perp)^\perp$, write $x = y + z$ for $y \in Y, z \in Y^\perp$. Then

$$0 = \langle x, z \rangle = \langle y + z, z \rangle = \langle y, z \rangle + \langle z, z \rangle = \langle z, z \rangle$$

so $z = 0$, and $x \in Y$, so $Y \subseteq (Y^\perp)^\perp$. ■

Remark

There exist Banach spaces X with $Y \subseteq X$ a closed subspace, but no closed subspace $Z \subseteq X$ such that $Y \oplus Z = X$.

Theorem 2.11 (Orthogonal Projection)

Let H be a Hilbert space with $Y \subseteq H$ a closed subspace. Define $P_Y : H \rightarrow H$ by $P_Y(h) = y$ for $h \in H$ written uniquely as $h = y + z$ for $y \in Y, z \in Y^\perp$. Then

(1) P_Y is bounded and linear with $\|P_Y\|_{op} = 1$ if $Y \neq \{0\}$.

(2) $P_Y^2 = P_Y$

(3) $P_Y(H) = Y$, $\ker P_Y = Y^\perp$.

Proof:

Linearity is straightforward (and for Alex, this is all on your old 450 assignment). To see boundedness, let $h \in H$ and write it as $h = y + z$ for $y \in Y, z \in Y^\perp$. Then, by the Pythagorean theorem,

$$\|P_Y(h)\|^2 = \|y\|^2 \leq \|y\|^2 + \|z\|^2 = \|h\|^2$$

so $\|P_Y\|_{op} \leq 1$. Conversely, if $y \neq 0 \in Y$, then

$$\left\| P \left(\frac{y}{\|y\|} \right) \right\| = \frac{1}{\|y\|} \|P(y)\| = \frac{1}{\|y\|} \|y\| = 1$$

so $\|P_Y\|_{op} \geq 1$, so we have equality.

(2) is left as an exercise (also on old 450).

For (3), we have $P_Y(H) \subseteq Y$ by definition, and is surjective onto y because $P_Y(y) = y$ for $y \in Y$. For the kernel, we have $Y^\perp \subseteq \ker P_Y$ since $z = 0 + z$ for $z \in Y^\perp, 0 \in Y$, and conversely if $z \in \ker P_Y$, then $P_Y(z) = 0$, so the Y -part of z is zero, its $Y^\perp$ -part must just be z , so $z \in Y^\perp$. ■

Theorem 2.12 (Riesz Representation Theorem)

Let H be a Hilbert space. Define $\varphi : H \rightarrow H^*$ by $\varphi(h)(z) = \langle z, h \rangle$ for $h, z \in H$. Then φ is conjugate linear, isometric, and surjective.

Proof:

The fact φ is conjugate linear is clear by properties of the inner product. To show it is isometric, for $h, z \in H$,

$$|\varphi(h)(z)| = |\langle z, h \rangle| \leq \|z\| \|h\|$$

so $\|\varphi(h)\|_{op} \leq \|h\|$. Conversely, if $h \neq 0$, then

$$\left| \varphi(h) \left(\frac{h}{\|h\|} \right) \right| = \left| \langle h, \frac{h}{\|h\|} \rangle \right| = \|h\|$$

and thus $\|\varphi(h)\|_{op} = \|h\|$. To prove φ is surjective, let $f \in H^*$. If $f = 0$, then $\varphi(0) = f$, so assume $f \neq 0$. Let $Y = \ker f$, which is closed. Then since $H = Y \oplus Y^\perp$, we have $\dim(Y^\perp) = \dim(H/Y) = 1$. Let $x \neq 0 \in Y^\perp$ and set $h = \frac{1}{f(x)}x$. Then if $z \in H$, write $z = y + \alpha h$ for $\alpha \in \mathbb{C}, y \in Y$. Then

$$f(z) = f(y + \alpha h) = \alpha f(h) = \alpha f \left(\frac{1}{f(x)}x \right) = \alpha$$

and

$$\varphi \left(\frac{1}{\|h\|^2} h \right) (z) = \langle y + \alpha h, \frac{1}{\|h\|^2} h \rangle = \frac{\alpha}{\|h\|^2} \langle h, h \rangle = \alpha$$

so $f = \varphi \left(\frac{1}{\|h\|^2} h \right)$, showing surjectivity. ■

Corollary

Suppose $A \subseteq \mathbb{R}^n$ is measurable. Then for each $\phi \in L^2(A)^*$, there exists a unique function $h \in L^2(f)$ such that $\phi(f) = \int_A f \bar{h} dm$, where m is Lebesgue measure, moreover $\|\phi\|_{op} = \|h\|_2$.

Orthonormal Bases

Definition 2.13

Let H be a Hilbert space. We say that $\{e_j \in H : j \in J\}$ is an orthonormal set if $\|e_j\| = 1$ for all j and $\langle e_i, e_j \rangle = 0$ for all $i \neq j$. We say that $\{e_j : j \in J\}$ is an orthonormal basis if its span is dense in H .

Examples

- (1) Let $H = \ell^2(X)$ for some set X . Then $e_x : X \rightarrow \mathbb{F}$ given by $e_x(y) = \delta_{x,y}$ for each $x \in X$ makes $\{e_x : x \in X\}$ an orthonormal basis.
- (2) Let $H \in L^2(\mathbb{T})$ with respect to Lebesgue measure. For each $n \in \mathbb{Z}$, let $e_n : \mathbb{T} \rightarrow \mathbb{C}$, $e_n(z) = \frac{z^n}{\sqrt{2\pi}}$. Then $\{e_n : n \in \mathbb{Z}\}$ is an orthonormal basis for $L^2(\mathbb{T})$.
- (3) We have for $n, m \in \mathbb{Z}$,

$$\langle e_m, e_n \rangle = \frac{1}{2\pi} \int_0^{2\pi} e^{i(m-n)t} dt = \delta_{m,n}$$

and its span being dense in $L^2(\mathbb{T})$ follows from Stone-Weierstrass (it is dense in continuous functions on the circle, which is dense in L^2). If we instead consider normalized Lebesgue measure on the circle, where $\int_{\mathbb{T}} f = \frac{1}{2\pi} \int_0^{2\pi} f(e^{it}) dt$, then these e_n 's without dividing by $\frac{1}{\sqrt{2\pi}}$ form an orthonormal basis, and $\langle f, e_n \rangle$ is the n th Fourier coefficient of a function f .

Proposition 2.14

Every orthonormal set in a Hilbert space can be extended to an orthonormal basis. In particular, every Hilbert space has an orthonormal basis.

Proof:

Apply Zorn's lemma. ■

Remark

An orthonormal basis for an infinite dimensional Hilbert space is never a Hamel basis.

Remark

Let H be a Hilbert space and $\{x_n : n \in \mathbb{N}\}$ a countable dense subset. We can make this into an orthonormal basis via Gram-Schmidt, throwing away linearly dependent vectors if necessary. In particular, any separable Hilbert space has a countable orthonormal basis.

Proposition 2.15

Let H and K be Hilbert spaces with orthonormal bases $\{e_i : i \in I\}$ and $\{f_j : j \in J\}$ respectively, where I and J have the same cardinality. Then there exists a bounded linear operator $U : H \rightarrow K$ such that $\langle Uh, Uz \rangle = \langle h, z \rangle$ for all $h, z \in H$.

Proof:

Let $\gamma : I \rightarrow J$ by a bijection. We define $U_0 : \text{Span}\{e_i : i \in I\} \rightarrow \text{Span}\{f_j : j \in J\}$ by

$$U_0 \left(\sum_{i \in I} \alpha_i e_i \right) = \sum_{i \in I} \alpha_i f_{\gamma(i)}$$

Then, if $h = \sum_{i \in I} \alpha_i e_i$ and $z = \sum_{i \in I} \beta_i e_i$, since γ is a bijection, we have

$$\langle U_0(h), U_0(z) \rangle \overset{f_j \text{'s orthonormal}}{=} \sum_{i \in I} \alpha_i \bar{\beta}_i \overset{\gamma \text{ bijection}}{=} \langle h, z \rangle$$

In particular, U_0 is an isometric linear operator on a dense set and hence it extends to a bounded linear operator $U : H \rightarrow K$. By continuity of the inner product, U satisfies $\langle U(h), U(z) \rangle = \langle h, z \rangle$ for all $h, z \in H$. ■

Corollary 2.16

- (a) Every Hilbert space of finite dimension is isometrically isomorphic to $\mathbb{F}^n$.
- (b) Every separable infinite dimensional Hilbert space is isometrically isomorphic to $\ell^2(\mathbb{N})$.
- (c) In general, every Hilbert space is isometrically isomorphic to $\ell^2(X)$ for some set X of cardinality equal to the cardinality of its orthonormal basis. ■

Theorem 2.17 (Bessel's Inequality)

Let $\{e_n : n \in \mathbb{N}\}$ be an orthonormal set in a Hilbert space H and $h \in H$. Then

$$\sum_{n=1}^{\infty} |\langle h, e_n \rangle|^2 \leq \|h\|^2$$

Proof:

For $n \in \mathbb{N}$ fixed, let $h_n = h - \sum_{k=1}^n \langle h, e_k \rangle e_k$. Then for each $k \in \{1, \dots, n\}$, we have $\langle h_n, e_k \rangle = \langle h, e_k \rangle - \langle h, e_k \rangle = 0$. By the Pythagorean theorem,

$$\|h\|^2 = \|h_n + \sum_{k=1}^n \langle h, e_k \rangle e_k\|^2 = \|h_n\|^2 + \left\| \sum_{k=1}^n \langle h, e_k \rangle e_k \right\|^2 \geq \sum_{k=1}^n |\langle h, e_k \rangle|^2$$

Letting $n \rightarrow \infty$, we obtain the result. ■

Corollary 2.18

Let $\{e_i : i \in I\}$ be an orthonormal set in H and $h \in H$. Then the set $\{i \in I : \langle h, e_i \rangle \neq 0\}$ is countable. In particular, by Bessel's inequality,

$$\sum_{i \in I} |\langle h, e_i \rangle|^2 \leq \|h\|^2$$

Proof:

For each $n \in \mathbb{N}$, let $E_n = \{i \in I : |\langle h, e_i \rangle|^2 \geq \frac{1}{n}\}$. By Bessel's inequality, E_n must be countable (Bessel's inequality tells us $\sum_{n=1}^{\infty} |\langle h, e_n \rangle|^2 < \infty$, but take a countable subset of E_n and sum over its indices, we will get something infinite). Then $\{i \in I : \langle h, e_i \rangle \neq 0\} = \bigcup_{n \in \mathbb{N}} E_n$ is countable, where the inequality then follows. ■

Theorem 2.19

Let H be a Hilbert space and $E = \{e_\lambda\}_{\lambda \in \Lambda}$ be an orthonormal set in H . The following are equivalent:

- (1) E is an orthonormal basis.
- (2) $E^\perp = \{0\}$.
- (3) For all $h \in H$ we have $h = \sum_{\lambda \in \Lambda} \langle h, e_\lambda \rangle e_\lambda$.
- (4) For every $h \in H$ we have $\|h\|^2 = \sum_{\lambda \in \Lambda} |\langle h, e_\lambda \rangle|^2$.

Proof:

For (1) implies (2), suppose E is an orthonormal basis. Then $\text{Span}(E)$ is dense, so $E^\perp = (\text{Span}(E))^\perp = \overline{\text{Span}(E)}^\perp = H^\perp = \{0\}$.

For (2) implies (3), let $h \in H$. by Corollary 2.18 the set $\{\lambda \in \Lambda : \langle h, e_\lambda \rangle \neq 0\}$ is countable. Let $\{\lambda_1, \lambda_2, \dots\}$ be an enumeration. It suffices to show $h - \sum_{n=1}^{\infty} \langle h, e_{\lambda_n} \rangle e_{\lambda_n} \in E^\perp$. Then, for all $k \in \mathbb{N}$,

$$\left\langle h - \sum_{n=1}^{\infty} \langle h, e_{\lambda_n} \rangle e_{\lambda_n}, e_{\lambda_k} \right\rangle = \langle h, e_{\lambda_k} \rangle - \langle h, e_{\lambda_k} \rangle = 0$$

Similarly, for all $\lambda \in \Lambda$ such that $\langle h, e_\lambda \rangle = 0$ we have $\langle h - \sum_{n=1}^{\infty} \langle h, e_{\lambda_n} \rangle e_{\lambda_n}, e_\lambda \rangle = 0$, so we see $h - \sum_{n=1}^{\infty} \langle h, e_{\lambda_n} \rangle e_{\lambda_n} \in E^\perp = \{0\}$.

For (3) implies (4), let $\varepsilon > 0$ and $N \in \mathbb{N}$ such that $\sum_{n=N+1}^{\infty} |\langle h, e_n \rangle|^2 < \varepsilon$. Let $F_0 = \{\lambda_1, \dots, \lambda_N\}$, and let $F \supseteq F_0$ be finite with $F \subseteq \Lambda$. Then,

$$\left\| \sum_{\lambda \in F} \langle h, e_\lambda \rangle e_\lambda - \sum_{m=1}^{\infty} \langle h, e_{\lambda_m} \rangle e_{\lambda_m} \right\|^2 = \left\| \sum_{\lambda \in \Lambda \setminus F} \langle h, e_\lambda \rangle e_\lambda \right\|^2 \leq \sum_{n=N+1}^{\infty} |\langle h, e_{\lambda_n} \rangle|^2 < \varepsilon$$

To see (4) implies (1), suppose $h \in (\overline{\text{Span}(E)})^\perp$. Then $\|h\|^2 = \sum_{\lambda \in \Lambda} |\langle h, e_\lambda \rangle|^2 = 0$. Thus $H^\perp = \{0\} = (\overline{\text{Span}(E)})^\perp$, so $\overline{\text{Span}(E)} = H$, so E is an orthonormal basis. ■

Operators on Hilbert Spaces

Let H be a separable Hilbert space with orthonormal basis $\{e_n\}_{n \in \mathbb{N}}$. It is helpful to view bounded linear operators T on H as infinite matrices with columns $T(e_1), T(e_2), \dots$.

Definition

A sesquilinear form on a vector space V is a function $B : V \times V \rightarrow \mathbb{F}$ that is linear in the first variable and conjugate-linear in the second variable. If V is a normed space, we say the form is bounded if

$$\|B\| = \sup_{\|x\| \leq 1, \|y\| \leq 1} |B(x, y)| < \infty$$

Lemma 2.20

Let H be a Hilbert space. There is a bijective isometric correspondence between operators in $\mathcal{B}(H)$ and bounded sesquilinear forms on H , given by $T \mapsto B_T$, where $B_T(x, y) = \langle x, Ty \rangle$.

Proof:

B_T is sesquilinear because T is linear and by properties of the inner product. It is bounded by Cauchy-Schwarz and the fact that T is bounded. In particular, $\|B_T\| \leq \|T\|_{op}$. To show we have equality, we note that

$$\|T(x)\|^2 = \langle T(x), T(x) \rangle = B_T(T(x), x) \leq \|B_T\| \|T(x)\| \|x\| \leq \|B_T\| \|T\|_{op} \|x\|^2$$

which gives $\|T\|_{op}^2 \leq \|B_T\| \|T\|_{op}$, so $\|T\|_{op} \leq \|B_T\|$, so we have an isometry. Since it is an isometry, we have injectivity, so we need only prove surjectivity. Let B be a bounded sesquilinear form on H . For fixed $y \in H$, the map $x \mapsto B(x, y)$ is a bounded linear functional, so by the Riesz representation theorem, we can find a unique vector $z \in H$ such that $\langle x, z \rangle = B(x, y)$ for all $x \in H$. We define $T : H \rightarrow H$ by $T(y) = z$, where the map is linear by properties of the inner product and sesquilinear forms. To show it is bounded, we have $\|T(y)\| = \|z\| \leq \|B\| \|y\|$, so $\|T\|_{op} \leq \|B\| < \infty$. ■

Theorem 2.21

Let H be a Hilbert space. For each $T \in \mathcal{B}(H)$, there exists a unique $T^* \in \mathcal{B}(H)$, called the adjoint of T , such that $\langle Tx, y \rangle = \langle x, T^*y \rangle$ for all $x, y \in H$. Moreover, the map $*$: $\mathcal{B}(H) \rightarrow \mathcal{B}(H)$ is conjugate linear and satisfies for all $T, S \in \mathcal{B}(H)$,

- (a) $\|T\|_{op} = \|T^*\|_{op}$
- (b) $\|T^*T\|_{op} = \|T\|_{op}^2$
- (c) $T^{**} = T$
- (d) $(ST)^* = T^*S^*$.

Proof:

Given $T \in \mathcal{B}(H)$, define a bounded sesquilinear form B on H by $B(x, y) = \langle Tx, y \rangle$ for all $x, y \in H$. Then $\|B\| = \|T\|_{op}$. By Lemma 2.20, there exists a unique $T^* \in \mathcal{B}(H)$ with $\|T^*\|_{op} = \|B\| = \|T\|_{op}$ such that $B(x, y) = \langle Tx, y \rangle = \langle x, T^*y \rangle$ for all $x, y \in H$, which shows existence, uniqueness, and (a).

To show (b), from A1 and (a), we have

$$\|T^*T\|_{op} \leq \|T^*\|_{op} \|T\|_{op} = \|T\|_{op}^2$$

For the reverse inequality, suppose we have $x \in H$ with $\|x\| = 1$. Then by Cauchy-Schwarz,

$$\|Tx\|^2 = \langle Tx, Tx \rangle = \langle x, T^*Tx \rangle \leq \|T^*T\|_{op}$$

giving the reverse inequality.

Part (c) and (d) follow by uniqueness of the adjoint. ■

Examples

- (a) The Volterra operator $V : L^2([0, 1]) \rightarrow L^2([0, 1])$ is given by $V(f)(t) = \int_0^t f(s)ds$. Then V is bounded,

and the adjoint of V is given by $V^*(f)(t) = \int_t^1 f(s)ds$ (use Fubini's theorem).

(b) The integral operator with kernel $k \in C([0, 1] \times [0, 1])$ on $L^2([0, 1])$ is given by

$$K(f)(x) = \int_0^1 k(x, y)f(y)dy$$

Then K is a bounded linear operator, and using Fubini's theorem we can show that K^* is given by the integral operator with kernel $k^*(x, y) = \overline{k(y, x)}$.

Definition

Let H be a Hilbert space, and let $T \in \mathcal{B}(H)$.

(a) T is unitary if $T^*T = TT^* = \text{Id}$.

(b) T is normal if $T^*T = TT^*$.

(c) T is self-adjoint if $T = T^*$.

(d) T is an isometry if $\|T(x)\| = \|x\|$ for all $x \in H$. By the polarization identity, this holds if and only if $\langle T(x), T(y) \rangle = \langle x, y \rangle$ for all $x, y \in H$, so taking adjoints, this holds if and only if $T^*T = \text{Id}$. We say T is a proper isometry if it is an isometry but not unitary (only possible in infinite dimensions).

Examples

(a) The map $U : \ell^2(\mathbb{Z}) \rightarrow \ell^2(\mathbb{Z})$ defined by $U(\delta_n) = \delta_{n+1}$ (the bilateral shift) is a unitary operator.

(b) If $f \in C(\mathbb{T})$ and $M_f : L^2(\mathbb{T}) \rightarrow L^2(\mathbb{T})$ is the multiplication by f operator, that is, $M_f(g) = fg$, its adjoint is $M_{\bar{f}}$, hence M_f is normal, and is self-adjoint if and only if f is real valued.

(c) The unilateral shift operator $S : \ell^2(\mathbb{N}) \rightarrow \ell^2(\mathbb{N})$ defined by $S(\delta_n) = \delta_{n+1}$ is a proper isometry.

3: Open Mapping Theorem, Inverse Mapping Theorem, Closed Graph Theorem, and Uniform Boundedness Principle

Recall:

Theorem (Baire Category Theorem)

Let X be a complete metric space. We have the following three equivalent statements:

(1) If (F_n) is a sequence of closed sets within X such that $\bigcup_{n=1}^{\infty} F_n$ has nonempty interior, then at least one of the F_n 's has nonempty interior.

(2) If (F_n) is a sequence of closed sets with empty interior, then $\bigcup_{n=1}^{\infty} F_n$ has empty interior.

(3) If (U_n) is a sequence of open dense sets, then $\bigcap_{n=1}^{\infty} U_n$ is dense in X .

Theorem 3.1 (Open Mapping Theorem)

Let V, W be Banach spaces and $T : V \rightarrow W$ a surjective bounded linear operator. Then T is an open map.

To prove this, we need the following lemma.

Lemma 3.2

Let V, W and $T : V \rightarrow W$ be as in the statement of the Open Mapping Theorem. Suppose that $B(0, r) \subseteq \overline{T(B(0, 1))}$ for some $r > 0$. Then for every $0 < \varepsilon < 1$, we have $B(0, (1 - \varepsilon)r) \subseteq T(B(0, 1))$.

Proof:

We first observe that

$$B(0, r) \subseteq \overline{T(B(0, 1))} \iff B(0, \varepsilon^n r) \subseteq \overline{T(B(0, \varepsilon^n))}$$

by linearity. The idea is that we will show that if $y \in B(0, r)$, then there exists $x \in B(0, \frac{1}{1-\varepsilon})$ such that $y = T(x)$, so $B(0, (1 - \varepsilon)r) \subseteq T(B(0, 1))$. Indeed, if we show $z := \frac{y}{1-\varepsilon} \in B(0, r)$ with $T(x) = z$ for $x \in B(0, \frac{1}{1-\varepsilon})$, then $(1 - \varepsilon)x \in B(0, 1)$ with $T((1 - \varepsilon)x) = y$, so $y \in T(B(0, 1))$, completing the proof.

Now, let $U = T(B(0, 1))$ and $y \in B(0, r)$. Let $0 < \varepsilon < 1$. Then there exists $y_1 \in U$ such that $\|y - y_1\| < \varepsilon r$ by hypothesis. Let $x_1 \in B(0, 1)$ with $T(x_1) = y_1$. Repeating the same argument (but with the displayed

inclusion), there exists $y_2 \in \varepsilon U$ such that $\|y - y_1 - y_2\| < \varepsilon^2 r$, so there exists x_2 with $\|x_2\| < \varepsilon$ such that $T(x_2) = y_2$. Continuing with this process, we obtain a sequence (y_k) in W and (x_k) in V , with $T(x_k) = y_k$ for all k , and

$$\|y - \sum_{k=1}^n y_k\| < \varepsilon^n r, \|x_n\| < \varepsilon^{n-1} r$$

for all $n > 1$. Then, $\sum_{n=1}^{\infty} x_n$ must converge since it absolutely converges and V is Banach:

$$\|x\| \leq \sum_{n=1}^{\infty} \|x_n\| \leq \sum_{n=1}^{\infty} \varepsilon^{n-1} r = \frac{1}{1 - \varepsilon} r$$

Since T is continuous and linear, we have

$$T(x) = \lim_{n \rightarrow \infty} \sum_{k=1}^n T(x_k) = \lim_{n \rightarrow \infty} \sum_{k=1}^n y_k = y$$

which completes the proof of the claim. ■

Proof of Theorem 3.1:

Let $T : V \rightarrow W$ be a surjective bounded linear map of Banach spaces, and let $U \subseteq V$. We claim that if 0 is an interior point of $T(B(0, 1))$, then $T(U)$ is open. Indeed, if this holds, by linearity, 0 is an interior point of $T(B(0, r))$ for every $r > 0$ since then $B(0, r\delta) \subseteq T(B(0, r))$ for some $\delta > 0$. Hence if U is open, and $0 \in U$, then 0 is an interior point of $T(U)$. Now, for $x \in U$ an arbitrary point, we have that $U - x$ is open (it is the preimage of U under the continuous translation map $z \mapsto x + z$), and $0 \in U - x$. Then, 0 is an interior point of $T(U - x) = T(U) - T(x)$ by the above argument, so it follows that $T(x)$ is an interior point of $T(U)$, so $T(U)$ is open.

It remains to prove the claim. Since T is surjective, we have $W = \bigcup_{n \in \mathbb{N}} \overline{T(B(0, n))}$. By Baire's theorem, there exists $n \in \mathbb{N}$ such that $\overline{T(B(0, n))}$ has nonempty interior. Then there exists $y \in W$ and $r' > 0$ such that $B(y, r') \subseteq \overline{T(B(0, n))}$, so by linearity, we have

$$B\left(\frac{y}{2n}, \frac{r'}{2n}\right) \subseteq \overline{T(B(0, 1/2))}$$

Recall that if $O \subseteq V$ is open, then $O - O = \bigcup_{x \in O} O - x$ is open, so by the above inclusion we have

$$\underbrace{B\left(\frac{y}{2n}, \frac{r'}{2n}\right) - B\left(\frac{y}{2n}, \frac{r'}{2n}\right)}_{\text{open, contains } 0} \subseteq \overline{T(B(0, \frac{1}{2}))} - \overline{T(B(0, \frac{1}{2}))} \subseteq \overline{T(B(0, 1))}$$

Thus there is $r > 0$ such that $B(0, r) \subseteq \overline{T(B(0, 1))}$, so by Lemma 3.2, we can find $\delta > 0$ such that $B(0, \delta) \subseteq T(B(0, 1))$, so 0 is an interior point of $T(B(0, 1))$, so we are done by the claim. ■

As a consequence of the open mapping theorem, we have the inverse mapping and closed graph theorems.

Theorem 3.3 (Inverse Mapping Theorem)

Let V and W be Banach spaces and $T : V \rightarrow W$ a bounded linear bijection. Then $T^{-1} : W \rightarrow V$ is a bounded linear map.

Proof:

Suppose $U \subseteq V$ is open. Then $(T^{-1})^{-1}(U) = T(U)$ is open by the Open Mapping Theorem, hence T^{-1} is continuous. ■

Theorem 3.4 (Closed Graph Theorem)

Let V and W be Banach spaces and $T : V \rightarrow W$ a linear operator. Suppose that the graph of T ,

$G(T) = \{(v, T(v)) : v \in V\} \subseteq V \oplus W$ is closed with respect to the ℓ^1 norm ($\|(v, T(v))\|_1 = \|v\| + \|T(v)\|$). Then T is bounded.

Proof:

The graph $G(T)$ is itself a Banach space because it is a closed subspace of $V \oplus W$. Consider the projection $\pi_1 : G(T) \rightarrow V$ onto the first coordinate, which is bounded and linear (it is bounded since $\|\pi_1(v, T(v))\| = \|v\| \leq \|(v, T(v))\|_1$), and a bijection $G(T) \rightarrow V$, so by the inverse mapping theorem, $\pi_1^{-1} : V \rightarrow G(T)$ is bounded. Similarly, the projection $\pi_2 : G(T) \rightarrow W$ is also bounded. Since $T = \pi_2 \circ \pi_1^{-1}$, we get that T is bounded. ■

Remark

(a) The converse of the closed graph theorem holds: if T is continuous, then its graph is closed (this holds without an assumption on V, W being Banach).

(b) An advantage of the closed graph theorem: to show that T is bounded, we need to show that if (x_n) is a sequence in V and $x_n \rightarrow x \in V$, then $T(x_n) \rightarrow T(x)$. Using the closed graph theorem, we can assume that $(T(x_n))$ converges, that is, we are left to show that $x_n \rightarrow x \in V$ and $T(x_n) \rightarrow y \in W$, then $y = T(x)$.

Example

Suppose $g : [0, 1] \rightarrow \mathbb{F}$ is measurable and $gL^2([0, 1]) \subseteq L^2([0, 1])$. Then $g \in L^\infty([0, 1])$, and if $M_g : L^2([0, 1]) \rightarrow L^2([0, 1])$ is the operator given by multiplication by g , then $\|M_g\|_{op} = \|g\|_\infty$. First, we show that M_g is bounded. Suppose (f_n) is a sequence in $L^2([0, 1])$ with $f_n \rightarrow f$ in $L^2([0, 1])$ and $gf_n \rightarrow h \in L^2([0, 1])$. By the closed graph theorem, it suffices to show $h = gf$. Passing to a subsequence if necessary, we may assume $f_n \rightarrow f$ a.e. and $gf_n \rightarrow h$ a.e. This implies that $gf_n \rightarrow gf$ a.e. and $gf_n \rightarrow h$ a.e., so $gf = h$ a.e. and thus $gf = h$ in $L^2([0, 1])$, so by the closed graph theorem, M_g is bounded, and so

$$\|gf\|_2 \leq \|M_g\|_{op} \|f\|_2$$

for all $f \in L^2([0, 1])$. Let $C = \|M_g\|_{op}$. Looking for a contradiction, suppose

$$A_C := \{x : |g(x)| > C\}$$

has positive measure. Then

$$\int_{A_C} |g|^2 dm = \|g\chi_{A_C}\|_2^2 \leq \|M_g\|_{op}^2 \|\chi_{A_C}\|_2^2 = C^2 m(A_C)$$

Since $|g|^2 > C^2$ on A_C , this shows $C^2 m(A_C) < C^2 m(A_C)$, a contradiction. Hence $g \in L^\infty$ and $\|g\|_\infty \leq \|M_g\|_{op}$, where the other inequality is easier, which gives $\|M_g\|_{op} = \|g\|_\infty$.

Theorem 3.5 (Uniform Boundedness Principle/Banach-Steinhaus Theorem)

Suppose V is a Banach space and W is a normed space. Let $\{T_\lambda : \lambda \in \Lambda\}$ be a collection of bounded linear functions $V \rightarrow W$. Suppose that for all $v \in V$, $\{\|T_\lambda(v)\| : \lambda \in \Lambda\}$ is bounded. Then $\sup_{\lambda \in \Lambda} \|T_\lambda\|_{op}$ is bounded.

Proof:

Viewing each T_λ as a linear map into the completion of W if necessary, we can assume W is Banach. For each $\lambda \in \Lambda$, let $W_\lambda := W$, and consider the ℓ^∞ -direct product:

$$Z = \left\{ (w_\lambda)_{\lambda \in \Lambda} \in \prod_{\lambda \in \Lambda} W_\lambda : \sup_{\lambda \in \Lambda} \|w_\lambda\| < \infty \right\}$$

Then Z is a Banach space with ∞ -norm. We define a linear operator $T : V \rightarrow Z$ by $T(v) = (T_\lambda(v))_{\lambda \in \Lambda}$, which is well-defined because $\sup_{\lambda \in \Lambda} \|T_\lambda(v)\| < \infty$. We have that $\|T\|_{op} = \sup_{\lambda \in \Lambda} \|T_\lambda\|_{op}$, since

$$\|T(v)\|_\infty = \sup_{\lambda \in \Lambda} \|T_\lambda(v)\| \leq \sup_{\lambda \in \Lambda} \|T_\lambda\|_{op} \|v\|$$

giving one inequality, and the other holds because for any $\lambda_0 \in \Lambda$, we have $\|T_{\lambda_0}(v)\| \leq \sup_{\lambda \in \Lambda} \|T_{\lambda}(v)\| = \|T(v)\|_{\infty}$. Thus, all we need to do is show T is bounded. Towards this, let (v_n) be a sequence in V converging to some $v \in V$ and such that $T(v_n) \rightarrow z$ for some $z \in Z$. By the closed graph theorem, it suffices to show $T(v) = z$. Write $z = (w_{\lambda})_{\lambda \in \Lambda}$. By definition of $\|\cdot\|_{\infty}$, because we have a uniform bound, we have that $T_{\lambda}(v_n) \rightarrow w_{\lambda}$ for all $\lambda \in \Lambda$. Hence $w_{\lambda} = T_{\lambda}(v)$ since T_{λ} is bounded. Thus, $z = (w_{\lambda})_{\lambda \in \Lambda} = (T_{\lambda}(v))_{\lambda \in \Lambda} = T(v)$, so by the closed graph theorem, T is bounded. ■

Corollary 3.6

Suppose V and W are Banach spaces and let (T_n) be a sequence in $\mathcal{B}(V, W)$. Suppose that for all $v \in V$, the sequence $(T_n(v))$ converges. Then there exists $T \in \mathcal{B}(V, W)$ such that $T(v) = \lim_{n \rightarrow \infty} T_n(v)$ for all $v \in V$.

Proof:

Define T as in the statement. Then T is linear by linearity of the T_n 's and definition of the limit. To see it is bounded, we have $\|T\|_{op} \leq \sup_{n \in \mathbb{N}} \|T_n\|_{op}$ which is finite by the Uniform Boundedness Principle. ■

4: Topology

Definition

Let X be a set. A topology τ on X is a collection of subsets of X (called open sets) such that:

- (1) $\emptyset, X \in \tau$.
- (2) τ is closed under arbitrary unions.
- (3) τ is closed under finite intersection.

When X is equipped with a topology, we call X a topological space.

Definition

Given a set X and two topologies τ_1, τ_2 on it, we call τ_1 weaker/coarser/smaller if $\tau_1 \subseteq \tau_2$, and call τ_2 stronger/finer/larger. Functions $X \rightarrow Y$ continuous with respect to a coarser topology τ_1 are continuous with respect to the finer topology τ_2 , but the converse need not hold.

Examples

- (a) Given a metric space X , the open sets form a topology called the metric topology. Similarly, a pseudometric induces a topology on X .
- (b) The topology on $\mathbb{R}^n$ induced by the Euclidean metric is called the usual topology on $\mathbb{R}^n$.
- (c) The discrete topology on a space X is $\tau_d = \mathcal{P}(X)$, which is the finest topology one can take. The trivial topology $\tau_t = \{\emptyset, X\}$ is the coarsest topology one can take.

Definition

Given a collection F of subsets of a set X , there is a smallest topology on X that contains F , called the topology generated by F , denoted $\tau(F)$. Indeed, take the intersection of all topologies containing F (there is at least one such topology, the discrete topology), plus $\emptyset$ and X . $\tau(F)$ is the collection of all unions of finite intersections of elements of F (and $\emptyset, X$).

Examples

- (a) The topology generated by all open balls in a metric space is the metric topology.
- (b) Let $F = \{(-\infty, b) : b \in \mathbb{R}\} \cup \{(b, \infty) : b \in \mathbb{R}\}$. Then $\tau(F)$ is the usual topology on $\mathbb{R}$.
- (c) $\tau(\emptyset)$ is the trivial topology.

Definition

Let X be a topological space. We say that $F \subseteq X$ is closed if $X \setminus F$ is open.

Proposition

Let (X, τ) be a topological space. Then,

- (1) $\emptyset$ and X are closed.
- (2) A union of finitely many closed sets is closed.
- (3) The intersection of arbitrarily many closed sets are closed.

Proof:

DeMorgan's Law. ■

Definition

Let (X, τ) be a topological space with $Y \subseteq X$. Then:

- (a) The closure of Y is:

$$\overline{Y} = \bigcap \{F \subseteq X : F \supseteq Y, F \text{ is closed}\}$$

$\overline{Y}$ is the smallest closed set containing Y .

- (b) A point in $\overline{Y}$ is called a limit point of Y .
- (c) A point $x \in Y$ is called an interior point if there exists $U \in \tau$ such that $x \in U \subseteq Y$.
- (d) The interior of Y is the set of interior points of Y , denoted Y° . We have

$$Y^\circ = \bigcup \{U \in \tau : U \subseteq Y\}$$

which is the largest open set contained in Y .

Definition

Let (X, τ) be a topological space and $x \in X$. We say that $N \subseteq X$ is a neighbourhood of x if $x \in N^\circ$. The collection $\mathcal{O}(x)$ of all neighbourhoods of x is called the neighbourhood filter of x .

Proposition 4.5

Let (X, τ) be a topological space and $Y \subseteq X$ with $x \in X$. Then $x \in \overline{Y}$ if and only if every neighbourhood N of x intersects Y (i.e. $N \cap Y \neq \emptyset$).

Proof:

In the forward direction, suppose $x \in \overline{Y}$. If there is some $N \in \mathcal{O}(x)$ such that $N \cap Y = \emptyset$, then setting $U = N^\circ$, we have $x \in U$ and $U \cap Y = \emptyset$, so U^c is closed and contains Y , so $\overline{Y} \subseteq U^c$, but $x \notin U^c$, contradicting $x \in \overline{Y}$.

In the reverse direction, suppose $N \cap Y = \emptyset$ for every neighbourhood of x . Then for every $F \subseteq X$ closed containing Y , we have that F^c is open and $F^c \cap Y = \emptyset$, implying $x \notin F^c$. Thus, $x \in F$, so $x \in F$ for all closed sets F containing Y , so $x \in \overline{Y}$. ■

Definition

A point $x \in \overline{Y} \setminus Y^\circ$ is called a boundary point of Y . The boundary of Y is $\partial Y = \overline{Y} \setminus Y^\circ$. Notice that if $x \in \partial Y$, then for every $N \in \mathcal{O}(x)$, we have $N \cap Y \neq \emptyset$ and $N \cap Y^c \neq \emptyset$.

Definition

We say a subset $Y \subseteq X$ is dense if $\overline{Y} = X$.

Countability Axioms

Fix a topological space (X, τ) .

Definition

A collection of sets $\mathcal{F} \subseteq \tau$ is a subbasis if $\tau = \tau(\mathcal{F})$. We say that $\mathcal{F}$ is a basis if every open set $U \in \tau$ is an arbitrary union of sets in $\mathcal{F}$.

Remark

If $\mathcal{F}$ is a subbasis, then $\mathcal{F}$ is a basis if and only if every finite intersection of sets in $\mathcal{F}$ is a union of sets in $\mathcal{F}$.

Examples

- (a) The collection of open balls in a metric space is a basis for the metric topology.
- (b) Let $X = \{1, 2, 3\}$ with the discrete topology. Let $\mathcal{F} = \{\{1, 2\}, \{2, 3\}, \{1, 3\}\}$. Then this is a subbasis but not a basis.
- (c) Let X be $\mathbb{R}$ with the usual topology and $\mathcal{F} = \{(-\infty, b) : b \in \mathbb{R}\} \cup \{(b, \infty) : b \in \mathbb{R}\}$ is a subbasis but not a basis.

Definition

- (a) Let $x \in X$. A neighbourhood basis for x is a collection of neighbourhoods $\nu(x)$ of x such that for every $N \in \mathcal{O}(x)$, there exists $N' \in \nu(x)$ with $N' \subseteq N$.
We say that (X, τ) is first countable if every point $x \in X$ has a countable neighbourhood basis. Note that if $\nu(x) = \{N_m : m \in \mathbb{N}\}$ is a neighbourhood basis for x , then letting $N'_m = \bigcap_{k=1}^m N_k$, we have that $\{N'_m : m \in \mathbb{N}\}$ is a nested neighbourhood basis for x .
- (b) We say (X, τ) is second countable if it has a countable basis.
- (c) We say X is separable if it has a countable dense subset.

Example

Metric spaces are first countable, since the balls centered at x of positive rational radii form a countable neighbourhood basis.

Remark

- (a) A second countable space is separable. Indeed, if $\mathcal{F} = \{U_n : n \in \mathbb{N}\}$ is a countable basis, choose some $x_n \in U_n$ for each n . Then $\{x_n : n \in \mathbb{N}\}$ is dense. The converse need not hold (Assignment 3).
- (b) A second countable space is first countable: if $x \in X$, let $\nu(x) = \{U \in \mathcal{F} : x \in U\}$ where $\mathcal{F}$ is a countable basis for X .

Proposition 4.8

A metric space is second countable if and only if it is separable.

Proof:

Second countable always implies separable, so conversely assume X is a separable metric space. Let $\{x_n : n \in \mathbb{N}\}$ be a countable dense subset of X . Then the collection $\{B(x_n, q) : n \in \mathbb{N}, q \in \mathbb{Q}^+\}$ is a countable basis for X . ■

Example

We will construct a topological space that is not first-countable. Let X be an uncountable set and let τ be the co-countable topology on X (a nonempty set U is open if and only if U^c is countable). Suppose for the sake of contradiction that X is first countable. Let $x \in X$ and $\nu(x) = \{U_n : n \in \mathbb{N}\}$ be a countable neighbourhood basis for x , where without loss of generality we may assume each U_n is open. Set $N' = \bigcap_{n=1}^{\infty} U_n$. Then $x \in N'$ and $(N')^c = \bigcup_{n=1}^{\infty} U_n^c$ is countable, so it is an open neighbourhood of x . Since X is uncountable, we can find $y \in N'$ with $y \neq x$ and set $N = N' \setminus \{y\}$. Then N is a neighbourhood of x , but there is no U_n with $U_n \subseteq N$.

Nets and convergence

Definition

- (a) A nonempty set Λ is a directed set if it is partially ordered and for each $\lambda_1, \lambda_2 \in \Lambda$ there exists $\lambda_3 \in \Lambda$ such that $\lambda_1 \leq \lambda_3, \lambda_2 \leq \lambda_3$.
- (b) Let X be a set. A net is a pair (Λ, x) , where Λ is a directed set and $x : \Lambda \rightarrow X$ is a function. We denote $x_\lambda := x(\lambda)$ for all $\lambda \in \Lambda$.

Example

Let $\mathbb{N}^k$ is a directed set, where the order is given by $x \leq y$ when $x_i \leq y_i$ for each $i \in \{1, \dots, k\}$.

Examples

- (a) A sequence is a net with $\Lambda = \mathbb{N}$.
- (b) Let $\emptyset \neq S$ be a set with $x : S \rightarrow \mathbb{R}$ a function. Let Λ be the set of nonempty finite subsets of S ordered by inclusion. For each $F \in \Lambda$, set $x_F := \sum_{s \in F} x(s)$. Then $(x_F)_{F \in \Lambda}$ is a net.
- (c) Let (X, τ) be a topological space and $x \in X$. Let $\mathcal{O}(x)$ be the neighbourhood filter of x with the partial order given by reverse inclusion $N_1 \leq N_2$ when $N_2 \subseteq N_1$. Then $\mathcal{O}(x)$ is direct since if $N_1, N_2 \in \mathcal{O}(x)$, then we can take $N_3 = N_1 \cap N_2$. For each $N \in \mathcal{O}(x)$, choose $x_N \in N$. Then $(x_N)_{N \in \mathcal{O}(x)}$ is a net.

Definition

Let X be a topological space. We say that a net $(x_\lambda)_{\lambda \in \Lambda}$ in X converges to $x \in X$ if for every neighbourhood N of x , $(x_\lambda)_{\lambda \in \Lambda}$ is eventually in N . We then write $x = \lim_{\lambda \in \Lambda} x_\lambda$.

Example

If $\mathcal{O}(x)$ is the neighbourhood filter at x and $x_N \in N$ for each $N \in \mathcal{O}(x)$, then the net $(x_N)_{N \in \mathcal{O}(x)}$ converges to x .

Proposition 4.11

Suppose $x \in \bar{Y}$. Then $x \in \bar{Y}$ if and only if there exists a net $(x_\lambda)_{\lambda \in \Lambda}$ converging to x .

Proof:

Suppose $x \in \bar{Y}$. Then $N \cap Y \neq \emptyset$ for every $N \in \mathcal{O}(x)$. For each $N \in \mathcal{O}(x)$ choose $x_N \in Y \cap N$. Then $(x_N)_{N \in \mathcal{O}(x)}$ is a net in Y converging to x .

Conversely, suppose $(x_\lambda)_{\lambda \in \Lambda}$ is a net in Y and let $x = \lim_{\lambda \in \Lambda} x_\lambda$. Let $N \in \mathcal{O}(x)$. Then there exists $\lambda_0 \in \Lambda$ such that $x_\lambda \in N$ for all $\lambda \geq \lambda_0$. Hence $Y \cap N \neq \emptyset$ and so $x \in \bar{Y}$ by Proposition 4.5. ■

Conclusion

The topology on X is characterized by its family of convergent nets: $Y \subseteq X$ is closed if and only if every convergent net $(x_\lambda)_{\lambda \in \Lambda}$ in Y has its limit in Y . The same is not true for sequences.

Example

Let X be an uncountable set with the cocountable topology (a set $A \subseteq X$ is open if it is empty, X , or its complement in X is countable). We claim a sequence $(x_n)_{n=1}^\infty$ converges to x if and only if this sequence is eventually constant. One direction is easy, since eventually constant sequences always converge. Conversely, suppose (x_n) converges to x . Consider the open neighbourhood set of x given by $U = X \setminus \{x_n : x_n \neq x\}$ which is open since it is cocountable. Since (x_n) converges to x there exists $k \in \mathbb{N}$ such that $x_n \in U$ for all $n \geq k$, so $x_n = x$ for all $n \geq k$.

Exercise

Show that if X is first-countable, then convergent sequences characterize its topology.

Continuity

Definition

Let (X, τ) and (Y, σ) be topological spaces. A function $f : X \rightarrow Y$ is continuous if $f^{-1}(U) \in \tau$ for every $U \in \sigma$. We say f is continuous at $x \in X$ if $f^{-1}(U) \in \mathcal{O}(x)$ for every $U \in \mathcal{O}(f(x))$.

Proposition 4.13

A function $f : X \rightarrow Y$ is continuous if and only if whenever $(x_\lambda)_{\lambda \in \Lambda}$ is a net in X converging to $x \in X$ then $(f(x_\lambda))_{\lambda \in \Lambda}$ converges to $f(x)$.

Proof:

First, suppose f is continuous. Let N be a neighbourhood of $f(x)$, and let $(x_\lambda)_{\lambda \in \Lambda}$ be a net in X converging to x . Since f is continuous, $f^{-1}(N) \in \mathcal{O}(x)$. Hence there exists λ_0 so that $x_\lambda \in f^{-1}(N)$ for $\lambda \geq \lambda_0$. Applying f we have $f(x_\lambda) \in N$ for all $\lambda \geq \lambda_0$, so $(f(x_\lambda))_{\lambda \in \Lambda}$ converges to $f(x)$. Conversely,

suppose $(f(x_\lambda))_{\lambda \in \Lambda}$ converges to $f(x)$ whenever $(x_\lambda)_{\lambda \in \Lambda}$ converges to x . Looking for a contradiction, suppose f is not continuous. Then there exists $V \subseteq Y$ open such that $f^{-1}(V)$ is not open, so there exists $x \in f^{-1}(V)$, $x \notin f^{-1}(V)^\circ$. Thus x is a boundary point of $f^{-1}(V)$. So for every $N \in \mathcal{O}(x)$ we can find $x_N \in f^{-1}(V)^c \cap N$. Then $(x_N)_{N \in \mathcal{O}(x)}$ converges to x but $(f(x_N))_{N \in \mathcal{O}(x)}$ does not converge to $f(x)$ since $f(x_N) \notin V$ for all $N \in \mathcal{O}(x)$. ■

Separation Axioms

Definition

We say that a topological space X is Hausdorff (or T_2) if for every $x, y \in X$ with $x \neq y$, there are disjoint open neighbourhoods U and V of x and Y respectively.

Example

Metric spaces are Hausdorff.

Remark

If continuous functions on X separate points, then X is Hausdorff. That is, if for every $x, y \in X$ with $x \neq y$ there exists $f : X \rightarrow \mathbb{R}$ continuous with $f(x) \neq f(y)$.

Proposition 4.15

A topological space X is Hausdorff if and only if every net $(x_\lambda)_{\lambda \in \Lambda}$ converges to at most one point.

Proof:

Exercise. ■

Remark

If X is Hausdorff, singletons are closed.

Definition

We say that a topological space X is normal if whenever A and B are disjoint closed sets in X , there are disjoint open sets U and V with $A \subseteq U$ and $B \subseteq V$.

Examples

(a) Metric spaces are normal. If A and B are disjoint closed sets, then the function $\rho : X \rightarrow [0, 1]$ given by $\rho(x) = \frac{d(x, A)}{d(x, B) + d(x, A)}$ is a continuous function and satisfies $\rho(A) = 0$ and $\rho(B) = 1$, so we can separate A and B with a continuous function and thus with open sets (for example $U = \rho^{-1}([0, 1/2])$ and $V = \rho^{-1}((1/2, 1])$).

(b) Similarly, pseudometric spaces are normal.

Definition

Let $f : X \rightarrow \mathbb{R}$ be a real-valued function.

(a) We say that f is lower semicontinuous if $f^{-1}((a, +\infty))$ is open for every $a \in \mathbb{R}$.

(b) f is upper semicontinuous if $f^{-1}((-\infty, a))$ is open for all $a \in \mathbb{R}$.

If f is extended real-valued, we must replace these intervals to be closed at $\pm\infty$.

Remark

A function $f : X \rightarrow \mathbb{R}$ is continuous if and only if f is both lower and upper semicontinuous.

Examples

(a) Let $U \subseteq X$ be open. The characteristic function of U is lower semicontinuous because $\chi_U^{-1}((a, +\infty)) =$

$$\begin{cases} \emptyset & a \geq 1 \\ U & 0 \leq a < 1. \end{cases} \text{ Similarly, if } s > r, \text{ then } g(x) = \begin{cases} s & x \in U \\ r & x \in U^c \end{cases} \text{ is lower semicontinuous.} \\ X & \text{else} \end{cases}$$

(b) If $F \subseteq X$ is closed, then χ_F is upper semicontinuous, because $\chi_F^{-1}((-\infty, a)) = \begin{cases} X & a > 1 \\ F^c & 0 \leq a < 1 \\ \emptyset & a < 0 \end{cases}$.

(c) If $\{f_\lambda\}_{\lambda \in \Lambda}$ is a collection of lower semicontinuous functions, then $f = \sup_{\lambda \in \Lambda} f_\lambda$ is lower semicontinuous since $f^{-1}((a, +\infty)) = \bigcup_{\lambda \in \Lambda} f_\lambda^{-1}((a, +\infty))$. Similarly, if $\{f_\lambda\}_{\lambda \in \Lambda}$ is a collection of upper semicontinuous functions, then $f = \inf_{\lambda \in \Lambda} f_\lambda$ is upper semicontinuous because $f^{-1}((-\infty, a)) = \bigcup_{\lambda \in \Lambda} f_\lambda^{-1}((-\infty, a))$.

With this, we move towards proving Urysohn's lemma.

Remark

Suppose X is normal with $A \subseteq U, B \subseteq V$ with U, V disjoint open sets and A and B closed sets. Then $\overline{U} \subseteq V^c$ since V^c is a closed set containing U , so $\overline{U} \cap B = \emptyset$.

Lemma 4.16

Suppose X is normal, $B \subseteq X$ is closed, and $B \subseteq U$, with U open. Then there exists $V \subseteq X$ open with $B \subseteq V \subseteq \overline{V} \subseteq U$.

Proof:

Apply normality of X with B and U^c together with the above remark to get $V \subseteq X$ open with $B \subseteq V \subseteq \overline{V}$, and $\overline{V} \cap U^c = \emptyset$, hence $\overline{V} \subseteq U$. ■

Theorem 4.17 (Urysohn's Lemma)

Let X be a normal topological space and let A and B be disjoint closed subsets of X . Then there exists a continuous function $f : X \rightarrow [0, 1]$ such that $f(B) = 1, f(A) = 0$.

Proof:

The idea is that we will use the lemma to get $B \subseteq V_0 \subseteq \overline{V_0} \subseteq A^c$ with V_0 open, then apply the lemma again to get V_1 open such that $B \subseteq V_1 \subseteq \overline{V_1} \subseteq V_0 \subseteq \overline{V_0} \subseteq A^c$, and then again to get $V_{1/2}$ open such that $B \subseteq V_1 \subseteq \overline{V_1} \subseteq V_{1/2} \subseteq \overline{V_{1/2}} \subseteq V_0 \subseteq \overline{V_0} \subseteq A^c$. On each piece of this splitting, we will define a lower semicontinuous function, and then keep doing this process for all rationals between $[0, 1]$, then take supremums to get a lower semicontinuous function that is also upper semicontinuous.

Onto the actual proof, use the lemma to get open sets V_0 and V_1 as above. Let $\{r_n : n \geq 0\}$ be an enumeration of $\mathbb{Q} \cap [0, 1]$ with $r_0 = 0, r_1 = 1$. Suppose by induction that for some $n \geq 1$ we have obtained open sets V_{r_i} for $i = 0, \dots, n$ with $\overline{V_0} \cap A = \emptyset$ and if $r_j > r_i$, we have $\overline{V_{r_j}} \subseteq V_{r_i}$. For $n + 1$, take $k \in \{0, \dots, n\}$ such that r_k is the least element with $r_{n+1} < r_k$, and take $\ell \in \{0, \dots, n\}$ such that r_ℓ is the largest element with $r_\ell < r_{n+1}$ (note that 0 and 1 are in $r_0, \dots, r_n$ already, so we can always make such choices). Since $\overline{V_{r_k}} \subseteq V_{r_\ell}$, we get from the above lemma an open set $V_{r_{n+1}}$ with $\overline{V_{r_k}} \subseteq V_{r_{n+1}} \subseteq \overline{V_{r_{n+1}}} \subseteq V_{r_\ell}$. By induction, we get a collection of open sets $\{V_r : r \in [0, 1] \cap \mathbb{Q}\}$ with $\overline{V_0} \cap A = \emptyset$ and for all r and s in $\mathbb{Q} \cap [0, 1]$ with $s > r$ we have $B \subseteq V_s \subseteq \overline{V_s} \subseteq V_r$. For each $r \in \mathbb{Q} \cap [0, 1]$,

define $f_r : X \rightarrow [0, 1]$ by $f_r(x) = \begin{cases} r & x \in V_r \\ 0 & \text{else} \end{cases}$. Then f_r is lower semicontinuous since $f_r = r\chi_{V_r}$. For

each $s \in \mathbb{Q} \cap [0, 1]$, define $g_s : X \rightarrow [0, 1]$ by $g_s(x) = \begin{cases} 1 & x \in \overline{V_s} \\ s & \text{else} \end{cases}$. Then g_s is upper semicontinuous.

Let $f = \sup_r f_r$ and $g = \inf_s g_s$. Then f is lower semicontinuous and g is upper semicontinuous with $g(B) = 1$ and $f(A) = 0$. We are done if we prove $f = g$, since then our function f is both upper and lower semicontinuous, hence continuous, and satisfies the desired properties. Looking for a contradiction, suppose $f(x) \neq g(x)$ for some $x \in X$. First, suppose $f(x) > g(x)$. Then $f_r(x) > g_s(x)$ for some $r, s \in \mathbb{Q} \cap [0, 1]$. Then $x \notin \overline{V_s}$ by definition of g_s , and $x \in V_r$ by definition of f_r , so this can only happen if $r = f_r(x) > g_s(x) = s$. But $r > s$ implies $\overline{V_r} \subseteq V_s$, so $f_r \leq g_s$, which is a contradiction.

Thus, we must have $f(x) < g(x)$. Choose rationals p, q with $f(x) < p < q < g(x)$. Since $f(x) = \sup_r f_r(x) < p$ it follows that $x \notin V_p$ since $f(x) \geq f_p(x) = p\chi_{V_p}(x)$. Since $g(x) = \inf_s g_s(x) > q$, we must have $x \in \overline{V_q}$. But $p < q$, which yields $x \in \overline{V_q} \subseteq V_p$, a contradiction. We thus get $f = g$. ■

Remark

If X is normal and $A, B \subseteq X$ are disjoint closed sets, it follows from Urysohn's Lemma that if $a < b$, there exists $f : X \rightarrow [a, b]$ continuous with $f(A) = a, f(B) = b$.

Definition

If (X, τ) is a topological space and $A \subseteq X$, the relative topology on A is $\tau_A = \{A \cap U : U \in \tau\}$.

Theorem 4.18 (Tietze's Extension Theorem)

Let X be a normal space and $A \subseteq X$ a nonempty closed set. Suppose $f : A \rightarrow \mathbb{R}$ is continuous with respect to the relative topology. Then there exists $F : X \rightarrow \mathbb{R}$ continuous extending f . Moreover, if f is bounded, then F is bounded with $\|F\|_\infty = \|f\|_\infty$.

Proof:

Suppose first $f(A) \subseteq [-1, 1]$, with $\|f\|_\infty = 1$. Let $E_1 = f^{-1}([-1, -1/3])$ and $F_1 = f^{-1}([1/3, 1])$. These are closed sets in X because A is closed and f is continuous, with $E_1 \cap F_1 = \emptyset$. By Urysohn's lemma, there exists $f_1 : X \rightarrow [-1/3, 1/3]$ with $f_1(E_1) = -1/3$ and $f_1(F_1) = 1/3$. Then, if $x \in E_1 \cup F_1$, we have $|f(x) - f_1(x)| \leq 2/3$. On the other hand, if $x \in A$ with $x \notin E_1 \cup F_1$, then $f(x) \in (-1/3, 1/3)$, so $|f(x) - f_1(x)| \leq 2/3$, so $|f(x) - f_1(x)| \leq 2/3$ for all $x \in A$. Thus, $f - f_1 : A \rightarrow [-2/3, 2/3]$ is continuous, so we can apply Urysohn's lemma again to get $f_2 : X \rightarrow [-2/3^2, 2/3^2]$ with $|f(x) - f_1(x)| \leq 2^2/3^2$. Continuing by induction, we obtain a sequence (f_n) of continuous functions on X with $f_n : X \rightarrow [-\frac{2^{n-1}}{3^n}, \frac{2^{n-1}}{3^n}]$ with $|f(x) - f_1(x) - \dots - f_n(x)| \leq \frac{2^n}{3^n}$ for all $x \in A$ and $n \in \mathbb{N}$.

Thus, define $F(x) = \sum_{n=1}^{\infty} f_n(x)$. We have

$$\sum_{n=1}^{\infty} |f_n(x)| \leq \sum_{n=1}^{\infty} \frac{2^{n-1}}{3^n} = 1$$

and so by the Weierstrass M-test, this series converges uniformly on X and hence F is well-defined and continuous, with $|F(x)| \leq 1$ for all $x \in X$, so $\|F\|_\infty \leq 1$, and F extends f by construction, which since $\|f\|_\infty = 1$, implies that $\|F\|_\infty = 1$.

In the case $f(A) \subseteq [a, b]$, first compose with a homeomorphism $[a, b] \rightarrow [0, 1]$ and apply the above step. In the case where f is not bounded, choose a homeomorphism $\mathbb{R} \rightarrow (-1, 1)$ and apply the above. ■

Compactness**Definition**

Let X be a topological space. We say that X is compact if every open cover $\mathcal{G} = \{G_\alpha\}_{\alpha \in I}$ has a finite subcover.

Proposition 4.20

Suppose X is Hausdorff and $K \subseteq X$ is compact. Let $x \in X$ with $x \notin K$. Then there exist disjoint open sets U and V with $K \subseteq U$ and $x \in V$.

Proof:

Suppose $x \notin K$. Then for each $y \in K$, by Hausdorffness there exist disjoint open sets V_y, U_y with $x \in V_y$ and $y \in U_y$. Then $\{U_y : y \in K\}$ is an open cover of K , so by compactness it has a finite subcover $\{U_{y_1}, \dots, U_{y_n}\}$. Define $V = \bigcap_{i=1}^n V_{y_i}$ and $U = \bigcup_{i=1}^n U_{y_i}$, which are the desired disjoint open sets. ■

Remark

- (a) If X is Hausdorff and $K \subseteq X$ is compact, then the above argument shows that K is closed.
- (b) If $K \subseteq X$ is compact and $F \subseteq K$ is closed, then F is compact.
- (c) If X is Hausdorff, $K \subseteq X$ is compact, and $F \subseteq X$ is closed, then $K \cap F$ is compact.

Theorem 4.21

Every compact Hausdorff space X is normal.

Proof:

Suppose A and B are disjoint closed subsets of X , hence compact. For each $x \in B$, use Proposition 4.20 to get disjoint open sets U_x and V_x with $A \subseteq U_x$ and $x \in V_x$. Then $\{V_x : x \in B\}$ is an open cover of B , so proceed as in the proof of Proposition 4.20 using compactness of B to get disjoint open sets U and V with $A \subseteq U$ and $B \subseteq V$. ■

Definition

- (a) We say that $x \in X$ is an accumulation point of a net $(x_\lambda)_{\lambda \in \Lambda}$ if for every neighbourhood $N \in \mathcal{O}(x)$ of X , the net $(x_\lambda)_{\lambda \in \Lambda}$ is in N frequently, that is, for every $\lambda \in \Lambda$, there exists $\mu \in \Lambda$ with $\mu \geq \lambda$ such that $x_\mu \in N$.
- (b) X has the finite intersection property (abbreviated FIP) if for every collection $\{F_\alpha : \alpha \in I\}$ of closed sets in X such that the intersection of every finite subcollection is nonempty, we have that the full intersection is nonempty, $\bigcap_{\alpha \in I} F_\alpha \neq \emptyset$.

Theorem 4.23

Let (X, τ) be a topological space. The following are equivalent:

- (1) X is compact.
- (2) X has the finite intersection property.
- (3) Every net in X has an accumulation point.

Proof:

- (1) if and only if (2) follows as for metric spaces.

To prove (2) implies (3), suppose X has the finite intersection property, and let $(x_\lambda)_{\lambda \in \Lambda}$ be a net in X . For each $\lambda \in \Lambda$, define $F_\lambda = \{x_\mu : \mu \geq \lambda\}$. Now, if $\lambda_1, \dots, \lambda_n \in \Lambda$, we have $\bigcap_{i=1}^n F_{\lambda_i} \neq \emptyset$, because we can take a $\mu \in \Lambda$ such that $\mu \geq \lambda_i$ for all i because Λ is directed. Thus, by the finite intersection property, $\bigcap_{\lambda \in \Lambda} F_\lambda \neq \emptyset$, so there exists $x \in \bigcap_{\lambda \in \Lambda} F_\lambda$. We claim this is an accumulation point. Indeed, let $N \in \mathcal{O}(x)$ and $\lambda_0 \in \Lambda$ be arbitrary. Then, we have $x \in F_{\lambda_0}$, so by definition of F_{λ_0} , there is $\mu \geq \lambda_0$ so that $x_\mu \in N$, proving (3).

To prove (3) implies (2), let $F = \{F_\lambda : \lambda \in \Lambda\}$ be a collection of closed sets for which every finite subcollection is nonempty. Define a net $(x_S)_{S \subseteq \Lambda}$ indexed by finite subsets S of Λ ordered by inclusion: for each $S \subseteq \Lambda$ finite, pick $x_S \in \bigcap_{\lambda \in S} F_\lambda$ (which exists by hypothesis). By (3), (x_S) has an accumulation point x . We claim $x \in \bigcap_{\lambda \in \Lambda} F_\lambda$. Indeed, if $N \in \mathcal{O}(x)$ is a neighbourhood of x , because x is an accumulation point, for any $\lambda_0 \in \Lambda$, considering the finite subset $\{\lambda_0\} \subseteq \Lambda$, there is a finite subset $S \subseteq \Lambda$ with $\{\lambda_0\} \subseteq S$ so that $x_S \in N$. But $x_S \in \bigcap_{\lambda \in S} F_\lambda \subseteq F_{\lambda_0}$, so $x_S \in N \cap F_{\lambda_0}$, which shows $x \in \overline{F_{\lambda_0}} = F_{\lambda_0}$ by closure. Since F_{λ_0} was arbitrary, this implies $x \in \bigcap_{\lambda \in \Lambda} F_\lambda$, as needed. ■

Definition

Let X be a set and let $F = \{f : X \rightarrow Y_f\}$ be a family of functions. If each Y_f has a topology τ_f , the initial topology on F is the weakest topology on X that makes all the functions in F continuous. Examples include the product topology and the weak topology.

Definition

Let X be a set and let $F = \{f : Y_f \rightarrow X\}$ be a family of functions. If each Y_f has a topology τ_f , the final topology on X is the strongest topology on X that makes all the functions in F continuous. Examples include the quotient topology.

Definition

Let $\{(X_\alpha, \tau_\alpha)\}_{\alpha \in A}$ be a collection of topological spaces. Let $X = \prod_{\alpha \in A} X_\alpha$ be the Cartesian product over all α . Let $\pi_\alpha : X \rightarrow X_\alpha$ be the projection onto the α coordinate. The product topology on X is the initial topology on X induced by the family $\{\pi_\alpha : \alpha \in A\}$.

We note that $\{\pi_\alpha^{-1}(U_\alpha) : \alpha \in I, U_\alpha \subseteq X_\alpha, U_\alpha \in \tau_\alpha\}$ is a subbasis of the product topology. The product is Hausdorff if each X_α is Hausdorff, and a function $f : Y \rightarrow X$ is continuous if and only if $\pi_\alpha \circ f$ is continuous for all α . Each π_α is continuous and open.

Remark

The Cartesian product $\{0, 1\}^{\mathbb{N}}$, where $\{0, 1\}$ has the discrete topology, is not compact with the box topology (the topology given by the Cartesian product of open sets), but it turns out that $\{0, 1\}^{\mathbb{N}}$ with the product topology is homeomorphic to the Cantor set (and thus compact).

Theorem 4.26 (Tychonoff's Theorem)

Let $\{(X_\alpha, \tau_\alpha) : \alpha \in A\}$ be a family of compact topological spaces. Then $X = \prod_{\alpha \in A} X_\alpha$ is compact with respect to the product topology.

For the proof, we will need:

Theorem (Alexander Subbasis Theorem)

Let (X, τ) be a topological space and S a subbasis. If every open cover of X consisting of elements of S has a finite subcover, then X is compact.

Proof of Tychonoff's Theorem:

Recall that a subbasis of the product topology is given by $S = \{\pi_\alpha^{-1}(U) : \alpha \in A, U \text{ open in } X_\alpha\}$. By the Alexander Subbasis Theorem, it suffices to prove an open cover consisting of elements of S has a finite subcover. Let $\mathcal{V}$ be an open cover of X consisting of elements of S . Thus, every element of $\mathcal{V}$ is of the form $O = \pi_{\alpha(O)}^{-1}(V(O))$ for some $\alpha(O) \in A$ and some $V(O)$ open in $X_{\alpha(O)}$.

We claim there is some $\alpha \in A$ so that $\{V(O) : O \in \mathcal{V}, \alpha(O) = \alpha\}$ covers X_α . Indeed, suppose not, so for every $\alpha \in A$, we can pick a point $x_\alpha \in X_\alpha$ such that $x_\alpha \notin V(O)$ for all $O \in \mathcal{V}$ with $\alpha(O) = \alpha$ (using the Axiom of Choice). Let $x = (x_\alpha)_{\alpha \in A}$. Then, for all $O \in \mathcal{V}$, x is not in O . Indeed, if $x \in O$, let $\alpha = \alpha(O)$ and let $V = V(O)$. Then $\pi_\alpha(x) = x_\alpha$ is in V , which it was chosen not to be in. This contradicts the fact $\mathcal{V}$ covers X , and so the claim follows.

Fix α from the previous claim. Then, $\{V(O) : O \in \mathcal{V}, \alpha(O) = \alpha\}$ covers the compact space X_α , so by compactness, it has a finite subcover $V(O_1), \dots, V(O_n)$. We claim $O_1, \dots, O_n$ cover X . Indeed, let $x \in X$. Then $\pi_\alpha(x) \in V(O_i)$ for some $i \in \{1, \dots, n\}$ and thus $x \in \pi_\alpha^{-1}(V(O_i)) = O_i$. Thus, $O_1, \dots, O_n$ is a finite subcover of $\mathcal{V}$, so by the Alexander Subbasis Theorem, X is compact. ■

Proof of the Alexander Subbasis Theorem

Suppose X is not compact, so it has an open cover without a finite subcover. By Zorn's lemma, there is a maximal open cover $\mathcal{U}$ without a finite subcover. Now, we note four properties of $\mathcal{U}$:

- (1) If $V \notin \mathcal{U}$ is open, then $\mathcal{U} \cup \{V\}$ must have a finite subcover by maximality, and it must include V (lest $\mathcal{U}$ have a finite subcover).
- (2) Conversely, if there are finitely many $U_1, \dots, U_n \in \mathcal{U}$ such that $U_1, \dots, U_n, V$ cover X , then $V \notin \mathcal{U}$.
- (3) If $V_1, \dots, V_n \notin \mathcal{U}$, then $V_1 \cap \dots \cap V_n \notin \mathcal{U}$. Indeed, by (1), for each $i \in \{1, \dots, n\}$, we have $n_i \in \mathbb{N}$ and $V_{i,j} \in \mathcal{U}$ for $j = 1, \dots, n_i$ so that $V_i \cup \bigcup_{j=1}^{n_i} V_{i,j} = X$. But then $x \in X$ is in some $V_{i,j}$ for $i = 1, \dots, n$, $j = 1, \dots, n_i$, or if not must be in V_i for each $i = 1, \dots, n$, so that

$$(V_1 \cap \dots \cap V_n) \cup \bigcup_{i=1, \dots, n, j=1, \dots, n_i} V_{i,j} = X$$

As all $V_{i,j}$'s are in $\mathcal{U}$, by (2), $V_1 \cap \dots \cap V_n \notin \mathcal{U}$.

- (4) If $V \notin \mathcal{U}$ and $V \subseteq W$, then $W \notin \mathcal{U}$. Indeed, by (1), there are $V_1, \dots, V_n \in \mathcal{U}$ such that $V \cup \bigcup_{i=1}^n V_i = X$, so $W \cup \bigcup_{i=1}^n V_i = X$, so by (2), $W \notin \mathcal{U}$.

Now, we claim $S \cap \mathcal{U}$ is an open cover of X .

Indeed, let $x \in X$. Then there exists $O \in \mathcal{U}$ such that $x \in O$ since $\mathcal{U}$ is a cover, so since S is a subbasis, we can write $x \in \bigcap_{i=1}^n S_i \subseteq O$ for $S_1, \dots, S_n \in S$. Then, there is $S_j \in \mathcal{U}$, since otherwise by (3) we have $\bigcap_{i=1}^n S_i \notin \mathcal{U}$, so by (4) we have $O \notin \mathcal{U}$. Thus, $x \in S_j \in \mathcal{U} \cap S$ for some $j = 1, \dots, n$, so $\mathcal{U} \cap S$ is an open cover of X . But $S \cap \mathcal{U}$ is an open cover of X consisting of elements of S , so by hypothesis, it has a finite subcover. This gives a finite subcover of $\mathcal{U}$, contradiction. ■

5: The Bidual and Weak Topologies

Reflexive Spaces, Weak, Weak-* Topologies

We have seen that the unit ball of a normed space V is not compact if V is infinite dimensional. We consider other topologies on V to overcome this. Let V be a normed space, and V^* its dual (which is a Banach space). We consider the dual $V^{**} = (V^*)^*$, called the bidual of V , which is again a Banach space. For each $x \in V$, define $\hat{x} : V^* \rightarrow \mathbb{F}$ by $\hat{x}(f) = f(x)$ for $f \in V^*$. Then $\hat{x}$ is linear and $|\hat{x}(f)| \leq \|f\|_{op} \|x\|$, so by Hahn-Banach, we have $\|\hat{x}\| = \|x\|$.

Definition

The isometric linear map $i : V \rightarrow V^{**}$ defined by $i(x) = \hat{x}$ is called the natural map. We say that a Banach space is reflexive if the natural map is an isomorphism.

Remark

It may happen that V is not reflexive but V is isometrically isomorphic to V^{**} via some other map (James spaces).

Examples

- (a) For $1 < p < \infty$, ℓ^p is reflexive, indeed if $\frac{1}{p} + \frac{1}{q} = 1$, we have $(\ell^p)^{**} \cong (\ell^q)^* \cong \ell^p$, so we need only check the natural map is an isomorphism: we have $\hat{x}(y) = \sum_{n=1}^{\infty} x_n y_n$, so under the isomorphism $(\ell^q)^* \cong \ell^p$, this is the identity map on ℓ^p .
- (b) Let $c = \{(x_n) \in \ell^\infty : \lim_{n \rightarrow \infty} x_n \text{ exists}\}$. Then $c^* \cong \ell^1(\{0\} \cup \mathbb{N})$ by the assignments, where the isomorphism is $y \mapsto f_y$, where $f_y((x_n)) = y_0 \lim_{n \rightarrow \infty} x_n + \sum_{n=1}^{\infty} x_n y_n$, so under the identification $(\ell^1)^* = \ell^\infty$, the natural map $i : c \rightarrow \ell^\infty$ is $i((x_n)) := (\lim_{n \rightarrow \infty} x_n, x_1, x_2, \dots)$.
- (c) If H is a Hilbert space, then H is reflexive by the Riesz representation theorem.

Definition

Let V be a normed space. The weak topology on V , denoted by $\sigma(V, V^*)$, is the initial topology on V^* , that is, the weakest (coarsest) topology on V making all the functions in V^* continuous. A basis for $\sigma(V, V^*)$ consists of sets of the form $f_1^{-1}(U_1) \cap \dots \cap f_n^{-1}(U_n)$, where each $f_j \in V^*$ and $U_j \subseteq \mathbb{F}$ is open.

Remark

- (a) A subbasis for $\sigma(V, V^*)$ consists of sets of the form $U_{f, \varepsilon}(x) = \{y \in V : |f(y) - f(x)| < \varepsilon\}$, where $f \in V^*$ and $\varepsilon > 0$.
- (b) A net $(x_\lambda)_{\lambda \in \Lambda}$ converges to $x \in V$ weakly, written $x_\lambda \xrightarrow{w} x$, if and only if $f(x_\lambda) \rightarrow f(x)$ for all $f \in V^*$.
- (c) A set is a neighbourhood of a point x if and only if it contains a set of the form $U_{(f_1, \dots, f_n), \varepsilon}(x) = \bigcap_{j=1}^n U_{f_j, \varepsilon}(x)$, using the notation as above.
- (d) The weak topology is Hausdorff by Hahn-Banach, as if $x \neq y$, there exists $f \in V^*$ such that $f(x) \neq f(y)$.

Definition

Let V be a normed vector space. We define the weak-* topology on V^* to be the coarsest topology on V^* making all functionals in the set $\{\hat{x} := \iota(x) : x \in V\}$ continuous.

Remark

A net $(\psi_\lambda)_{\lambda \in \Lambda}$ converges to ψ in the weak-* topology if $\psi_\lambda(x) \rightarrow \psi(x)$ for all $x \in X$.

Theorem (Banach-Alaoglu)

Let X be a normed vector space. The closed unit ball B of X^* is compact in the weak-* topology:

Proof:

For every $x \in X$, let

$$D_x = \{z \in \mathbb{F} : |z| \leq \|x\|\}$$

Then D_x is compact in $\mathbb{F}$ for each $x \in X$, so by Tychonoff's theorem, $D = \prod_{x \in X} D_x$ is compact in the product topology. Define a map $\Phi : B \rightarrow D$ by $\Phi(f) = (f(x))_{x \in X}$. Then, Φ is injective, and it is a homeomorphism onto its image, since given a net $(f_\lambda)_{\lambda \in \Lambda}$ in B , it converges to $f \in B$ if and only if

$$\lim_{\lambda} f_\lambda(x) = \lim_{\lambda} \widehat{x}(f_\lambda) = \widehat{x}(f) = f(x)$$

for all $x \in X$, so it converges to f if and only if $\lim_{\lambda} \Phi(f_\lambda) = \Phi(f)$. This shows Φ is a homeomorphism $B \rightarrow \Phi(B)$. Thus, if $\Phi(B)$ is closed in D (and thus compact), we are done. Suppose that $(f_\lambda)_{\lambda \in \Lambda}$ is a net in B , and that $(\Phi(f_\lambda))_{\lambda \in \Lambda}$ converges to $d \in D$. Then,

$$\lim_{\lambda \in \Lambda} f_\lambda(x) = d_x$$

Define $f(x) := d_x$ for each $x \in X$. Then f is linear since each f_λ is, and $|f(x)| = |d_x| \leq \|x\|$ for all $x \in X$ (by definition of D_x), so $f \in B$. Then $\Phi(f) = \lim_{\lambda \in \Lambda} (\Phi(f_\lambda))$ by construction, so the image of Φ is closed in D and we are done. ■

Proposition

Let $\mathcal{H}$ be a Hilbert space. The closed unit ball B of $\mathcal{H}$ is weakly compact.

Proof:

Let $(\xi_\alpha)_{\alpha \in I}$ be a net in B . For each $\alpha \in I$, define $\psi_\alpha \in \mathcal{H}^*$ be defined by

$$\psi_\alpha(y) = \langle y, \xi_\alpha \rangle$$

By the Cauchy-Schwarz inequality, we have $|\psi_\alpha(y)| \leq \|y\|$, so $\|\psi_\alpha\|_{op} \leq 1$ for all $\alpha \in I$. By Banach-Alaoglu, (ψ_α) admits a subnet $(\psi_\beta)_{\beta \in J}$ converging weak-* to some $\psi \in \mathcal{H}^*$ with $\|\psi\|_{op} \leq 1$. By the Riesz Representation Theorem, there exists a unique vector $\xi_\psi \in B$ such that $\psi(y) = \langle y, \xi_\psi \rangle$ for all $y \in \mathcal{H}$, and thus

$$\langle \xi_\beta, y \rangle = \overline{\langle y, \xi_\beta \rangle} = \overline{\psi_\beta(y)} \rightarrow \overline{\psi(y)} = \overline{\langle y, \xi_\psi \rangle} = \langle \xi_\psi, y \rangle$$

for all $y \in \mathcal{H}$. Thus $(\xi_\beta) \rightarrow \xi_\psi$ weakly, and so B is weakly compact. ■

Hahn-Banach Separation

Definition

A topological vector space (X, τ) is a (real or complex) vector space equipped with a topology τ such that the maps $f : X \times X \rightarrow X$ defined by $f(x, y) = x + y$ and $g(\alpha, x) = \alpha x$ are continuous with respect to the product topologies on $X \times X$ and $\mathbb{F} \times X$. In this course, topological vector spaces are assumed to be Hausdorff.

Definition

Let X be a (real or complex) vector space. Then $K \subseteq X$ is convex if $tx + (1 - t)y \in K$ for all $x, y \in K$ and $t \in [0, 1]$.

Definition

Let $E \subseteq X$. The convex hull of E is the set

$$\text{conv}(E) = \left\{ \sum_{j=1}^n t_j x_j : x_1, \dots, x_n \in E, t_1, \dots, t_n \in [0, 1], \sum_{j=1}^n t_j = 1 \right\}$$

Proposition

Let (X, τ) be a topological vector space. If $K \subseteq X$ is convex, then $\overline{K}$ is convex.

Proof:

Exercise (use continuity of vector addition and scalar multiplication).

Lemma

Let (X, τ) be a topological vector space and let $C \subseteq X$ be a convex open set containing 0. The function $p : X \rightarrow \mathbb{R}$ defined by

$$p(x) = \inf\{t > 0 : \frac{1}{t}x \in C\}, \quad x \in X$$

(called the Minkowski functional of C) is a sublinear functional on C .

Proof:

Let $N \subseteq C$ be a neighbourhood of 0. Choosing a sequence (t_n) with $t_n > 0$ for all $n \in \mathbb{N}$ with $t_n \rightarrow \infty$, we see that $\frac{1}{t_n}x \rightarrow 0$ by continuity of scalar multiplication, so $(\frac{1}{t_n}x)$ is eventually contained in N , thus $\{t > 0 : \frac{1}{t}x \in C\} \neq \emptyset$, so p is well-defined.

Next, to see p is positively homogeneous, fix $\lambda > 0$. Then,

$$p(\lambda x) = \inf\{t > 0 : \frac{1}{t}(\lambda x) \in C\} = \lambda \inf\{\frac{t}{\lambda} > 0 : \frac{\lambda}{t}x \in C\} = \lambda p(x)$$

and it is clear that when $\lambda = 0$ that $p(\lambda x) = 0$. To see subadditivity, fix $x, y \in X$ and let $s, t > 0$ be such that $\frac{1}{s}x \in C, \frac{1}{t}y \in C$. Then

$$\frac{s}{s+t} + \frac{t}{s+t} = 1$$

so since C is convex, it follows that

$$\frac{1}{s+t}(x+y) = \frac{s}{s+t}\frac{1}{s}x + \frac{t}{s+t}\frac{1}{t}y \in C$$

Thus $s+t \in \{r > 0 : \frac{1}{r}(x+y) \in C\}$, hence $s+t \geq \inf\{r > 0 : \frac{1}{r}(x+y) \in C\} = p(x+y)$, so taking infimums, we get $p(x) + p(y) \geq p(x+y)$. ■

Theorem (Hahn-Banach Separation 1)

Let (X, τ) be a topological vector space $U \subseteq X$ a convex open set, and $K \subseteq X$ a convex set. If $K \cap U = \emptyset$, there exists a continuous linear functional $\psi : X \rightarrow \mathbb{F}$ and $\alpha \in \mathbb{R}$ such that $\operatorname{Re}(\psi(x)) \leq \alpha < \operatorname{Re}(\psi(y))$ for all $x \in K, y \in U$.

Proof:

First, consider the case $\mathbb{F} = \mathbb{R}$. Without loss of generality suppose $K, U \neq \emptyset$. Fix points $x_0 \in U, y_0 \in K$, and let $z_0 = y_0 - x_0$. Let $C = U - K + z_0$. Then C is convex (exercise). Also, $0 = x_0 - y_0 + z_0 \in C$. Since

$$C = \bigcup_{y \in K} (U - y + z_0)$$

we see that C is open. Moreover, since $U \cap K = \emptyset$, we have $z_0 \notin C$. Define $\varphi : \operatorname{Span}\{z_0\} \rightarrow \mathbb{R}$ by $\varphi(\lambda z_0) = \lambda$. We note that $p(w) < 1$ if and only if $w \in C$. This follows by definition of the Minkowski functional p . Since $z_0 \notin C$, we have $p(z_0) \geq 1$, so

$$\lambda \geq 0 \implies \varphi(\lambda z_0) = \lambda < p(\lambda z_0)$$

$$\lambda < 0 \implies \varphi(\lambda z_0) = \lambda < 0 \leq p(\lambda z_0)$$

By the Hahn-Banach extension theorem, there exists a linear functional $\psi : X \rightarrow \mathbb{R}$ with $\psi|_{\operatorname{Span}\{z_0\}} = \varphi$ and $\psi(w) \leq p(w)$ for all $w \in X$. Since C is open with $0 \in C$, the previous inequality implies ψ is continuous at 0, so by linearity, ψ is continuous on X . Then for $x \in U, y \in K$,

$$\psi(x-y) + 1 = \psi(x-y+z_0) \leq p(x-y+z_0) < 1$$

Thus, $\psi(x - y) < 0$ for all $x \in U, y \in K$. By convexity of K and U , it follows that $\psi(K), \psi(U)$ are disjoint intervals in $\mathbb{R}$. Moreover, $\psi(U)$ is open, so there exists $\beta \in \mathbb{R}$ such that $\psi(x) < \beta \leq \psi(y)$ for all $x \in U, y \in K$. In the case $\mathbb{F} = \mathbb{C}$, we can view X as an $\mathbb{R}$ -vector space to get an $\mathbb{R}$ -linear functional $\varphi : X_{\mathbb{R}} \rightarrow \mathbb{R}$ where the desired inequality holds. Let $\psi : X \rightarrow \mathbb{C}$ be defined by

$$\psi(x) = \varphi(x) - i\varphi(ix)$$

Then the claimed inequality holds for ψ . ■

Definition

A topological vector space (X, τ) is locally convex if there exists a collection P of seminorms on X witnessing the topology. That is, $(x_\alpha) \rightarrow x$ if and only if $p(x_\alpha - x) \rightarrow 0$ for all $p \in P$.

Theorem (Hahn-Banach Separation 2)

Let (X, τ) be a locally convex topological vector space, with $F \subseteq X$ be a closed convex set, and $K \subseteq X$ a compact convex set. If $K \cap F = \emptyset$, then there exists a continuous linear functional $\psi : X \rightarrow \mathbb{F}$ and $\alpha \in \mathbb{R}$ such that $\operatorname{Re}(\psi(x)) < \alpha < \operatorname{Re}(\psi(y))$ for all $x \in F, y \in K$.

Proof:

By local convexity, we can find $U \subseteq X \setminus F$ open and convex with $K \subseteq U$ (let P be a collection of seminorms on X . Then $U_{H,\varepsilon} = \{x \in X : p(x) < \varepsilon, \forall p \in H\}$ is an open and convex set for all $\varepsilon > 0$ and every finite subset $H \subseteq P$, so we get an open cover to which we can apply compactness). Thus, by Hahn-Banach separation 1, there exists a continuous linear functional $\psi : X \rightarrow \mathbb{F}$ and $\beta \in \mathbb{R}$ such that $\operatorname{Re}(\psi(x)) \leq \beta < \operatorname{Re}(\psi(y))$ for all $x \in F, y \in U$. Since $K \subseteq U$ is compact, it follows that $\gamma = \inf\{\operatorname{Re}(\psi(y)) : y \in K\} > \beta$. It suffices to let $\alpha = \frac{1}{2}(\beta + \gamma)$. ■

Remark

If F is a cone (closed under positive scalar multiplication), then ψ can be chosen such that $\operatorname{Re}(\psi(x)) \leq 0 < \operatorname{Re}(\psi(y))$. If F is a subspace, then ψ can be chosen such that $\psi|_F = 0$. If X is Hausdorff, then $\{x\}$ is closed, so if F is closed, convex, and $x \notin F$, then there exists ψ such that $\operatorname{Re}(\psi(y)) < \alpha < \operatorname{Re}(\psi(x))$ for all $y \in F$.

Theorem (Mazur)

Let X be a normed vector space. If $K \subseteq X$ is a convex set, then its weak closure and norm closure coincide.

Proof:

It is clear that $\overline{K} \subseteq \overline{K}^{weak}$. Without loss of generality, assume that K is norm closed. Let $(x_\alpha)_{\alpha \in I}$ be a net in K weakly converging to $x \in X$. Suppose $x \notin K$. Then by Hahn-Banach separation 2, we can find $\psi \in X^*$ and $\beta \in \mathbb{R}$ such that $\operatorname{Re}(\psi(y)) < \beta < \operatorname{Re}(\psi(x))$ for all $y \in K$. As $(x_\alpha) \rightarrow x$ weakly, we must have $(\psi(x_\alpha)) \rightarrow \psi(x)$. However, $\operatorname{Re}(\psi(x_\alpha)) < \beta < \operatorname{Re}(\psi(x))$ for all α , so we can't have convergence. Thus K is weakly closed. ■

Proposition

Let X be a locally convex topological vector space. If X is Hausdorff, then the continuous linear functionals $X \rightarrow \mathbb{F}$ separate the points of X .

Proof:

This is the remark above with Hahn-Banach separation 2, since singletons are closed, compact, and convex. ■

More on Weak and Weak-* Topologies

Theorem (Goldstine)

If X is a normed vector space and $\iota : X \rightarrow X^{**}$ is the canonical embedding, then for $B \subseteq X$ the closed unit ball, $\iota(B)$ is weak-* dense in $B_1(X^{**})$.

Proof:

Suppose that there exists $f \in B_1(X^{**}) \setminus \overline{\iota(B)}^{w*}$. Then by Hahn-Banach separation 2, there exists $\psi \in X^*$ and $\beta \in \mathbb{R}$ such that

$$\operatorname{Re} \psi(x) = \operatorname{Re} \iota(x)(\psi) < \beta < \operatorname{Re} f(\psi)$$

for all $x \in B$. (Hahn-Banach separation 2 tells us there is $\psi' : X^{**} \rightarrow \mathbb{F}$ weak-* continuous with the desired inequality, but this implies $\psi' = \iota(\psi)$ for some $\psi \in X^*$, see the Proposition at the end of the subsection). Let $x \in B$ be arbitrary. Then, there exists $\theta \in [0, 2\pi)$ such that $\psi(x) = e^{i\theta} |\psi(x)|$, and so $|\psi(x)| = e^{-i\theta} \psi(x) = \psi(e^{-i\theta} x)$, which implies $\|\psi\|_{op} = \sup\{\operatorname{Re} \psi(x) : \|x\| \leq 1\}$. However, then

$$\|\psi\|_{op} \leq \beta < \operatorname{Re} f(\psi) \leq \|f\|_{op} \|\psi\|_{op} \quad \overset{f \in B_1(X^{**})}{\leq} \quad \|\psi\|_{op}$$

a contradiction. ■

Theorem

Let X be a normed vector space. The closed unit ball B is weakly compact if and only if X is reflexive.

Proof:

First suppose X is reflexive, and let (x_α) be a net in B . Then $(\iota(x_\alpha))$ is a net in the closed unit ball of X^{**} , so by Banach-Alaoglu, it admits a subnet $(\iota(x_\beta))$ converging weak-* to some f in the closed unit ball in X^{**} . Moreover, as the canonical embedding is surjective, there exists $x \in X$ such that $\iota(x) = f$, and as ι is isometric, $\|x\| = \|f\|_{op} \leq 1$, so $x \in B$. As

$$\psi(x_\beta) = \iota(x_\beta)(\psi) \rightarrow f(\psi) = \iota(x)(\psi) = \psi(x)$$

for all $\psi \in X^*$, it follows that $(x_\beta) \rightarrow x$ weakly, and so B is weakly compact.

Conversely, suppose now that B is weakly compact and let $f \neq 0 \in X^{**}$ be arbitrary. By Goldstine's theorem, there exists a net (x_α) in $B' := \{x \in X : \|x\| \leq \|f\|_{op}\}$ such that $(\iota(x_\alpha)) \rightarrow f$ weak-*. As B' is weakly compact since B is, (x_α) admits a subnet (x_β) such that $(x_\beta) \rightarrow x$ weakly for some $x \in B'$. Then, for all $\psi \in X^*$, $\iota(x_\beta)(\psi) = \psi(x_\beta) \rightarrow \psi(x) = \iota(x)(\psi)$, so $(\iota(x_\beta)) \rightarrow \iota(x)$ weak-*. As the weak-* topology is Hausdorff, by uniqueness of limits we get $\iota(x) = f$, and so ι is surjective, so X is reflexive. ■

Proposition

Let (X, τ) be a locally convex topological vector space, with P a collection of seminorms witnessing the topology, and let $\psi : X \rightarrow \mathbb{F}$ be a linear functional.

The following are equivalent:

- (1) ψ is continuous.
- (2) $\ker \psi$ is closed.
- (3) There exists $\lambda > 0$ and seminorms $p_1, \dots, p_n \in P$ such that

$$|\psi(x)| \leq \lambda(p_1(x) + \dots + p_n(x))$$

for all $x \in X$.

Proof:

(1) implies (2) is immediate. Suppose (2) holds and assume without loss of generality that $\ker \psi \neq X$. Let $\mathcal{F}$ be the collection of finite subsets of P and define a direction on $I = \mathcal{F} \times (0, \infty)$ by letting $(E, \beta) \leq (F, \alpha)$ if $E \subseteq F$ and $\beta \leq \alpha$. Suppose for all $\lambda > 0$ and every finite set $F = \{p_1, \dots, p_n\} \subseteq P$ that there exists $x_{F,\lambda} \in X$ such that

$$|\psi(x_{F,\lambda})| > \lambda(p_1(x_{F,\lambda}) + \dots + p_n(x_{F,\lambda}))$$

Then let $y_{F,\lambda} = |\psi(x_{F,\lambda})|^{-1} x_{F,\lambda}$ (the inverse exists since the above strict inequality tells us its positive), so $|\psi(y_{F,\lambda})| = 1$ and $p_1(y_{F,\lambda}) + \dots + p_n(y_{F,\lambda}) < \frac{1}{\lambda}$. Let $p \in P$ be an arbitrary seminorm and let $\varepsilon > 0$. Then,

$$(F, \alpha) \geq (\{p\}, \frac{1}{\varepsilon})$$

implies $p(y_{f,\lambda}) \leq \sum_{q \in F} q(y_{F,\alpha}) < \alpha^{-1} \leq \varepsilon$, and so it follows that $p(y_{F,\alpha}) \rightarrow 0$ for all $p \in F$, and therefore $(y_{F,\alpha}) \rightarrow 0$. Thus, fix $z \notin \ker(\psi)$ and let $(z_{F,\alpha})$ be the net defined by

$$z_{F,\alpha} = z - \frac{\psi(z)}{\psi(y_{F,\alpha})} y_{F,\alpha}$$

Then, $z_{F,\alpha} \in \ker \psi$ for all $F \in \mathcal{F}$ and $\alpha > 0$, and moreover, $p(z_{F,\alpha} - z) = |\psi(z)|p(y_{F,\alpha}) \rightarrow 0$, so $(z_{F,\alpha}) \rightarrow 0$, so since $\ker \psi$ is closed, we get $z \in \ker \psi$, a contradiction. Thus (3) holds. Finally, if (3) holds, then $(x_\alpha) \rightarrow x$ implies

$$|\psi(x_\alpha) - \psi(x)| \leq \lambda(p_1(x_\alpha - x) + \cdots + p_n(x_\alpha - x)) \rightarrow 0$$

for some $\lambda > 0$ and $p_1, \dots, p_n \in P$, so (1) holds. ■

Lemma

Let X be a vector space and $\psi_1, \dots, \psi_n : X \rightarrow \mathbb{F}$ linear functionals. If $\psi : X \rightarrow \mathbb{F}$ is a linear functional such that $\bigcap_{j=1}^n \ker(\psi_j) \subseteq \ker(\psi)$, then $\psi \in \text{Span}\{\psi_1, \dots, \psi_n\}$.

Proof:

Assignment 4 (Question 6). ■

Proposition

Let (X, τ) be a locally convex topological vector space. If W is a subspace of the algebraic dual of X^* with the property that $(x_\alpha) \rightarrow x$ if and only if $(\psi(x_\alpha)) \rightarrow \psi(x)$ for all $\psi \in W$, then W is the subspace of continuous linear functionals on X .

Proof:

Assignment 4 (Question 7). ■

More on Convexity

Definition

Let V be a vector space and $C \subseteq V$ convex.

- (a) A nonempty convex subset F of C is called a face if whenever $tx + (1-t)y \in F$ for $x, y \in C$ and $t \in (0, 1)$, then $(x, y) \in F$.
- (b) A point $x \in C$ is an extreme point of C if $\{x\}$ is a face. We denote by ∂C the set of all extreme points of C .

Remark

If $C \neq \emptyset$, then C is a face.

Examples

- (a) Let C be the unit disk in $\mathbb{R}^2$. The extreme points of C are S^1 , since if $(x, y) \in C$ with $s \neq 0$ and $\|s\| < 1$, then $s = (1 - \|s\|)0 + \|s\|\frac{s}{\|s\|}$. The only proper face of C is thus S^1 .
- (b) Let $C = \{(x, y) \in \mathbb{R}^2 : x \leq 0\}$. Then C has no extreme points, and the y -axis is a face of C .
- (c) Let $V = \mathbb{R}^2$ endowed with the 1-norm $\|\cdot\|_1$. Let C be the unit ball of V (which is a diamond centered at the origin). Then $\partial C = \{(0, 1), (1, 0), (-1, 0), (0, -1)\}$. On the other hand, for V endowed with the p -norm $\|\cdot\|_p$ for $1 < p < \infty$ and C the unit ball of V , we have $\partial C = \{(x, y) \in \mathbb{R}^2 : \|(x, y)\|_p = 1\}$. For V endowed with the ∞ -norm $\|\cdot\|_\infty$ and C the unit ball of V , we have $\partial C = \{(1, 1), (1, -1), (-1, 1), (-1, -1)\}$.
- (d) Let $C = B_1(L^1([0, 1]))$. Then $\partial C = \emptyset$. Indeed, suppose $f \in B_1(L^1([0, 1]))$ is an extreme point. Then $\|f\|_1 = 1$. We will write f as a convex combination $f = tg + (1-t)h$ for $h, g \in B_1(L^1([0, 1]))$ and $t \in (0, 1)$. To see this, notice that the function $x \in [0, 1] \mapsto \int_0^x |f(t)| dm(t)$ is continuous, so by the intermediate value theorem there exists $x \in (0, 1)$ such that $\int_0^x |f(t)| dm(t) = \frac{1}{2}$. Let $g = 2f\chi_{[0, x]}$ and $h = 2f\chi_{[x, 1]}$. Then $f = \frac{1}{2}g + \frac{1}{2}h$ with $f \neq g, h$. Thus we are left to see $g, h \in B_1(L^1([0, 1]))$. But this

holds since

$$\|g\|_1 = 2 \int_0^x |f(t)| dm(t) = 1 = 2 \int_x^1 |f(t)| dm(t) = \|h\|_1$$

Thus f is not an extreme point.

(e) For $1 < p < \infty$ and $C = \{f \in L^p([0, 1]) : \|f\|_p \leq 1\}$, we have $\partial C = \{f \in L^p([0, 1]) : \|f\|_p = 1\}$. For $p = \infty$, the extreme points of the unit ball are functions $f \in L^\infty([0, 1])$ with $|f(x)| = 1$ a.e.

Theorem 5.23 (Krein-Milman)

Let V be a vector space equipped with the weak topology induced by a vector space E of linear functionals on V that separates the points of V . Let $C \subseteq V$ be a compact convex subset of V . Then the convex hull of ∂C is dense in C . In particular, $\partial C \neq \emptyset$ if $C \neq \emptyset$.

Proof:

Suppose $C \neq \emptyset$, and let C_0 be a closed face of C (one exists since C is a closed face of itself). Then, C_0 is compact since C is compact and C_0 is closed. Also, if $F \subseteq C_0$ is a face, then F is a face of C , since if $tx + (1-t)y \in F$ for $x, y \in C$ and $t \in (0, 1)$, then we must have $x, y \in C_0$ because C_0 is a face of C that contains F , and from there we must have $x, y \in F$ because F is a face of C_0 . We will now attempt to find a minimal closed face of C with Zorn's lemma.

Consider the collection Λ of all closed faces of C_0 , ordered by reverse inclusion ($F_1 \leq F_2$ when $F_2 \subseteq F_1$). We claim Λ has a maximal element, which is a minimal closed face of C . Looking to do some Zorn's lemma shenanigans, let $\{F_i : i \in I\} \subseteq \Lambda$ be a totally ordered subset of Λ . Then $\{F_i : i \in I\}$ has the finite intersection property by compactness of C_0 , so $F' = \bigcap_{i \in I} F_i \neq \emptyset$. Then $F' \subseteq C_0$ is closed as the intersection of closed sets, and is a face because each F_i is. Thus $F' \in \Lambda$ is an upper bound of $\{F_i : i \in I\}$, so by Zorn's lemma, C_0 contains a minimal closed face.

Let F be a minimal closed face of C_0 . We claim F is a singleton. Indeed, let $f \in E$ be arbitrary. Since $\text{Re } f$ is continuous on F and F is compact, by the extreme value theorem there exists $x \in F$ such that $\text{Re } f(x) = \inf_{x \in F} \text{Re } f(x) =: s$. Define $F_f := \{x \in F : \text{Re } f(x) = s\}$. Then F_f is nonempty by the above, closed by continuity of $\text{Re } f$, convex by linearity of f and Re , and a face of F because s is the minimum of f . Thus, F_f is a closed face of C_0 , so by minimality of F , $F_f = F$. This tells us $\text{Re } f(x) = \text{Re } f(y)$ for all $x, y \in F$. Since $f \in E$ was arbitrary and E is separating, we get that F is a singleton.

Now, let $K = \overline{\text{conv}(\partial C)}$, which is a compact and convex subset of C . Suppose for the sake of contradiction that there is $y \in C \setminus K$. Let U be an open convex neighbourhood of y with $U \cap K = \emptyset$. Such a U exists because the basic open neighbourhoods of y , $\{x \in V : |f_i(y) - f_i(x)| < \varepsilon, f_i \in E, i = 1, \dots, n\}$ are convex. By Hahn-Banach separation, there exists a continuous linear functional $f : V \rightarrow \mathbb{R}$ (so $f \in E$ by A4Q7) and $t \in \mathbb{R}$ such that $\text{Re}(f(z)) < t \leq \text{Re}(f(x))$ for all $z \in U, x \in K$.

Let $s = \min_{x \in C} \text{Re}(f(x))$, which exists by compactness of C and continuity of f , and define the face $F_f := \{x \in C : \text{Re}(f(x)) = s\}$, which is closed by continuity, and nonempty by the Extreme Value Theorem. Since $\text{Re}(f(z)) < t \leq \text{Re}(f(x))$ for all $x \in K$, $F_f \cap K = \emptyset$. However, we showed above every face contains extreme points of C , but every extreme point of C is contained in K , which is disjoint from F_f . Thus, we have a contradiction, so $K = C$. ■

Remark

A particular case of interest is when V is a Banach space and we equip V^* with the weak-* topology. We know that $B_1(V^*)$ is convex, so Krein-Milman says $\partial B_1(V^*)$ is nonempty, which is a common trick to show that a space is isometrically isomorphic to a dual of a Banach space.

Example

Recall that $B_1(L^1[0, 1])$ has no extreme points, so $L^1[0, 1]$ is not the dual space of any Banach space. Similarly, c_0 is not the dual of any Banach space: we claim $\partial B_1(c_0) = \emptyset$:

Indeed, suppose $x = (x_n) \in c_0$, with $\|x\| = 1$. Let $N \in \mathbb{N}$ be such that $n > N$ implies $|x_n| < \frac{1}{2}$. Define $a = (a_n), b = (b_n)$ such that $a_n = b_n = x_n$ for $n \leq N$, $a_n = x_n + \frac{1}{2^n}$, $b_n = x_n - \frac{1}{2^n}$ for $n > N$. Then $x = \frac{1}{2}a + \frac{1}{2}b$ and $\|a\| = \|b\| = 1$, so x is not an extreme point.

Proposition 5.24

Let X be a compact Hausdorff space and $C(X)$ the space of continuous functions on X with the supremum norm. Let $M(X)$ be the dual of $C(X)$, and $P(X) = \{\mu \in M(X) : \|\mu\|_{op} \leq 1, \mu(1) = 1\}$. Then $P(X)$ is a weak-* compact convex set with extreme points given by the Dirac measures δ_x for $x \in X$, where $\delta_x(f) = f(x)$ for $f \in C(X)$.

Note: $M(X)$ corresponds to the space of complex Radon measures on X .

Proof:

The fact that $P(X)$ is weak-* compact follows because the evaluation map $\mu \mapsto \mu(1)$ is weak-* continuous and an application of Banach-Alaoglu. Moreover, $P(X)$ is convex since it is the intersection of the (convex) plane $\{\mu : \mu(1) = 1\}$ and the closed unit ball.

Claim 1: $P(X)$ consists of positive self-adjoint linear functionals.

Let $f \in C(X)$ such that $\bar{f} = f$. Write $\mu(f) = s + ti$ for $s, t \in \mathbb{R}$. For $m \geq 1$,

$$|\mu(f + tim)|^2 = |s + it(m+1)|^2 = s^2 + t^2(m+1)^2$$

Then, since $\|\mu\|_{op} \leq 1$, we have

$$|\mu(f + itm)|^2 \leq \|f + itm\|^2 = \|f\|_\infty^2 + t^2 m^2$$

and so combining the above, we have

$$s^2 + t^2 + 2mt^2 \leq \|f\|_\infty^2$$

Since m is arbitrary, we must have $t = 0$, so $\mu(f) \in \mathbb{R}$. To show μ is positive, suppose $f \geq 0$. Then $\mu(f) \in \mathbb{R}$ by the above. Thus, we have $0 \leq \|f\|_\infty - f$ and since $f \geq 0$, we have $\|f\|_\infty - f \leq \|f\|_\infty$ which implies $|\|f\|_\infty - f| \leq \|f\|_\infty$, and so

$$\|f\|_\infty - \mu(f) = \mu(\|f\|_\infty - f) \leq \| \|f\|_\infty - f \|_\infty \leq \|f\|_\infty$$

which shows $\mu(f) \geq 0$. Since μ is self-adjoint, $|\mu(f)| \leq \mu(|f|)$ for all $f \in C(X)$ since f is real-valued, which proves the claim.

Claim 2: If μ is an extreme point of $P(X)$, then $\mu(fg) = \mu(f)\mu(g)$ for $f, g \in C(X)$.

Suppose $0 \leq f \leq 1$, and thus by Claim 1, $0 \leq \mu(f) \leq 1$. First, consider the case $0 < \mu(f) < 1$. Write $\alpha = \mu(f)$ and define linear functionals φ, ψ on $C(X)$ by $\varphi(g) = \frac{1}{\alpha}\mu(fg)$, and $\psi(g) = \frac{1}{1-\alpha}\mu((1-f)g)$ for $g \in C(X)$. Then,

$$|\varphi(g)| = \frac{1}{\alpha}|\mu(fg)| \leq \frac{1}{\alpha}\mu(|fg|) \leq \frac{\mu(f)\|g\|_\infty}{\alpha} \leq \|g\|_\infty$$

Thus $\|\varphi\|_{op} \leq 1$ and $\varphi(1) = 1$. Thus $\varphi \in P(X)$. Similarly, since $1 - f \geq 0$, and $\mu(1 - f) = 1 - \alpha$, we get $\psi \in P(X)$. Because μ is an extreme point, we have $\mu = \varphi = \psi$, and so $\mu(g) = \frac{1}{\alpha}\mu(fg)$, so $\mu(fg) = \mu(f)\mu(g)$.

In the case $\mu(f) = 0$, then

$$|\mu(fg)| \leq \mu(|fg|) \leq \mu(f\|g\|_\infty) = \|g\|_\infty \mu(f) = 0$$

and so $\mu(fg) = 0 = 0\mu(g) = \mu(f)\mu(g)$. If $\mu(f) = 1$, then $\mu(1 - f) = 0$, and since $1 - f \geq 0$, we get as above $\mu((1 - f)g) = 0$, hence $\mu(fg) = \mu(g) = 1\mu(g) = \mu(f)\mu(g)$. Since non-negative functions in the unit ball of $C(X)$ span $C(X)$ and μ is linear, we deduce that μ is multiplicative.

Claim 3: If μ is an extreme point of $P(X)$, then there exists $x \in X$ so that $\ker \mu \subseteq \ker \delta_x$. To prove this, suppose not, so for each $x \in X$ there exists $\bar{f}_x \in C(X)$ such that $f_x \in C(X)$ with $f_x \in \ker \mu$ but $f_x(x) \neq 0$. Since μ is multiplicative, we have $\mu(\bar{f}_x \bar{f}_x) = 0$, so we may assume $f_x \geq 0$ (by replacing it with $|f_x|^2$ if necessary). For each $x \in X$, set $U_x = \{y \in X : f_x(y) \neq 0\}$, so that $\mathcal{G} = \{U_x : x \in X\}$ is an open cover of X . By compactness of X , it has a finite subcover $\{U_{x_1}, \dots, U_{x_n}\}$ for some $n \in \mathbb{N}$. Set $f = \sum_{i=1}^n f_{x_i}$. Then $f(y) \neq 0$ for all $y \in X$, $f \geq 0$, and $f \in C(X)$. By the extreme value theorem, there exists $\alpha > 0$ so that $\alpha = \inf_{y \in X} f(y)$. Then $f \geq \alpha$, so by positivity, $\mu(f) \geq \mu(\alpha) = \alpha > 0$ (since $\mu(1) = 1$), contradicting the fact $\mu(f) = 0$.

Thus, there is $x \in X$ so that $\ker \mu \subseteq \ker \delta_x$. Then, $f - \mu(f) \in \ker \mu$, so $\delta_x(f - \mu(f)) = 0$ implies $\mu(f) = f(x) = \delta_x(f)$. This gives $\partial P(X) \subseteq \{\delta_x : x \in X\}$.

For the reverse inclusion, we must show $\delta_x \in \partial P(X)$ for all $x \in X$. Suppose $\varphi, \psi \in P(X)$, $t \in (0, 1)$, and $\delta_x = t\varphi + (1 - t)\psi$. If $f \in C(X)$, we have

$$t|\varphi(f)| \leq t\varphi(|f|) \leq t\varphi(|f|) + (1 - t)\psi(|f|) = \delta_x(|f|)$$

which shows that $\ker \delta_x \subseteq \ker \varphi$. Then, $f - f(x) \in \ker \delta_x \subseteq \ker \varphi$, so $f - f(x) \in \ker \varphi$, implying $\varphi(f) = f(x) = \delta_x(f)$, so $\varphi = \delta_x$. Similarly, $\psi = \delta_x$, proving that $\delta_x \in \partial P(X)$. ■

Remark

(a) Applying the Krein-Milman theorem, it follows that probability measures on $C(X)$ can be approximated pointwise as convex combinations of the Dirac measures, that is, approximated by pointwise by measures of finite support.

(b) The above proof shows that if $\mu \in M(X)$ and $\mu(1) = \|\mu\|_{op}$, then μ is positive.

6: Compact Operators and the Spectral Theorem

Let H be a separable Hilbert space with orthonormal basis $\{e_n : n \in \mathbb{N}\}$. We may view a bounded operator on H as an infinite matrix $(a_{ij})_{i,j \in \mathbb{N}}$. If $T \in \mathcal{B}(H)$, under which conditions on T does H admit an orthonormal basis such that $[T]$ is diagonal? If such a basis exists, each vector in the basis is an eigenvector of T and each diagonal entry of $[T]$ is an eigenvalue of T :

Definition

Let H be a Hilbert space and $T \in \mathcal{B}(H)$.

(a) We say $\lambda \in \mathbb{F}$ is an eigenvalue of T if $\ker(\lambda I - T) \neq \{0\}$, i.e., there exists some $0 \neq v \in H$ such that $Tv = \lambda v$. We call a nonzero vector $v \in \ker(\lambda I - T)$ a λ -eigenvector, and $\ker(\lambda I - T)$ is called the λ -eigenspace. We let $\sigma_p(T)$ be the set of eigenvalues of T , which is called the point spectrum of T .

(b) The spectrum of T is $\sigma(T) = \{\lambda \in \mathbb{F} : \lambda I - T \text{ is not invertible}\}$, where we note that this coincides with $\sigma_p(T)$ if H is finite dimensional.

Examples

(a) Let $u : \ell^2(\mathbb{Z}) \rightarrow \ell^2(\mathbb{Z})$ be the right shift. Recall that u is unitary. We claim $\sigma_p(u) = \emptyset$. Indeed, suppose $\lambda \in \mathbb{F}$ and we have $0 \neq x \in \ell^2(\mathbb{Z})$ such that $u(x) = \lambda x$. Since u is injective, we have $\lambda \neq 0$. Then, $x_n = \lambda x_{n+1}$ for all $n \in \mathbb{Z}$, and so $x_n = \frac{1}{\lambda^n} x_0$ for all $n \in \mathbb{N}$. Since $\lim_{n \rightarrow \infty} |x_n| = 0$, and we must have $x_0 \neq 0$, we get $|\lambda| > 1$. But $x_0 = \frac{1}{\lambda} x_{-1}$, or more generally, $x_0 = \frac{1}{\lambda^n} x^{-n}$, which taking the limit $n \rightarrow \infty$ shows $x_0 = 0$, a contradiction.

(b) If we consider the right shift $S : \ell^2(\mathbb{N}) \rightarrow \ell^2(\mathbb{N})$ given by $S(x_1, x_2, \dots) = (0, x_1, x_2, \dots)$. Then $\sigma_p(S) = \emptyset$.

(c) Let $S^* : \ell^2(\mathbb{N}) \rightarrow \ell^2(\mathbb{N})$ be the left shift, $S^*(x_1, x_2, \dots) = (x_2, x_3, \dots)$. We claim $\sigma_p(S^*) = \{\lambda \in \mathbb{F} : |\lambda| < 1\}$. Indeed, suppose $|\lambda| < 1$. Define $x_\lambda = (1, \lambda, \lambda^2, \dots)$, where $x_\lambda \in \ell^2$ since $|\lambda| < 1$, and $x_\lambda \neq 0$. We have $S^*(x_\lambda) = (\lambda, \lambda^2, \dots) = \lambda x_\lambda$, giving the inclusion $B_1(0) \subseteq \sigma_p(S^*)$. For the reverse inclusion, suppose $\lambda \in \sigma_p(S^*)$ and let $x_\lambda = (x_1, x_2, \dots) \in \ell^2$ be an eigenvector. Then we have $x_n = \lambda^n x_1$ for $n \geq 1$, hence $x_1 \neq 0$, and so $\lim_{n \rightarrow \infty} |\lambda|^n = 0$ because $x_\lambda \in \ell^2(\mathbb{N})$, and so $|\lambda| < 1$, giving the other inclusion. In fact, $\sigma(S^*) = \{\lambda \in \mathbb{F} : |\lambda| \leq 1\}$.

In the first two examples, we have no eigenvalues (even though u is normal), and in the third example, we have uncountable many eigenvalues in a separable space, so in any of these cases, we cannot expect these operators to be “diagonalizable”, in the sense that they can’t possibly admit an orthonormal basis of eigenvectors. The key is to restrict our focus.

Compact Operators

Definition

Let H be a Hilbert space. We say $T \in \mathcal{B}(H)$ has finite rank if $\text{ran } T$ is finite dimensional. We write $F(H)$ for the set of finite rank operators on H . We say that $T \in \mathcal{B}(H)$ is compact if $\overline{T(B_1(H))}$ is norm compact. We write $K(H)$ for the set of compact operators.

Examples

- (a) If $\{e_n : n \in \mathbb{N}\}$ is an orthonormal basis for a separable Hilbert space H , and $A = (a_{i,j})_{i,j}$ is a countably infinite matrix, then $A \in F(H)$ if and only if there exists N such that $a_{i,j} = 0$ for $i, j \geq N$.
(b) Every finite rank operator is compact. Indeed, if $T \in F(H)$, then $\overline{T(B_1(H))} \subseteq \overline{B_{\|T\|}(H)} \cap \text{ran } T$, which is compact by Heine-Borel since $\text{ran } T$ is finite dimensional.

Proposition 6.4

For every $T \in \mathcal{B}(H)$, we have $\ker T^* = (T(H))^\perp$.

Proof:

Suppose $z \in \ker T^*$. Then for $h \in H$, we have $\langle z, T(h) \rangle = \langle T^*(z), h \rangle = 0$, so $z \in (T(H))^\perp$.

Conversely, suppose $z \in (T(H))^\perp$. Then for all $h \in H$, we get $\langle T^*(z), h \rangle = \langle z, T(h) \rangle = 0$, so $z \in \ker T^*$. ■

Corollary 6.5

Let H be a Hilbert space. Then $F(H)$ is a two-sided, self-adjoint ideal of $\mathcal{B}(H)$ (i.e. $F(H)$ is a subspace such that $T \in F(H)$ implies $T^* \in F(H)$ and $T \in F(H), S \in \mathcal{B}(H)$ implies $ST, TS \in F(H)$).

Proof:

Clearly $F(H)$ is a subspace. To see that $F(H)$ is self-adjoint, let $T \in F(H)$. Then $T(H)$ is closed because it has finite dimension, so by Proposition 6.4 we get the decomposition $H = T(H) \oplus \ker T^*$. Thus, $T^*(H) = T^*(T(H))$, which is the image of a finite dimensional subspace under a linear operator, and thus finite dimensional.

Now, suppose $S \in \mathcal{B}(H)$ and let $T \in F(H)$. Then $TS(H) \subseteq T(H)$ is finite dimensional, so $TS \in F(H)$. Since $T^*S^* \in F(H)$ by the above and $F(H)$ is self-adjoint, we get $ST \in F(H)$. ■

Lemma 6.6

Let H be a Hilbert space with orthonormal basis $\{e_j : j \in J\}$. Let Λ be the family of all finite subsets of J ordered by inclusion and for each $F \in \Lambda$, let $P_F : H \rightarrow \text{Span}\{e_j : j \in F\}$ be the orthogonal projection. Then for all $h \in H$, we have $\lim_F \|P_F(h) - h\| = 0$.

Proof:

Recall from Section 2 that for all $h \in H$, we can write $h = \sum \langle h, e_j \rangle e_j$. Hence given any $\varepsilon > 0$, let $F_0 \subseteq J$ be a finite set such that

$$\|h - \sum_{j \in F_0} \langle h, e_j \rangle e_j\| < \varepsilon$$

for all F finite with $F_0 \subseteq F \subseteq J$. Then for all such F , $\|h - P_F(h)\| = \|h - \sum_{j \in F} \langle h, e_j \rangle e_j\| < \varepsilon$. ■

Remark

If V is a finite dimensional normed space, then the weak topology on V coincides with the norm topology. Indeed, since all norms on V are equivalent, we may assume $V = \mathbb{F}^n$ with the Euclidean norm. Then $x_\lambda \rightarrow x$ weakly if and only if $\langle x_\lambda, e_j \rangle \rightarrow \langle x, e_j \rangle$ for all $j \in \{1, \dots, n\}$, so $x_\lambda = \sum_{j=1}^n \langle x_\lambda, e_j \rangle e_j \rightarrow x = \sum_{j=1}^n \langle x, e_j \rangle e_j$ in norm.

Theorem 6.7 (Characterization of Compact Operators)

Let H be a Hilbert space and let $T \in \mathcal{B}(H)$. The following are equivalent:

- (1) $T \in K(H)$.
- (2) T is weak-norm continuous on $B_1(H)$ (it is a continuous map from $B_1(H)$ with the relative weak topology to H with the norm topology).

(3) $T(B_1(H))$ is compact.

(4) T is compact.

Proof:

To prove (1) implies (2), let $(x_\lambda)_{\lambda \in \Lambda}$ be a net in $B_1(H)$ converging weakly to $x \in B_1(H)$. Given $\varepsilon > 0$, let $S \in F(H)$ be such that $\|T - S\|_{op} < \varepsilon/3$. Then for each $\lambda \in \Lambda$, we have

$$\begin{aligned} \|T(x_\lambda) - T(x)\| &\leq \|T(x_\lambda) - S(x_\lambda)\| + \|S(x_\lambda) - S(x)\| + \|S(x) - T(x)\| \\ &\leq \|T - S\|_{op}\|x_\lambda\| + \|S(x_\lambda) - S(x)\| + \|S - T\|_{op}\|x\| \underbrace{\qquad}_{x, x_\lambda \in B_1(H)} < 2\varepsilon/3 + \|S(x_\lambda) - S(x)\| \end{aligned}$$

Since $\langle S(x_\lambda), z \rangle \rightarrow \langle S(x), z \rangle$ for all $z \in H$ (because S is continuous and $x_\lambda \rightarrow x$ weakly), we have $S(x_\lambda) \rightarrow S(x)$ weakly in $S(H)$, but $S(H)$ is finite dimensional, so its norm and weak topologies coincide, and thus $S(x_\lambda) \rightarrow S(x)$ in norm, so there is $\lambda_0 \in \Lambda$ such that $\|S(x_\lambda) - S(x)\| < \varepsilon/3$ for all $\lambda \geq \lambda_0$. Thus, the above estimate implies for all $\lambda \geq \lambda_0$, we have

$$\|T(x_\lambda) - T(x)\| < 2\varepsilon/3 + \|S(x_\lambda) - S(x)\| < \varepsilon$$

proving (2).

To see (2) implies (3), since H is reflexive, we know that $B_1(H)$ is weakly compact, so since T is weak-norm continuous, we get that $\overline{T(B_1(H))}$ is norm compact.

To see (3) implies (4), by (3), $\overline{T(B_1(H))} = T(B_1(H))$ since $T(B_1(H))$ is norm compact.

To see (4) implies (1), suppose T is compact and let $\{e_j : j \in J\}$ be an orthonormal basis for H . Let $(P_F)_{F \in \Lambda}$ be as in Lemma 6.6. Then, $P_F(h) \rightarrow h$ in norm for all $h \in H$ by Lemma 6.6. We claim that $P_F T \rightarrow T$ in operator norm. Indeed, suppose not, so there exists $\varepsilon > 0$ such that for every finite subset $F \subseteq J$, we can find a finite set $K_F \supseteq F$ and $x_F \in H$ with $\|x_F\| = 1$ so that $\|P_{K_F} T(x_F) - T(x_F)\| \geq \varepsilon$. Since $\overline{T(B_1(H))}$ is compact, we can find an accumulation point of $(T(x_F))_{F \in \Lambda}$, say $y \in H$. Then,

$$\|P_{K_F} T(x_F) - T(x_F)\| \leq \|P_{K_F} T(x_F) - P_{K_F}(y)\| + \|P_{K_F}(y) - y\| + \|y - T(x_F)\|$$

For the middle term, since $K_F \supseteq F$ and the fact $P_F(h) \rightarrow h$ in norm for all $h \in H$, we have $P_{K_F}(y) \rightarrow y$ in norm, so we can make the middle term arbitrarily small. Moreover, since $\|P_{K_F}\|_{op} \leq 1$ (it's a projection), we have $\|P_{K_F} T(x_F) - P_{K_F}(y)\| \leq \|P_{K_F}\|_{op} \|T(x_F) - y\| \leq \|T(x_F) - y\|$. Since y is an accumulation point, there is some $F_0 \in \Lambda$ such that $\|P_{K_{F_0}}(y) - y\| < \varepsilon/3$ and $\|T(x_{F_0}) - y\| < \varepsilon/3$, implying

$$\varepsilon \leq \|P_{K_{F_0}} T(x_{F_0}) - T(x_{F_0})\| \leq \|P_{K_{F_0}} T(x_{F_0}) - P_{K_{F_0}}(y)\| + \|P_{K_{F_0}}(y) - y\| + \|y - T(x_{F_0})\| < 3\varepsilon/3 = \varepsilon$$

a contradiction. Thus $P_F T \rightarrow T$ in operator norm, and thus $T \in \overline{F(H)}$. ■

Exercise

If $T \in K(H)$, then $T(H)$ is closed if and only if $T \in F(H)$.

Theorem 6.8

$K(H)$ is a closed, self-adjoint, two-sided ideal of $B(H)$.

Proof:

This follows from Corollary 6.5 since $F(H)$ is a self-adjoint, two-sided ideal of $\mathcal{B}(H)$, and $K(H) = \overline{F(H)}$ by Theorem 6.7. ■

Spectral Theorem

Lemma 6.9

Let H be a Hilbert space, and let $T \in \mathcal{B}(H)$. Suppose that T is diagonal with respect to an orthonormal basis $\{e_j : j \in J\}$, with corresponding eigenvalues λ_j for each $j \in J$. Then T is compact if and only if $(\lambda_j)_{j \in J} \in c_0(J)$ (functions that vanish at infinity with the discrete topology on J).

Proof:

Suppose T is compact. For $\varepsilon > 0$ we let $J_\varepsilon = \{j \in J : |\lambda_j| \geq \varepsilon\}$. We must show J_ε is finite. Suppose for the sake of contradiction that it is infinite. If $h \in H$, then

$$h = \sum_{j \in J} \langle h, e_j \rangle e_j$$

so

$$T(h) = \sum_{j \in J} \langle h, e_j \rangle \lambda_j e_j = \sum_{j \in J} \langle \lambda h, e_j \rangle e_j = \sum_{j \in J} \langle T(h), e_j \rangle e_j$$

Consider the collection $\Lambda = \{I \subseteq J_\varepsilon : J_\varepsilon \setminus I \text{ is finite}\}$ ordered by reverse inclusion. For each $I \in \Lambda$, choose $j \in I$ and let $f_I := e_j$. Then $(f_I)_{I \in \Lambda}$ is a net in $B_1(H)$. By Parseval's identity, $\|h\|^2 = \sum_{j \in J} |\langle h, e_j \rangle|^2$, implying $|\langle h, e_j \rangle| \rightarrow 0$ which implies $(f_I)_{I \in \Lambda}$ weakly converges to 0. Since T is weak-norm continuous, $T(f_I) \rightarrow 0$. But by construction, $T(f_I) = \lambda_j e_j$ for some $j \in I \subseteq J_\varepsilon$. Thus, $\|T(f_I)\| \geq \varepsilon$, a contradiction. Conversely, suppose $(\lambda_j)_{j \in J} \in c_0(J)$. Given $\varepsilon > 0$, the set $J_\varepsilon = \{j \in J : |\lambda_j| \geq \varepsilon\}$ is finite. Define $T_\varepsilon : H \rightarrow H$ by

$$T_\varepsilon(h) = T_\varepsilon \left(\sum_{j \in J} \langle h, e_j \rangle e_j \right) = \sum_{j \in J_\varepsilon} \langle h, e_j \rangle \lambda_j e_j$$

for all $h \in H$. We note that $T_\varepsilon \in F(H)$. Then,

$$\|T(h) - T_\varepsilon(h)\|^2 = \sum_{j \in J \setminus J_\varepsilon} |\lambda_j|^2 |\langle h, e_j \rangle|^2 \leq \varepsilon^2 \|h\|^2$$

so $\|T - T_\varepsilon\|_{op} \leq \varepsilon$, implying $T \in \overline{F(H)} = K(H)$. ■

Lemma 6.10

Let H be a Hilbert space, and $T \in \mathcal{B}(H)$. Then T is normal if and only if $\|T^*(x)\| = \|T(x)\|$ for all $x \in H$.

Proof:

Suppose T is normal. Let $x \in H$. Then,

$$\|T(x)\|^2 = \langle T(x), T(x) \rangle = \langle T^*T(x), x \rangle = \langle TT^*(x), x \rangle = \langle T^*(x), T^*(x) \rangle = \|T^*(x)\|^2$$

Conversely, if $\|T(x)\| = \|T^*(x)\|$ for all $x \in H$, by polarization it follows that $\langle T(x), T(y) \rangle = \langle T^*(x), T^*(y) \rangle$, so $\langle T^*T(x) - TT^*(x), y \rangle = 0$ for all $y \in H$, so $T^*T = TT^*$, so T is normal. ■

Lemma 6.11

Let H be a Hilbert space and let $T \in \mathcal{B}(H)$ be normal. Then,

- (1) If $\lambda \in \sigma_p(T)$ and h is a λ -eigenvector, then $\bar{\lambda}$ is an eigenvalue of T^* and h is an $\bar{\lambda}$ -eigenvector.
- (2) If λ_1, λ_2 are distinct eigenvalues of T and h_1, h_2 are distinct eigenvectors with $h_1 \in \ker(\lambda_1 - T), h_2 \in \ker(\lambda_2 - T)$, then h_1 and h_2 are orthogonal.

Proof:

- (1) Let $\lambda \in \sigma_p(T)$ and $h \neq 0 \in \ker(\lambda - T)$. Notice that $(\lambda - T)^* = \bar{\lambda} - T^*$, so $\lambda - T$ is also normal. Then $\|(\lambda - T)(h)\| = 0 = \|(\bar{\lambda} - T^*)(h)\|$ by Lemma 6.10, which shows $T^*(h) = \bar{\lambda}h$.
- (2) Suppose $\lambda_1 \neq \lambda_2$. Without loss of generality, suppose $\lambda_1 \neq 0$. Then

$$\langle h_1, h_2 \rangle = \frac{1}{\lambda_1} \langle \lambda_1 h_1, h_2 \rangle = \frac{1}{\lambda_1} \langle T(h_1), h_2 \rangle = \frac{1}{\lambda_1} \langle h_1, T^*(h_2) \rangle = \frac{\lambda_2}{\lambda_1} \langle h_1, h_2 \rangle$$

Since $\lambda_1 \neq \lambda_2$, this forces $\langle h_1, h_2 \rangle = 0$. ■

Corollary 6.12

Suppose $T \in \mathcal{B}(H)$ is compact and $\{e_j : j \in J\}$ is an orthonormal basis of H such that T is diagonal with respect to it. For each $j \in J$, let λ_j be the corresponding eigenvalue to e_j . Then,

- (1) $\sigma_p(T) = \{\lambda_j : j \in J\}$
- (2) If $\lambda_j \neq 0$, then $\ker(\lambda_j - T)$ is finite dimensional.
- (3) $\sigma_p(T)$ is countable and $\lim_{n \rightarrow \infty} \lambda_n = 0$.

Proof:

(1) Observe that T is normal because T is diagonal. If $\lambda \in \sigma_p(T)$ and $\lambda \neq \lambda_j$ for all $j \in J$, then by Lemma 6.11, if $h \neq 0$ is a λ -eigenvector, then $h \in \{e_j : j \in J\}^\perp$, which is impossible because $\{e_j : j \in J\}$ is an orthonormal basis and $h \neq 0$. Thus, $\sigma_p(T) = \{\lambda_j : j \in J\}$.

(2) Since T is compact, by Lemma 6.9, $(\lambda_j)_{j \in J} \in c_0(J)$. Hence for each $\varepsilon > 0$, the set $J_\varepsilon = \{j \in J : |\lambda_j| \geq \varepsilon\}$ is finite, so it follows that $\dim(\ker \lambda - T) < \infty$ for $\lambda \neq 0$. Then

$$\sigma_p(T) \setminus \{0\} = \{j \in J : \lambda_j \neq 0\} = \bigcup_{n \in \mathbb{N}} \{\lambda \in J : |\lambda_j| \geq \frac{1}{n}\}$$

is a countable union of finite sets and thus countable, so $\sigma_p(T)$ is countable, and $\lim_{n \rightarrow \infty} \lambda_n$ follows from Lemma 6.9. ■

We need to show that if $T \in \mathcal{B}(H)$ is compact and normal, and H is complex, then T has an eigenvalue.

Proposition 6.13

Let H be a complex Hilbert space. Define the numerical radius of $T \in \mathcal{B}(H)$ by $\mathcal{R}(T) = \sup_{\|x\|=1} |\langle T(x), x \rangle|$. Then $\frac{1}{2}\|T\|_{op} \leq \mathcal{R}(T) \leq \|T\|_{op}$.

Proof:

The inequality $\mathcal{R}(T) \leq \|T\|_{op}$ follows by Cauchy-Schwarz. For the other, let $x, y \in H$ with $\|x\| = \|y\| = 1$. Consider the bounded sesquilinear form $B_T : H \times H \rightarrow \mathbb{C}$ given by $B_T(x, y) = \langle T(x), y \rangle$. By polarization,

$$4\langle T(x), y \rangle = \sum_{k=0}^3 i^k \langle T(x + i^k y), x + i^k y \rangle$$

and so

$$\begin{aligned} 4|\langle T(x), y \rangle| &\leq \sum_{k=0}^3 |\langle T(x + i^k y), x + i^k y \rangle| \\ &\leq \sum_{k=0}^3 \|B_T\| \|x + i^k y\|^2 = \|B_T\| (\|x + y\|^2 + \|x + iy\|^2 + \|x - iy\|^2 + \|x - y\|^2) \end{aligned}$$

By the parallelogram law, we get

$$\leq \|B_T\| (2\|x\|^2 + \|y\|^2 + 2\|x\|^2 + \|iy\|^2) = 8\|B_T\|$$

implying

$$|\langle T(x), y \rangle| \leq 2\|B_T\| = 2\mathcal{R}(T)$$

Taking $y = \frac{T(x)}{\|T(x)\|}$ for $\|T(x)\| \neq 0$, we get $\|T(x)\| \leq 2\mathcal{R}(T)$, so taking operator norm, we have $\|T\|_{op} \leq 2\mathcal{R}(T)$. ■

Remark

- (a) Every $T \in \mathcal{B}(H)$ can be written uniquely as $T = A + iB$ where $A, B \in \mathcal{B}(H)$ are self-adjoint, where $A = \frac{T+T^*}{2}, B = \frac{T-T^*}{2i}$.
- (b) If $T \in \mathcal{B}(H)$ is self-adjoint, we get $\|T^2\|_{op} = \|T^*T\|_{op} = \|T\|_{op}^2$, and similarly by induction for all $p \geq 1$, $\|T^{2^p}\|_{op} = \|T\|_{op}^{2^p}$.
- (c) If $\mathbb{F} = \mathbb{R}$, then $\mathcal{R}(T)$ may be zero, even if T is normal, for example, $T = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \in \mathcal{B}(\mathbb{R}^2)$ has $\mathcal{R}(T) = 0$.

Lemma 6.14

Let H be a Hilbert space, where $\mathbb{F} = \mathbb{R}$ or $\mathbb{F} = \mathbb{C}$, and let $T \in \mathcal{B}(H)$ be self-adjoint. Then $\mathcal{R}(T) = \|T\|_{op}$.
Proof:

Assume $T \neq 0$. Because T is self-adjoint, we have $\langle T(x), x \rangle \in \mathbb{R}$ for all $x \in H$, since $\overline{\langle T(x), x \rangle} = \langle x, T(x) \rangle = \langle T(x), x \rangle$. Now let $x, y \in H$ be of norm 1. We have

$$\langle T(x+y), (x+y) \rangle = \langle T(x), x \rangle + 2\operatorname{Re}\langle T(x), y \rangle + \langle T(y), y \rangle$$

and so taking this same expression but for $\langle T(x-y), (x-y) \rangle$ and subtracting, we get

$$\langle T(x+y), (x+y) \rangle - \langle T(x-y), (x-y) \rangle = 4\operatorname{Re}\langle T(x), y \rangle$$

and so

$$4\operatorname{Re}\langle T(x), y \rangle \leq \mathcal{R}(T)(\|x+y\|^2 + \|x-y\|^2) = 2\mathcal{R}(T)(\|x\|^2 + \|y\|^2) = 4\mathcal{R}(T)$$

and so $\operatorname{Re}\langle T(x), y \rangle \leq \mathcal{R}(T)$ whenever $\|x\| = \|y\| = 1$. Thus, for x of norm 1 such that $T(x) \neq 0$, for $y = \frac{T(x)}{\|T(x)\|}$, we get

$$\|T(x)\| = \operatorname{Re}\langle T(x), \frac{T(x)}{\|T(x)\|} \rangle \leq \mathcal{R}(T)$$

which implies $\|T\|_{op} \leq \mathcal{R}(T)$. However, $\mathcal{R}(T) \leq \|T\|_{op}$ always, so they are equal. ■

Proposition 6.15

Let H be a Hilbert space over $\mathbb{C}$. Then for all $T \in \mathcal{B}(H)$ we have $\mathcal{R}(T^2) \leq \mathcal{R}(T)^2$. If T is normal, then $\mathcal{R}(T) = \|T\|_{op}$.

Proof:

Write $T = A + iB$ for $A, B \in \mathcal{B}(H)$ self-adjoint. We claim $\mathcal{R}(T)^2 \geq \|A\|_{op}^2$.

Indeed, suppose $x \in H$, $\|x\| = 1$. Then

$$\begin{aligned} \mathcal{R}(T)^2 &\geq |\langle T(x), x \rangle|^2 = \left| \underbrace{\langle A(x), x \rangle}_{\in \mathbb{R}} + i \underbrace{\langle B(x), x \rangle}_{\in \mathbb{R}} \right|^2 \\ &= |\langle A(x), x \rangle|^2 + |\langle B(x), x \rangle|^2 \geq |\langle A(x), x \rangle|^2 \end{aligned}$$

Taking supremums and applying Lemma 6.14 gives

$$\mathcal{R}(T)^2 \geq \mathcal{R}(A)^2 = \|A\|_{op}^2$$

Now, write $\operatorname{Re}(T) = A$. We claim $\langle \operatorname{Re}(T^2)(x), x \rangle \leq \mathcal{R}(T)^2$ for all $x \in H$ of norm 1. Indeed, notice that

$$\operatorname{Re}(T^2) = \operatorname{Re}(A^2 - B^2 + i(AB + BA)) = A^2 - B^2$$

and so

$$\langle \operatorname{Re}(T^2)(x), x \rangle = \langle A^2(x), x \rangle - \langle B^2(x), x \rangle \leq \langle A^2(x), x \rangle$$

where the last inequality follows since B is self-adjoint, so $\langle B^2(x), x \rangle = \|B(x)\|^2 \geq 0$. Thus

$$\langle \operatorname{Re}(T^2)(x), x \rangle \leq \langle A^2(x), x \rangle \leq \|A^2\|_{op} = \|A\|_{op}^2 \leq \mathcal{R}(T)^2$$

by the first claim. Now, suppose $\|x\| = 1$. Let $z \in \mathbb{C}$, $|z| = 1$ such that $z\langle T^2(x), x \rangle = |\langle T^2(x), x \rangle|$, and let $\alpha = z^{1/2}$. Then

$$\langle (\alpha T)^2(x), x \rangle = |\langle T^2(x), x \rangle| \in \mathbb{R}$$

implying

$$\langle (\alpha T)^2(x), x \rangle = \langle \operatorname{Re}((\alpha T)^2)(x), x \rangle$$

and thus by the above claim,

$$|\langle T^2(x), x \rangle| = \langle \operatorname{Re}((\alpha T)^2)(x), x \rangle \leq \mathcal{R}(\alpha T)^2 = \sup_{\|y\|=1} |\langle \alpha T(y), y \rangle|^2 \stackrel{|\alpha|=1}{=} \sup_{\|y\|=1} |\langle \alpha T(y), \alpha y \rangle \alpha|^2 = \mathcal{R}(T)^2$$

Therefore, $|\langle T^2(x), x \rangle| \leq \mathcal{R}(T)^2$ for all $x \in H$ of norm 1. Thus, $\mathcal{R}(T^2) \leq \mathcal{R}(T)^2$. Now, assume T is normal. Then, the above shows by induction that $\mathcal{R}(T^n) \leq \mathcal{R}(T)^n$ for all $n = 2^p$ for $p \in \mathbb{N}$. By normality, for $n = 2^p$, we then get

$$\|T^n\|_{op}^2 = \|(T^n)^* T^n\|_{op} = \|(T^* T)^n\|_{op} = \|T^* T\|_{op}^n = \|T\|_{op}^{2n}$$

where the last two inequalities use the fact that $T^* T$ is self-adjoint. Then, by Proposition 6.13,

$$\frac{\|T\|_{op}^{2n}}{4} = \frac{\|T^n\|}{2} \frac{\|T^n\|}{2} \leq \mathcal{R}(T^n)^2 \leq \mathcal{R}(T)^{2n} \leq \|T\|_{op}^{2n}$$

which implies

$$\frac{\|T\|_{op}}{4^{1/2n}} \leq \mathcal{R}(T) \leq \|T\|_{op}$$

taking the limit $n \rightarrow \infty$, we obtain $\|T\|_{op} \leq \mathcal{R}(T) \leq \|T\|_{op}$, so $\mathcal{R}(T) = \|T\|_{op}$. ■

Lemma 6.16

Let H be a Hilbert space over $\mathbb{C}$ and $T \in \mathcal{B}(H)$ be a compact normal operator. Then T has an eigenvalue λ with $|\lambda| = \|T\|_{op}$.

Proof:

Since T is compact, the function $T : B_1(H) \rightarrow H$ is weak-norm continuous. We claim $x \in B_1(H) \mapsto |\langle T(x), x \rangle|$ is also weakly continuous. Suppose $(x_\lambda)_{\lambda \in \Lambda}$ is a net in $B_1(H)$ that converges to $x \in B_1(H)$. Then $T(x_\lambda) \rightarrow T(x)$ in norm. We have

$$|\langle T(x_\lambda), x_\lambda \rangle - \langle T(x), x \rangle| = |\langle T(x_\lambda) - T(x), x_\lambda \rangle + \langle T(x), x_\lambda - x \rangle| \leq \|T(x_\lambda) - T(x)\| + |\langle T(x), x_\lambda - x \rangle| \rightarrow 0$$

because $x_\lambda \rightarrow x$ weakly and $T(x_\lambda) \rightarrow T(x)$ in norm. Since H is reflexive, $B_1(H)$ is weakly compact. Hence $x \in B_1(H) \mapsto |\langle T(x), x \rangle|$ attains its maximum by the extreme value theorem, so by Proposition 6.15, this maximum is $\|T\|_{op}$ since T is normal. Let $x \in B_1(H)$ such that $|\langle T(x), x \rangle| = \|T\|_{op}$. Then by Cauchy-Schwarz,

$$\|T\|_{op} = |\langle T(x), x \rangle| \leq \|T(x)\| \|x\| \leq \|T\|_{op}$$

In particular, we have $|\langle T(x), x \rangle| = \|T(x)\| \|x\|$, so since we have achieved the equality in Cauchy-Schwarz, we get that $T(x) = \lambda x$ for some $\lambda \in \mathbb{C}$, and the above inequality shows $|\lambda| = \|T\|_{op}$. ■

Remark

If $\mathbb{F} = \mathbb{R}$, and T is compact and self-adjoint, the proof of Lemma 6.16 with Lemma 6.14 also shows T has an eigenvalue λ with $|\lambda| = \|T\|_{op}$.

Theorem 6.17 (Spectral Theorem for Compact Normal Operators)

Let H be a complex Hilbert space and let $T \in \mathcal{B}(H)$ be a compact normal operator. Then T is diagonalizable. Moreover, if $(\lambda_j)_{j \in J}$ are the corresponding eigenvalues, then $(\lambda_j)_{j \in J}$ vanishes at infinity.

Proof:

This is immediate for $T = 0$, so suppose $T \neq 0$. Consider the collection S of orthonormal sets consisting of eigenvectors of T , ordered by inclusion. Then S is nonempty by the previous lemma. By taking unions over every element of the chain, every chain in S has an upper bound. Thus, by Zorn's lemma, there is some maximal element $B = \{e_j : j \in J\}$. Let $\{\lambda_j : j \in J\}$ be the corresponding eigenvalues.

We will show that $B^\perp = \{0\}$ to show it is an orthonormal basis. Indeed, let $H_1 = \overline{\operatorname{Span} B}$, and let

$\pi : H \rightarrow H_1$ be the orthogonal projection onto H_1 . Then, we claim π commutes with T . Indeed, since π is self-adjoint,

$$\begin{aligned} T\pi(x) &= T \left(\sum_{j \in J} \langle \pi(x), e_j \rangle e_j \right) = T \left(\sum_{j \in J} \langle x, \pi(e_j) \rangle e_j \right) = T \left(\sum_{j \in J} \langle x, e_j \rangle e_j \right) \\ &= \sum_{j \in J} \langle x, e_j \rangle \lambda_j e_j = \sum_{j \in J} \langle x, \bar{\lambda} e_j \rangle e_j \stackrel{\text{Lemma 6.11}}{=} \sum_{j \in J} \langle x, T^*(e_j) \rangle e_j = \sum_{j \in J} \langle T(x), e_j \rangle e_j = \sum_{j \in J} \langle \pi T(x), e_j \rangle e_j = \pi T(x) \end{aligned}$$

Thus, $T\pi = \pi T$, and taking adjoints, $T^*\pi = \pi T^*$. Thus, $T(I - \pi) = (I - \pi)T$ and $T^*(I - \pi) = (I - \pi)T^*$. Now, looking for a contradiction, suppose $\pi \neq I$, so that $H_1^\perp = \text{ran}(I - \pi) \neq \{0\}$. Then, $T(I - \pi)$ is a nonzero compact (by Theorem 6.8) normal operator, so by Lemma 6.16, it has an eigenvalue λ with $|\lambda| = \|T(I - \pi)\|_{op} \neq 0$. Let $h \in H$ be an eigenvector, so $T(I - \pi)(h) = \lambda h$. Since $\lambda \neq 0$ and $T(I - \pi) = (I - \pi)T$, this shows $h \in \text{ran}(I - \pi) = H_1^\perp$. Thus, $T(I - \pi)(h) = T(h) = \lambda h$, so h is an eigenvector for T with $h \in B^\perp$, contradicting maximality of B . Thus, $\pi = I$, so B is an orthonormal basis of H . The fact the eigenvalues vanish at infinity is Lemma 6.9. ■

Remark

If we write $\sigma_p(T) = \{\lambda_m : m \in \mathbb{N}\}$, let $P : H \rightarrow \ker(\lambda_m I - T)$ be the orthogonal projection. Then $T = \sum_{m=1}^\infty \lambda_m P_m$. If $T \in \mathcal{B}(H)$ is compact, then T^*T is self-adjoint and compact, hence diagonalizable. When $\mathbb{F} = \mathbb{R}$, we need the stronger assumption that T is self-adjoint, not just normal, to get the spectral theorem.

Examples (Integral Operators)

Proposition 6.18

Let $H = L^2([0, 1])$. For $k \in L^2([0, 1] \times [0, 1])$, define $K : L^2([0, 1]) \rightarrow L^2([0, 1])$ by $K(f)(y) = \int_0^1 f(x)k(x, y)dx$ for $y \in [0, 1]$. Then K is a compact operator with $\|K\|_{op} \leq \|k\|_2$, called the Hilbert-Schmidt operator with kernel k .

Proof:

That K is well-defined a.e. with $\|K\|_{op} \leq \|k\|_2$ follows from Fubini's theorem and Hölder's inequality. To see that K is compact, suppose that $k(x, y) = g(x)h(y)$ with $g, h \in C([0, 1])$. Then

$$K(f)(y) = \int_0^1 f(x)k(x, y)dx = h(y) \int_0^1 f(x)g(x)dx$$

implying $K_f = c_1 h$, with $c_1 \int_0^1 f(x)g(x)dx$. Since $\text{ran}(K) = \mathbb{C}h$ is finite-dimensional, K has finite-rank, hence is compact. Since finite linear combinations $\sum_{i=1}^n g_i h_i$ are dense in $C([0, 1]^2)$ by Stone-Weierstrass and $C([0, 1]^2)$ is dense in $L^2([0, 1]^2)$, it follows that k is compact for every $k \in L^2([0, 1]^2)$. ■

Example

The Volterra operator $V : L^2([0, 1]) \rightarrow L^2([0, 1])$ given by

$$V(f)(y) = \int_0^y f(x)dx$$

is a Hilbert-Schmidt operator with kernel $k = \chi_S$, $S = \{(x, y) \in [0, 1]^2 : x \leq y\}$. Its adjoint is

$$V^*(f)(y) = \int_y^1 f(x)dx$$

Then V is compact, so V^*V is compact and self-adjoint. By the spectral theorem, V^*V is diagonalizable. Let us find an orthonormal basis of eigenvectors. We have that

$$(V^*V)(f)(y) = \int_y^1 V(f(x))dx = \int_y^1 \int_0^x f(t)dt$$

We note that V^*V is injective, since if $V^*V(f) = 0$, then $\int_a^b \int_0^x f(t) dt dx = 0$ for all $a < b$, implying $\int_0^x f(t) dt = 0$ for all $x \in [0, 1]$, implying $f = 0$ a.e.

Hence if $V^*V(f) = \lambda f$ for $\lambda \in \mathbb{R}$, then $\lambda > 0$ by injectivity and the fact $\langle V^*V(f), f \rangle = \langle V(f), V(f) \rangle \geq 0$. Thus, we want to solve

$$\int_y^1 \int_0^x f(t) dt dx = \lambda f(y)$$

for $\lambda > 0$. We see that f must be continuous and differentiable, with $\lambda f'(y) = -\int_0^y f(t) dt$. From this, we see we may differentiate again to obtain the ODE

$$\lambda f''(y) = -f(y)$$

with initial values $f(1) = 0, f'(0) = 0$. Solutions are of the form $f(y) = c_1 \cos(y/\sqrt{\lambda}) + c_2 \sin(c_2/\sqrt{\lambda})$ for $c_1, c_2 \in \mathbb{F}$. Applying the condition $f'(0) = 0$, we get $c_2 = 0$. Applying the condition $f(0) = 1$, we have that the eigenvalues of V^*V are λk for $k \in \mathbb{N} \cup \{0\}$, where

$$\frac{1}{\sqrt{\lambda_k}} = \frac{(2k+1)\pi}{2} \implies \lambda_k = \left(\frac{2}{(2k+1)\pi} \right)^2$$

where the associated eigenvector is $f_k(y) = \cos\left(\frac{(2k+1)\pi y}{2}\right)$. Then $(f_k)_{k=0}^\infty$ are all mutually orthogonal since they're associated to distinct eigenvalues, and we can compute $\|f_k\|_2 = \frac{1}{\sqrt{2}}$, so we can get an orthonormal basis $(j_k)_{k=1}^\infty$, $j_k = \sqrt{2}f_k$, for $L^2([0, 1])$, which diagonalizes V^*V .

Since V^*V is compact and self-adjoint (hence normal), we can use this to calculate $\|V^*V\|_{op}$, which is the maximum of its eigenvalues. The maximum is achieved for $k = 0$, $\lambda_0 = (2/\pi)^2$. By Theorem 2.21, $\|V\|_{op} = \sqrt{\|V^*V\|_{op}} = \frac{2}{\pi}$.

Now, using the exponential form of cosine, we can express

$$j_k = \frac{1}{\sqrt{2}} \left(e^{\frac{(2k+1)\pi y i}{2}} + e^{-\frac{(2k+1)\pi y i}{2}} \right)$$

for each $k \in \mathbb{N} \cup \{0\}$, so that we can define an operator $U : L^2([0, 1]) \rightarrow L^2([0, 1])$ by

$$U(f)(y) = \frac{1}{\sqrt{2}} \left(f(y)e^{i\pi y/2} + f(1-y)e^{-i\pi y/2} \right)$$

Then we can compute the adjoint of U :

$$\begin{aligned} \langle U(f), g \rangle &= \frac{1}{\sqrt{2}} \int_0^1 f(y)e^{i\pi y/2} \overline{g(y)} dy + \frac{1}{\sqrt{2}} \int_0^1 f(1-y)e^{-i\pi y/2} \overline{g(y)} dy \\ &= \frac{1}{\sqrt{2}} \int_0^1 f(y) \overline{e^{-i\pi y/2} g(y)} dy + \frac{1}{\sqrt{2}} \int_0^1 f(y) \overline{e^{i\pi(1-y)/2} g(y)} dy = \langle f, U^*(g) \rangle \end{aligned}$$

so

$$U^*(g)(y) = \frac{1}{\sqrt{2}} \left(g(y)e^{-i\pi y/2} + e^{i\pi(1-y)/2} g(1-y) \right)$$

We find that U is unitary, so $U^*U = UU^* = \text{id}$. We can consider the basis $(e_n)_{n \in \mathbb{Z}}$ where $e_n(y) = e^{2\pi n i y}$. For $n \geq 0$, we have

$$\begin{aligned} U(e_n)(y) &= \frac{1}{\sqrt{2}} \left(e^{2(2n+1)\pi i y/2} + e^{2n i \pi(1-y)} e^{-i\pi y/2} \right) \\ &= \frac{1}{\sqrt{2}} \left(e^{2(2n+1)\pi i y/2} + e^{-2(2n+1)\pi i y/2} \right) \\ &= j_{2n} \end{aligned}$$

For $n \leq -1$, we have $U(e_n) = j_{2n+1}$.

Exercise

Suppose $T \in \mathcal{B}(H)$ is compact and normal. Let $0 \neq \lambda \in \mathbb{C}$ with $\lambda \notin \sigma_p(T)$. Show that $\lambda \text{id} - T$ is invertible.

This in particular implies that for $k \in L^2([0, 1]^2)$ and $K \in \mathcal{B}(L^2([0, 1]^2))$ the compact operator with kernel k , and suppose $\overline{K(x, y)} = K(y, x)$. Then if $\lambda \neq 0$ with $\lambda \notin \sigma_p(T)$, then for each g , the equation

$$\int_0^1 f(x)K(x, y)dx - \lambda f(y) = g(y)$$

has a unique solution in $g \in L^2([0, 1])$.

Functional Calculus

Suppose H is a separable, infinite-dimensional complex Hilbert space. Let $T \in \mathcal{B}(H)$ be a normal operator with spectrum $\sigma(T)$. Let $C^*(T)$ be the closed self-adjoint subalgebra of $\mathcal{B}(H)$ generated by $\{T, \text{id}\}$. Then there exists a unique isometric isomorphism $\varphi : C(\sigma(T)) \rightarrow C^*(T)$ that preserves adjoints (called a $*$ -isomorphism) such that the inclusion $\iota : \sigma(T) \rightarrow \mathbb{C}$ is sent to T and the constant function 1 is sent to id . We call φ the functional calculus at T . Note that $C^*(T) = \overline{\text{Span}\{T^n(T^*)^m : m, n \geq 0\}}$.

Remark

If $p \in C(\sigma(T))$ is a polynomial, $p = \sum_{i=0}^n \alpha_i z^i$, $\varphi(p) = \sum_{i=0}^n \alpha_i T^i$. We similarly write $f(T) = \varphi(f)$ for $f \in C(\sigma(T))$.

Definition

Let H be a complex Hilbert space. We say that $T \in \mathcal{B}(H)$ is positive if $\langle T(x), x \rangle \geq 0$ for all $x \in H$ (when $\mathbb{F} = \mathbb{R}$, we further require self-adjointness, but over $\mathbb{C}$ this implies self-adjointness). Note that T^*T is always positive for $T \in \mathcal{B}(H)$.

Theorem

Let H be a complex Hilbert space. For every positive $T \in \mathcal{B}(H)$, there exists a unique positive operator $S \in \mathcal{B}(H)$ with $S^2 = T$. We call S the square root of T .

Existence is shown using the spectral theorem, uniqueness is done by supposing there is another one and applying the functional calculus.

This completes the course. ■